

On Weak-to-Strong Framework in Decision Making

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We study whether weak-to-strong (W2S) learning phenomenon, previously studied for prediction tasks, exists in decision-making problems. This question is motivated by modern settings where labeled outcomes are scarce but unlabeled contexts are abundant, and practitioners have access to representations of different strengths (e.g., pre-trained representations of different sizes). We formulate this question through contextual stochastic optimization and propose a W2S framework tailored to downstream decision risk, in which a weak model trained on labeled data provides predicted outcome distributions that supervise a stronger model on unlabeled contexts. We establish a non-asymptotic upper bound on the excess decision risk of W2S, and a lower bound for the strong model trained on labeled data only, which serves as a benchmark. The comparison yields explicit conditions under which W2S strictly outperforms in terms of downstream decision performance. To be specific, W2S gains arise when the correlation dimension between the weak and strong feature representations is small. A controlled synthetic experiment and a comment moderation study based on the a real-world dataset corroborate the theory and illustrate regimes in which W2S outperforms strong-only training.

Key words: weak-to-strong; contextual stochastic optimization

1. Introduction

In many operational settings, decision-making relies on predictive models that estimate uncertain outcomes conditional on observable contexts. Examples include inventory planning with demand forecasts (Ban and Rudin 2019), including feature-rich newsvendor and inventory-learning models (Chang et al. 2025, Chen et al. 2021), dynamic pricing and revenue management with demand prediction (Cohen et al. 2020, Javanmard et al. 2024), including online retail pricing, contextual pricing, and joint pricing-inventory learning (Ferreira et al. 2016, Chen et al. 2026, 2022), routing with travel-time estimation (Guo et al. 2023), and portfolio allocation with return distribution modeling (Gu et al. 2020), alongside broader data-driven robust-optimization and offline-policy-learning settings (Wang et al. 2016, Zhou et al. 2023); see also (Bastani et al. 2022) for an overview of machine learning in operations management. All these settings can be formulated as a *contextual stochastic optimization* problem.

A growing literature studies how predictive models should be trained when their outputs feed into an optimization problem. The key insight is simple: predictive accuracy alone does not guar-

antee decision quality. Errors that are small under standard statistical losses can induce large downstream cost if they distort decision-critical directions. This observation has motivated decision-aware (Elmachtoub and Grigas 2022) and integrated learning frameworks (Qi et al. 2025) that explicitly align predictive training with operational objectives. In this work, we adopt the integrated conditional estimation-optimization (ICEO) framework proposed by Qi et al. (2025).

Yet an increasingly common practical challenge lies outside the standard formulation, which assumes sufficiently many labeled outcomes. In many applications, labeled outcomes are scarce and/or costly while contextual covariates are abundant. At the same time, practitioners often have access to multiple models or feature representations of different strengths. For example, they may have access to a smaller model that can be reliably pre-trained on limited labeled data and a larger model that is more expressive.

This raises a natural question which we aim to answer rigorously:

*Can we leverage a weak model trained on limited labeled data to guide the training of a stronger model using abundant unlabeled data, thereby improving **downstream decision** performance in contextual stochastic optimization?*

To be specific, suppose we are given two pre-trained models for a generic purpose. By *pre-training*, we mean training a model on a broad auxiliary objective before adapting it to the downstream decision task of interest. For example, large language models are often trained on broad text corpora using next-token prediction (Brown et al. 2020), which can produce broadly useful representations but does not specify how the model’s predictions should be used inside an operational decision problem. In this paper, we focus on the post-training stage. By *post-training*, we mean the downstream adaptation stage after pre-training. How should one use limited labeled decision data and abundant unlabeled contexts to adapt them for the downstream contextual stochastic optimization task? Consistent with empirical scaling laws for pre-training, under which model quality improves predictably with data and compute (Kaplan et al. 2020, Hoffmann et al. 2022), we focus on the regime where the available pre-training data and compute are sufficient to obtain a useful weak model, but not sufficient to obtain a strong model that is already reliable for the downstream decision task.

At a high level, the question above can be viewed as a problem of knowledge transfer under heterogeneous supervision. We have a weak but supervised model that captures part of the signal from limited labeled data, and a stronger representation that has higher expressive capacity but lacks sufficient supervision. In recent practice, one natural approach is to use the weak model’s outputs on abundant unlabeled contexts (i.e., pseudo-labels) as auxiliary supervision for training the stronger representation. Empirically, such procedures have sometimes been observed to improve the performance of the stronger model beyond that of its weak supervisor. This phenomenon has

recently been studied in machine learning literature under the name *weak-to-strong generalization* (Burns et al. 2024, Dong et al. 2025). While both empirical evidence and theoretical explanation have been provided in the context of classification and language modeling, it remains an open question whether such a phenomenon, if the algorithm is designed properly, can also be observed in the context of *decision-making*.

1.1. Our contribution

We summarize our contributions as follows:

1. **W2S formulation in decision making.** Existing W2S theory has largely focused on predictive objectives such as classification and regression, leaving open whether and when W2S can improve *decision making* problems. We address this gap by formulating W2S in the broader realm of decision making and instantiating it through contextual stochastic optimization, a canonical framework in Operations Research. This formulation is a first step toward a broader theory of W2S for decision making, and it provides a foundation for future extensions to richer settings such as sequential decision making and reinforcement learning.
2. **Algorithm and its theoretical understanding.** We propose a W2S algorithm that uses a weak model trained on limited labeled data to supervise a strong model on abundant unlabeled contexts. We analyze its downstream decision performance in finite samples in Theorem 1 and, through the comparison developed in Section 4, identify regimes in which this algorithm is certified to outperform training the strong model directly on the labeled data alone in Corollary 1. Intuitively, W2S gains arise when the weak and strong representations are sufficiently different, so that the weak model’s errors appear as noise to the strong model and can be averaged out with many pseudo-labels.
3. **Empirical evidence.** We complement the theory with two empirical studies. We first use a controlled synthetic experiment to test whether the regimes predicted by the theory are observed in simulation. We then evaluate W2S on a real-world comment moderation task, where predictions guide automatic and human-review moderation decisions. In both studies, the qualitative patterns corroborate the theory: W2S delivers its largest gains when labeled data are scarce and unlabeled contexts are plentiful, and these benefits attenuate as labels accumulate.

1.2. Relevant literature

Our work advances the frontier of several lines of research.

Contextual optimization. There has been a surge of interest in contextual stochastic optimization in the operations research community in recent years. This line of work studies how predictive models and optimization methods can be combined to improve decision-making under uncertainty. The survey paper (Sadana et al. 2025) provides a comprehensive review of the literature on this

topic, and classifies the existing works into three broad categories: decision rule optimization (Donti et al. 2017, Zhang and Gao 2017, Ban and Rudin 2019, Bertsimas and Koduri 2022, Huber et al. 2019), sequential learning and optimization (Deng and Sen 2022, Wang et al. 2026) and integrated learning and optimization (Elmachtoub and Grigas 2022, Qi et al. 2025, Loke et al. 2020). Our work falls within the third category. Elmachtoub and Grigas (2022) introduce the SPO/SPO+ framework and formalize the idea that predictive accuracy alone may be poorly aligned with downstream decision quality. El Balghiti et al. (2019) complement this line by establishing out-of-sample guarantees for decision-aware learning, while Elmachtoub et al. (2020) develop interpretable decision trees trained directly for optimization performance. More recently, Elmachtoub et al. (2023) compare estimate-then-optimize, integrated-estimation-optimization, and sample average approximation, helping clarify the relative strengths of these paradigms. Among these works, Qi et al. (2025) is closest to ours. They propose the integrated conditional estimation-optimization framework, which learns the conditional distribution of the uncertain outcome and evaluates it through the downstream decision risk of the induced plug-in policy. They show that this framework enjoys strong statistical guarantees. Our paper builds on this perspective and studies how a weak-to-strong training pipeline can be incorporated when labeled data are scarce but unlabeled contexts are abundant. Interested readers can also refer to the survey papers Qi and Shen (2022) and Sadana et al. (2025).

Weak-to-strong generalization. Burns et al. (2024) first formalize weak-to-strong generalization as the empirical phenomenon that a strong pretrained model, finetuned on labels produced by a much weaker model, can outperform its weak supervisor across tasks (NLP benchmarks, chess puzzles, and reward modeling). Lang et al. (2025) show that a debate stage, where strong models generate competing answers and a weak supervisor selects between them, improves pseudo-label quality. Interestingly, Goel et al. (2025) find that as models become more capable they make increasingly correlated mistakes, which can reduce the benefits of W2S training and pose risks for AI oversight because similar models have less complementary knowledge to transfer.

There is also a line of works focusing on theoretical understanding of W2S generalization. Our work adopts the perspective of Dong et al. (2025) and Liu et al. (2026). They propose a discrepancy-based framework to analyze the W2S generalization phenomenon. They find that W2S gains arise from variance reduction in low-dimensional finetuning. When teacher and student rely on different feature directions, the teacher’s errors appear as noise to the student and can be averaged out with many pseudo-labels. Hence, moderate discrepancy can improve W2S. A relevant intuition is also provided in Charikar et al. (2024), where they prove that the gain of W2S comes from the misfit between the weak and the strong model, i.e., the erroneous knowledge of the strong model is not inherited from the weak model. Other theoretical explanations include Lang et al. (2024), who

focuses more on a geometry perspective. They show that under an expansion condition (Cai et al. 2021) on the data graph, any student classifier that agrees with a large set of teacher mistakes must also disagree on many neighboring points (collateral mistakes), making it impossible to maintain low error while preserving those mistakes.

We note that the W2S generalization is also closely related to broader paradigms of learning from imperfect supervision, such as weakly supervised learning (Zhang et al. 2025), semi-supervised learning (Yang et al. 2022), self-training (Wei et al. 2021), etc. Semi-supervised learning (SSL) uses unlabeled samples to improve learning when labels are scarce. Classical approaches include co-training, graph-based label propagation, and entropy minimization (Blum and Mitchell 1998, Zhu et al. 2003, Grandvalet and Bengio 2004); modern deep SSL develops pseudo-labeling, self-training, and consistency regularization (Lee 2013, Wei et al. 2021, Sohn et al. 2020, Yang et al. 2022). Related work also studies weak supervision, prompt-based or weakly supervised labeling, robustness gains from unlabeled data, and sample-complexity limits (Zhang et al. 2025, Chen et al. 2025, Ben-David et al. 2008, Carmon et al. 2019, Wegel et al. 2025), with teacher-student transfer surveyed in the knowledge-distillation literature (Yang et al. 2025). Unlike these prediction-oriented methods, our W2S framework uses a weak model’s pseudo-distributions over outcomes to train a stronger representation and evaluates the induced policy through downstream contextual stochastic optimization risk. For broader background on teacher-student transfer mechanisms that overlap with W2S, we refer readers to a survey on knowledge distillation for LLMs (Yang et al. 2025).

Organization. In Section 2, we introduce the problem setting and the W2S procedure. Section 3 presents the main W2S decision-risk upper bound and its proof. Section 4 develops the benchmark and teacher-side estimates needed to interpret the main theorem, and uses them in a W2S performance case study. Section 5 presents numerical experiments on synthetic and text-based operational data to validate our theoretical findings.

Notation. We use $\lesssim$ to denote inequality up to a universal constant factor, independent of any problem parameters. For a positive integer m , let $[m] \stackrel{\text{def}}{=} \{1, \dots, m\}$. For a vector a , we use either a_k or $[a]_k$ to denote its k -th entry. For a symmetric matrix A , we write $A \succeq 0$ if A is positive semidefinite. Expectation subscripts indicate the source of randomness; for example, $\mathbb{E}_x[\cdot]$ denotes expectation with respect to x .

2. The Setting

In Section 2.1, we first overview our decision making problem at a high level. Then in Section 2.2, we explain in detail our W2S training framework.

2.1. Problem Setup

Contextual Stochastic Optimization. We consider a convex contextual stochastic optimization framework, which arises in many operations research applications. The feasible region for the decision variable w is a convex set $\mathcal{A} \subseteq \mathbb{R}^d$. The form of the cost function $c(\cdot, \xi) : \mathcal{A} \rightarrow \mathbb{R}$ is fixed and known to the decision maker; while c also depends on a random parameter $\xi \in \Xi$. In this work, we consider the case where the random parameter ξ has finite discrete support, namely, $\Xi = \{z_1, \dots, z_K\}$. The realized outcome ξ is not known at the time of decision-making. However, we assume that the decision-maker has access to a context vector $x \in \mathcal{X} \subseteq \mathbb{R}^{d_x}$ at the time of decision-making, and the goal is to make a decision w based on this context. We denote by $\mathcal{D}$ the joint distribution of (x, ξ) , and by $\mathcal{D}_x$ the marginal distribution of x . To emphasize when x denotes a random variable, rather than its realization, we write x in the subscript of the expectation operator.

Formally, the decision maker's goal is, when given a context vector x , to solve the contextual stochastic optimization problem:

$$\min_{w \in \mathcal{A}} \mathbb{E}_\xi [c(w, \xi) \mid x], \quad (1)$$

where the expectation is taken with respect to the conditional distribution of ξ given x . Hence, a policy $\pi : \mathcal{X} \rightarrow \mathcal{A}$ is a function that maps each context to a decision.

The labeled and unlabeled dataset. Since the conditional distribution of ξ given x is unknown, it must be learned from data. We consider a setting in which two datasets are available: one labeled data set $\tilde{\mathcal{B}} = \{\tilde{x}_i, \tilde{\xi}_i\}_{i=1}^n$ of size n , and one unlabeled dataset $\mathcal{B} = \{x_j\}_{j=1}^N$ of size N . We assume N is much larger than n , as is common in practice. The labeled dataset $\tilde{\mathcal{B}}$ is generated i.i.d. according to the ground-truth joint distribution $\mathcal{D}$, while the unlabeled dataset $\mathcal{B}$ is generated i.i.d. according to the ground-truth marginal distribution $\mathcal{D}_x$ of contexts, and hence $\mathcal{B}$ is independent of $\tilde{\mathcal{B}}$.

The two models. A central question, then, is how to leverage the large unlabeled dataset to improve decision quality. Inspired by Dong et al. (2025), we consider two (pre-trained) models that process the same context in different ways. Concretely, let us fix a common feature dimension d_ϕ . We consider two feature mappings (a.k.a. models) from the context space to this common feature space: a *strong* one $\phi_s : \mathcal{X} \rightarrow \mathbb{R}^{d_\phi}$ and a *weak* one $\phi_w : \mathcal{X} \rightarrow \mathbb{R}^{d_\phi}$. Given a parameter matrix $\Theta \in \mathbb{R}^{d_\phi \times K}$, these feature mappings can induce two parametric conditional models $P_\Theta^s, P_\Theta^w : \mathcal{X} \rightarrow \Delta(\Xi)$, obtained by applying a softmax to the logits $\Theta^\top \phi_s(x)$ and $\Theta^\top \phi_w(x)$, respectively, where $\Delta(\Xi)$ is the set of all probability distributions over Ξ . In our framework, the strong feature mapping $\phi_s(\cdot)$ is required to be more *expressive* in the sense that it can reproduce any representation induced by the weak feature mapping $\phi_w(\cdot)$. We will make it precise in Section 3.1.

In practice, the two feature mappings can arise from different pretrained models available to the decision maker. For example, a smaller model trained with limited resources and a larger

model trained with more data, compute, or capacity may process the same context but encode different information: the weak representation may capture coarse decision-relevant signals, while the strong representation may contain richer features that still require downstream adaptation. In our framework, we abstract these pretrained representations as fixed feature mappings ϕ_w and ϕ_s , and study how the downstream W2S procedure uses limited labeled data and abundant unlabeled contexts to adapt them for decision-making.

We now introduce the notion of *correlation dimension*, which turns out to be a key quantity that characterizes the W2S phenomenon in our setting. We denote $\Sigma_w \stackrel{\text{def}}{=} \mathbb{E}_{x \sim \mathcal{D}_x} [\phi_w(x) \phi_w(x)^\top]$ and $\Sigma_s \stackrel{\text{def}}{=} \mathbb{E}_{x \sim \mathcal{D}_x} [\phi_s(x) \phi_s(x)^\top]$.

DEFINITION 1 (INTRINSIC AND CORRELATION DIMENSIONS). We let $d_s = \text{rank}(\Sigma_s)$ and $d_w = \text{rank}(\Sigma_w)$ be the *intrinsic dimensions* of the strong and weak feature mappings, respectively. Consider spectral decompositions $\Sigma_s = V_s \Lambda_s V_s^\top \in \mathbb{R}^{d_\phi \times d_\phi}$ and $\Sigma_w = V_w \Lambda_w V_w^\top \in \mathbb{R}^{d_\phi \times d_\phi}$, where $\Lambda_s \in \mathbb{R}^{d_s \times d_s}$ and $\Lambda_w \in \mathbb{R}^{d_w \times d_w}$ are diagonal matrices with positive eigenvalues in non-increasing order; while $V_s \in \mathbb{R}^{d_\phi \times d_s}$ and $V_w \in \mathbb{R}^{d_\phi \times d_w}$ consist of the corresponding orthonormal eigenvectors. We denote $\Sigma_s^{-\frac{1}{2}} = V_s \Lambda_s^{-\frac{1}{2}} V_s^\top$ and $\Sigma_w^{-\frac{1}{2}} = V_w \Lambda_w^{-\frac{1}{2}} V_w^\top$. Let

$$d_{s \wedge w} \stackrel{\text{def}}{=} \left\| \Sigma_s^{-\frac{1}{2}} \mathbb{E}_{x \sim \mathcal{D}_x} [\phi_s(x) \phi_w(x)^\top] \Sigma_w^{-\frac{1}{2}} \right\|_F^2 \quad (2)$$

be the *correlation dimension* between ϕ_s and ϕ_w such that $0 \leq d_{s \wedge w} \leq \min\{d_s, d_w\}$.

The correlation dimension measures the alignment between the strong and weak features. Intuitively, a larger correlation dimension indicates that the two features share more common information. The fact that $0 \leq d_{s \wedge w} \leq \min\{d_s, d_w\}$ is proved in Lemma 6 in the Appendix.

Decision-making pipeline. To study the decision-making pipeline induced by these feature-based models, we next specify how a conditional model over outcomes generates a decision. For any conditional model $P : \mathcal{X} \rightarrow \Delta(\Xi)$, we denote by $\tilde{w} : \Delta(\Xi) \rightarrow \mathcal{A}$ the oracle policy (i.e., plug-in policy) that selects the optimal action for each given context x

$$\tilde{w}(P(x)) \in \arg \min_{w \in \mathcal{A}} \mathbb{E}_{\xi \sim P(x)} [c(w, \xi) | x]. \quad (3)$$

Such a decision pipeline is studied, for example, in Qi et al. (2025), where the induced empirical risk minimization (ERM) training is shown to enjoy favorable statistical guarantees in terms of decision risk.

Performance measure. Given a policy $\pi : \mathcal{X} \rightarrow \mathcal{A}$, we define its expected decision risk to be

$$R(\pi) = \mathbb{E}_{x, \xi \sim \mathcal{D}} [c(\pi(x), \xi)]. \quad (4)$$

Here, $\mathbb{E}_{x,\xi \sim \mathcal{D}}[\cdot]$ denotes the expectation with respect to $(x, \xi) \sim \mathcal{D}$. The benchmark decision risk is the optimal decision risk achieved by the oracle policy induced by the ground-truth conditional distribution P^* :

$$R^* = \mathbb{E}_{x,\xi \sim \mathcal{D}}[c(\tilde{w}(P^*(x)), \xi)], \quad (5)$$

where $P^*(x) \in \Delta(\Xi)$ denotes the ground-truth conditional distribution of ξ given x . It is shown in Qi et al. (2025) that the ground-truth conditional distribution P^* is the distribution that yields the lowest value for expression (5).

2.2. The W2S Framework

In the previous section, we have introduced the two feature mappings $\phi_s(\cdot)$ and $\phi_w(\cdot)$, acquired from pre-training. However, they are not yet specialized to the downstream decision problem. Next, we will explain how W2S pipeline can be used to adapt these two representations for the downstream decision problem.

At a high level, the W2S framework first trains a weak model $P_{\Theta_w}^w$ on the labeled dataset $\tilde{\mathcal{B}}$, and then uses its predicted outcome distributions as *supervision* to train a strong model $P_{\Theta_{w2s}}^s$ on the unlabeled dataset $\mathcal{B}$. The resulting strong model then induces a plug-in decision policy $x \mapsto \tilde{w}(P_{\Theta_{w2s}}^s(x))$.

Weak model. Within our W2S framework, a weak model $P_{\Theta}^w : \mathcal{X} \rightarrow \Delta(\Xi)$ is used to generate pseudo-distributions for the unlabeled data. We do not impose a specific (post-)training procedure for this weak model, except that it must be constructed solely from the labeled dataset $\tilde{\mathcal{B}}$ and based on the weak feature map $\phi_w(\cdot)$. We assume that its estimation error for the ground-truth model parameter is controlled in mean squared error, in a sense to be specified precisely below. One natural way to obtain such a weak model is via maximum likelihood estimation (MLE).

W2S model. Given a weak model $P_{\Theta_w}^w$ trained on the labeled dataset $\tilde{\mathcal{B}}$, the W2S framework trains the strong model by solving the following optimization problem:

$$\hat{\Theta}_{w2s} \in \arg \min_{\Theta \in \mathbb{R}^{d_{\phi} \times K}, \|\Theta\|_F \leq B} \frac{1}{N} \sum_{j=1}^N \mathbb{E}_{\xi \sim P_{\Theta_w}^w(\cdot | x_j)} [c(\tilde{w}(P_{\Theta}^s(x_j)), \xi)], \quad (6)$$

where $B > 0$ is some constant. Here, $P_{\Theta_w}^w(\cdot | x_j)$ denotes the conditional distribution over ξ given x_j , induced by the weak model. Thus, the weak model provides more than a scalar prediction: it specifies how likely each outcome is at a given context. The strong model then uses this soft information to learn a decision rule that behaves well across the outcomes deemed plausible by the teacher, thereby extracting supervision from unlabeled contexts.

We let

$$\hat{\pi}_{w2s}(x) \stackrel{\text{def}}{=} \tilde{w}(P_{\hat{\Theta}_{w2s}}^s(x)) \quad (7)$$

be the policy induced by the W2S training framework. We measure its generalization error via the expected decision risk over both $\tilde{\mathcal{B}}$ and $\mathcal{B}$. Namely,

$$\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} [R(\hat{\pi}_{\text{w2s}}) - R^*]. \quad (8)$$

Here, the expectation is taken with respect to the randomness from both the labeled data set $\tilde{\mathcal{B}}$ and the unlabeled data set $\mathcal{B}$.

Strong model. A natural benchmark is to train the strong model directly on the labeled sample $\tilde{\mathcal{B}}$, as empirical scaling laws motivate using stronger pretrained models when sufficient data and compute are available (Kaplan et al. 2020, Hoffmann et al. 2022, Hernandez et al. 2021). For the downstream decision-making problem, we can train the strong model by directly minimizing the empirical decision risk on the labeled data:

$$\hat{\Theta}_s \in \arg \min_{\Theta \in \mathbb{R}^{d_\phi \times K}} \frac{1}{n} \sum_{i=1}^n c\left(\tilde{w}(P_\Theta^s(\tilde{x}_i)), \tilde{\xi}_i\right). \quad (9)$$

Let $\hat{\pi}_s(x) := \tilde{w}(P_{\hat{\Theta}_s}^s(x))$ be the policy induced by the strong model trained on labeled data only. We measure its generalization error via the expected decision risk over $\tilde{\mathcal{B}}$ only. Namely,

$$\mathbb{E}_{\tilde{\mathcal{B}}} [R(\hat{\pi}_s) - R^*]. \quad (10)$$

Metric for W2S performance. We use the outperforming ratio (OPR) to compare the expected excess decision risk of the strong model and the W2S model:

$$\text{OPR} = \frac{\mathbb{E}_{\tilde{\mathcal{B}}} [R(\hat{\pi}_s) - R^*]}{\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} [R(\hat{\pi}_{\text{w2s}}) - R^*]}. \quad (11)$$

A higher OPR implies better performance of the W2S model compared to the strong model: $\hat{\pi}_{\text{w2s}}$ outperforms $\hat{\pi}_s$ when $\text{OPR} > 1$.

3. Analysis of the W2S Framework

This section presents the main theoretical result of the paper: a non-asymptotic upper bound on the expected decision risk of the W2S framework. We first state the structural assumptions and then prove the bound by decomposing the W2S risk into imitation, generalization, teacher-estimation, and approximation terms.

3.1. Blanket assumptions

Underlying problem structure We use $P^*(\cdot | x)$ to denote the ground-truth conditional distribution of ξ given x . Throughout this work, we make the following assumption on realizability.

ASSUMPTION 1 (**realizability**). *There exists a measurable logit mapping $\eta^* : \mathcal{X} \rightarrow \mathbb{R}^K$ such that*

$$P^*(\xi = z_k | x) = \frac{\exp(\eta_k^*(x))}{\sum_{j=1}^K \exp(\eta_j^*(x))}, \quad k = 1, \dots, K. \quad (12)$$

In particular, the ground-truth conditional probabilities are represented by finite softmax logits, and we assume $\mathbb{E} [\|\eta^(x)\|_2^4] < \infty$. Moreover, without loss of generality, we assume that $\eta^*(x)$ is centered, i.e., $\eta^*(x)^\top \mathbf{1} = 0$ for all $x \in \mathcal{X}$.*

Based on the realizability assumption, we hence consider a particular parametric model for the conditional distributions. Given a feature mapping $\phi : \mathcal{X} \rightarrow \mathbb{R}^{d_\phi}$, we let $\Theta \in \mathbb{R}^{d_\phi \times K}$ be the model parameter, and define the conditional model P_Θ to be

$$P_\Theta(\xi = z_k | x) = \frac{\exp([\eta_\Theta(x)]_k)}{\sum_{k'=1}^K \exp([\eta_\Theta(x)]_{k'})}, \quad \text{for } k = 1, \dots, K. \quad (13)$$

with linear feature mapping $\eta_\Theta(x) = \Theta^\top \phi(x) \in \mathbb{R}^K$. Note that linear feature mapping is commonly used in literature and it represents a broad class of models (Jacot et al. 2018, Woodworth et al. 2020, Aouad et al. 2025).

Moreover, we make the following assumption on the cost function.

ASSUMPTION 2 (**cost function**). 1. *For any action $a \in \mathcal{A}$, and outcome $z_k \in \Xi$, the cost function $c(a, z_k)$ is μ -strongly convex and L -smooth in the first argument, i.e.,*

$$\mu I \preceq \nabla_a^2 c(a, z_k) \preceq LI.$$

2. *Let $\mathcal{W} = \text{conv}\{\tilde{w}(p) : p \in \Delta(\Xi)\}$ be the convex hull of all plug-in decisions. The gradients of the scenario costs are uniformly bounded on $\mathcal{W}$: for all $a \in \mathcal{W}$ and $k \in [K]$,*

$$\|\nabla_a c(a, z_k)\|_2 \leq G.$$

In particular, since $\mathcal{W}$ is convex, for all $w, w' \in \mathcal{W}$ and $k \in [K]$,

$$|c(w, z_k) - c(w', z_k)| \leq G \|w - w'\|_2.$$

While the strong convexity assumption makes the analysis cleaner and more tractable, it is not essential regarding the outperforming phenomenon of W2S. It can be relaxed to convexity with some additional technical efforts, at the cost of slightly worse rates. Moreover, in Section 5, we show numerically that W2S outperforms the strong model even when the cost function is not strongly convex.

We consider the feasible region to be the whole space, i.e., $\mathcal{A} = \mathbb{R}^d$. This assumption is not essential regarding the outperforming phenomenon of W2S. It can be relaxed to a bounded convex set with some additional technical efforts, at the cost of slightly worse rates.

The two models. Recall that we are given two feature mappings, a strong one $\phi_s : \mathcal{X} \rightarrow \mathbb{R}^{d_\phi}$ and a weak one $\phi_w : \mathcal{X} \rightarrow \mathbb{R}^{d_\phi}$ with different model capacities. We can define the *approximation error* for the strong and weak feature model, respectively. Namely, we define

$$\rho_s = \min_{\Theta \in \mathbb{R}^{d_\phi \times K}, \Theta \mathbf{1} = 0} \mathbb{E}_{x \sim \mathcal{D}_x} \left[\left\| \Theta^\top \phi_s(x) - \eta^*(x) \right\|_2^2 \right], \quad (14)$$

with Θ_s^* being a minimizer. Similarly, we define

$$\rho_w = \min_{\Theta \in \mathbb{R}^{d_\phi \times K}, \Theta \mathbf{1} = 0} \mathbb{E}_{x \sim \mathcal{D}_x} \left[\left\| \Theta^\top \phi_w(x) - \eta^*(x) \right\|_2^2 \right], \quad (15)$$

with Θ_w^* being a minimizer.

We introduce some notation for the two models. For a parameter Θ , we denote the logits of the two feature mappings by $\eta_\Theta^s(x) = \Theta^\top \phi_s(x) \in \mathbb{R}^K$ and $\eta_\Theta^w(x) = \Theta^\top \phi_w(x) \in \mathbb{R}^K$. In addition, we denote $P_\Theta^s(\cdot | x) \stackrel{\text{def}}{=} \text{softmax}(\Theta^\top \phi_s(x))$ (and $P_\Theta^w(\cdot | x)$) to be the conditional model obtained by composing softmax with $\eta_\Theta^s(x)$ (and $\eta_\Theta^w(x)$), in the same way as in (13). We denote $\mathcal{H}_B \stackrel{\text{def}}{=} \{\Theta \in \mathbb{R}^{d_\phi \times K}, \|\Theta\|_F \leq B\}$.

We then quantify how well the strong feature class can reproduce the logits induced by the learned weak model. Specifically, given a weak model estimator $\hat{\Theta}_w$, we define the weak-to-strong logit approximation error to be

$$\rho_{s \rightarrow w}(\hat{\Theta}_w) \stackrel{\text{def}}{=} \inf_{\Theta \in \mathcal{H}_B} \mathbb{E}_x \left\| \Theta^\top \phi_s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2. \quad (16)$$

Namely, it measures the extent to which the strong feature representation can emulate the weak model at the logit level. As we will see in Theorem 1, we need the strong model to be more expressive than the weak model in the sense that $\rho_{s \rightarrow w}(\hat{\Theta}_w)$ is small.

To facilitate the analysis, we consider the following least-squares formulation to train the strong model based on the pseudo-labels (more precisely, pseudo-distributions over outcomes) generated by the weak model:

$$\hat{\Theta}_{\text{ls}} \in \arg \min_{\Theta \in \mathbb{R}^{d_\phi \times K}} \frac{1}{N} \sum_{j=1}^N \left\| \Theta^\top \phi_s(x_j) - \hat{\Theta}_w^\top \phi_w(x_j) \right\|_2^2. \quad (17)$$

ASSUMPTION 3. *The least-squares estimate $\hat{\Theta}_{\text{ls}}$ is bounded in Frobenius norm, i.e., $\left\| \hat{\Theta}_{\text{ls}} \right\|_F \leq B$ almost surely.*

Assumption 3 is made merely for technical convenience. It allows us to place the least-squares estimator $\hat{\Theta}_{\text{ls}}$ in the bounded class $\mathcal{H}_B$, which simplifies the empirical process arguments used in the proof. The assumption is not central to the weak-to-strong effect itself, and could be relaxed with additional technical efforts. For example, we can show that $\hat{\Theta}_{\text{ls}}$ is bounded with high probability when the features are subgaussian.

Moreover, we need the following assumption on the feature mappings, which controls the tail behavior of the features.

ASSUMPTION 4 (subgaussian features). *Let $\phi(x) = [\phi_s(x); \phi_w(x)] \in \mathbb{R}^{2d_\phi}$ be the concatenation of $\phi_s(x)$ and $\phi_w(x)$, with covariance matrix Σ_ϕ . There exists a constant $c > 0$ such that for any $u \in \mathbb{R}^{2d_\phi}$, we have*

$$\mathbb{E}_x [\exp(u^\top \phi(x))] \leq \exp\left(\frac{1}{2}c^2 u^\top \Sigma_\phi u\right). \quad (18)$$

This notion of subgaussianity is commonly used when it comes to covariance estimation. See, for example, Theorem 4.7.1 and Theorem 9.2.4 in (Vershynin 2018), as well as Lobo et al. (2024). The zero-mean assumption (implied by (18)) is merely for technical convenience, as we can always center the features without loss of generality.

3.2. W2S decision-risk upper bound

Now we are ready to present the main result of our work, which is a non-asymptotic upper bound on the expected decision risk of the W2S framework.

THEOREM 1. *Under Assumptions 1, 2, 3, 4, whenever $N \gtrsim d_s$, we have*

$$\begin{aligned} & \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} [R(\hat{\pi}_{w2s}) - R^*] \\ & \lesssim \left(\frac{L^2 K G^2}{\mu^3} + \frac{L K G^2}{\mu^2} \right) \cdot \left(\mathbb{E}_{\tilde{\mathcal{B}}} [\rho_{s \rightarrow w}(\hat{\Theta}_w)] \right. \\ & \quad + \frac{(B_\Theta^w)^2}{\sqrt{N}} \left(\mathbb{E}_x \|\phi_w(x)\|_2^4 \right)^{1/2} + \frac{B^2}{\sqrt{N}} \left(\mathbb{E}_x \|\phi_s(x)\|_2^4 \right)^{1/2} + \frac{B B_\Theta^w}{\sqrt{N}} \left(\mathbb{E}_x [\|\phi_s(x)\|_2^2 \|\phi_w(x)\|_2^2] \right)^{1/2} \\ & \quad + \frac{L K G^2}{\mu^2} \cdot \left(\frac{1}{\sqrt{N}} \left(\mathbb{E}_x [\|\eta^*(x)\|_2^4] \right)^{1/2} + B \sqrt{\text{tr}(\Sigma_s)} \frac{1}{\sqrt{N}} + \frac{B}{\sqrt{N}} \left(\mathbb{E}_x [\|\eta^*(x)\|_2^4] \right)^{1/4} \left(\mathbb{E}_x [\|\phi_s(x)\|_2^4] \right)^{1/4} \right. \\ & \quad \left. \left. + \lambda_{\max}(\Lambda_w) \mathbb{E}_{\tilde{\mathcal{B}}} \left[\left\| \hat{\Theta}_w - \Theta_w^* \right\|_F^2 \right] \left(d_{s \wedge w} + \frac{1}{N} d_s (d_w - d_{s \wedge w}) \right) + \rho_w + \rho_s \right) + \frac{L C_{\text{unif}}}{\mu} \frac{1}{\sqrt{N}}. \right. \end{aligned} \quad (19)$$

We recall that N is the number of unlabeled samples, K is the support size of the outcome variable ξ , and μ, L, G are the curvature and gradient parameters from Assumption 2. The quantity B is the F-norm bound on the least squares fit, and $d_{s \wedge w}$ is the correlation dimension between the strong and weak features. Here $C_{\text{unif}} \stackrel{\text{def}}{=} \max_{k \in [K]} |c(\tilde{w}(\text{softmax}(0)), z_k)|$.

To interpret the upper bound, it is useful to decompose the bound into four parts:

1. Imitation approximation error: $\mathcal{E}_{\text{imit}} \stackrel{\text{def}}{=} \left(\frac{L^2 K G^2}{\mu^3} + \frac{L K G^2}{\mu^2} \right) \cdot \mathbb{E}_{\tilde{\mathcal{B}}} [\rho_{s \rightarrow w}(\hat{\Theta}_w)]$. It measures how well the strong feature class can reproduce the logits induced by the weak model.

2. Unlabeled-sample statistical error: Let $\mathcal{E}_N$ denote the following unlabeled-sample statistical-error term:

$$\begin{aligned} \mathcal{E}_N \stackrel{\text{def}}{=} & \left(\frac{L^2 K G^2}{\mu^3} + \frac{L K G^2}{\mu^2} \right) \left[\frac{(B_{\Theta}^w)^2}{\sqrt{N}} \left(\mathbb{E}_x \|\phi_w(x)\|_2^4 \right)^{1/2} + \frac{B^2}{\sqrt{N}} \left(\mathbb{E}_x \|\phi_s(x)\|_2^4 \right)^{1/2} \right. \\ & \left. + \frac{B B_{\Theta}^w}{\sqrt{N}} \left(\mathbb{E}_x \left[\|\phi_s(x)\|_2^2 \|\phi_w(x)\|_2^2 \right] \right)^{1/2} \right] \\ & + \frac{L K G^2}{\mu^2} \left[\frac{1}{\sqrt{N}} \left(\mathbb{E}_x \left[\|\eta^*(x)\|_2^4 \right] \right)^{1/2} + B \sqrt{\text{tr}(\Sigma_s)} \frac{1}{\sqrt{N}} \right. \\ & \left. + \frac{B}{\sqrt{N}} \left(\mathbb{E}_x \left[\|\eta^*(x)\|_2^4 \right] \right)^{1/4} \left(\mathbb{E}_x \left[\|\phi_s(x)\|_2^4 \right] \right)^{1/4} \right] + \frac{L C_{\text{unif}}}{\mu \sqrt{N}}. \end{aligned}$$

This term captures the finite-sample fluctuation from replacing population expectations over x with empirical averages over the unlabeled sample $\mathcal{B}$; for fixed problem parameters, it decays at the rate $N^{-1/2}$. It comes from generalization error with respect to the unlabeled data.

3. Weak-to-strong term: $\mathcal{E}_{\text{teacher}} \stackrel{\text{def}}{=} \frac{L K G^2}{\mu^2} \lambda_{\max}(\Lambda_w) \mathbb{E}_{\mathcal{B}} \left[\left\| \hat{\Theta}_w - \Theta_w^* \right\|_F^2 \right] (d_{s \wedge w} + \frac{1}{N} d_s (d_w - d_{s \wedge w}))$. It captures how the weak teacher's error propagates to the final W2S decision risk. The persistent component is governed by the overlap dimension $d_{s \wedge w}$, whose contribution is not reduced by increasing either the unlabeled sample size N or the labeled sample size n . A smaller overlap dimension can reduce the error inherited from the weak teacher and thereby create room for the student to outperform the teacher. Intuitively, when ways of thinking of the teacher and the student are more different, from the student's perspective the teacher's mistakes behave more like random noise than like a systematic bias. The remaining component $\frac{1}{N} d_s (d_w - d_{s \wedge w})$ quantifies the residual contribution from the non-overlapping directions, and vanishes when the unlabeled sample size N is sufficiently large.
4. Model misspecification error: $\mathcal{E}_{\text{approx}} \stackrel{\text{def}}{=} \frac{L K G^2}{\mu^2} \cdot (\rho_w + \rho_s)$. It measures the approximation error of the strong and weak feature class, respectively. Unlike the unlabeled-sample statistical error, this term does not vanish with larger N unless the feature classes themselves are enriched.

REMARK 1 (NO FREE LUNCH). At first glance, when $\mathcal{E}_{\text{init}} = \mathcal{E}_{\text{approx}} = 0$, the bound may seem to suggest that, if $d_{s \wedge w} = 0$, increasing N can wash out the weak-teacher estimation error $\mathbb{E}_{\mathcal{B}} \left[\left\| \hat{\Theta}_w - \Theta_w^* \right\|_F^2 \right]$ and drive the W2S error arbitrarily low. This is not the case; the formal statement and proof are given in Appendix B.11.

3.3. Proof of Theorem 1

This section is devoted to the proof of Theorem 1. Proofs of the supporting technical lemmas are deferred to the appendix.

To start off the analysis, we first present a useful result, which is a consequence of the strong convexity and smoothness of the cost function.

LEMMA 1. *Under Part (1) of Assumption 2, consider two conditional distributions $P, P' : \mathcal{X} \rightarrow \Delta(\Xi)$. For any x , we have*

$$\frac{\mu}{2} \|\tilde{w}(P' | x) - \tilde{w}(P | x)\|_2^2 \leq \mathbb{E}_{\xi \sim P(\cdot | x)} [c(\tilde{w}(P' | x), \xi) - c(\tilde{w}(P | x), \xi)] \leq \frac{L}{2} \|\tilde{w}(P' | x) - \tilde{w}(P | x)\|_2^2. \quad (20)$$

The proof appears in Appendix B.1. Moreover, the following observation is also useful for the analysis.

LEMMA 2. *Under Assumption 2 and the assumption that $\mathcal{A} = \mathbb{R}^d$, given two logits $\eta, \eta' : \mathcal{X} \rightarrow \mathbb{R}^K$, let P, P' be the corresponding softmax distributions, namely $P = \text{softmax}(\eta)$ and $P' = \text{softmax}(\eta')$. For any given $x \in \mathcal{X}$, we have*

$$\|\tilde{w}(P_\eta | x) - \tilde{w}(P_{\eta'} | x)\|_2 \leq \frac{G\sqrt{K}}{2\mu} \|\eta(x) - \eta'(x)\|_2. \quad (21)$$

The proof appears in Appendix B.2.

Consider a policy $\pi = \tilde{w} \circ P$, where $P : \mathcal{X} \rightarrow \Delta(\Xi)$ is some conditional model. In view of Lemma 1, we conclude that

$$R(\pi) - R^* = \mathbb{E}_x [\mathbb{E}_{\xi | x} [c(\tilde{w}(P | x), \xi) - c(\tilde{w}(P^* | x), \xi)]] \quad (22)$$

$$\begin{aligned} &= \mathbb{E}_x \left[\sum_{k=1}^K P^*(\xi = z_k | x) [c(\tilde{w}(P | x), z_k) - c(\tilde{w}(P^* | x), z_k)] \right] \\ &\leq \frac{L}{2} \mathbb{E}_x [\|\tilde{w}(P | x) - \tilde{w}(P^* | x)\|_2^2]. \end{aligned} \quad (23)$$

Therefore, to control $\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} [R(\hat{\pi}_{w2s}) - R^*]$, it suffices to upper bound $\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{w2s}}^s | x \right) - \tilde{w} (P^* | x) \right\|_2^2 \right] \right]$. To this end, we recall that $\hat{\Theta}_{1s}$, defined in (17), is obtained by the multivariate least-squares procedure trained using the strong model based on the pseudo-labels generated by the weak model. We proceed to use $\hat{\Theta}_{1s}$ as an intermediate quantity for the analysis. Hence, we consider the following decomposition

$$\begin{aligned} &\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{w2s}}^s | x \right) - \tilde{w} (P^* | x) \right\|_2^2 \right] \\ &\leq 2\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{w2s}}^s | x \right) - \tilde{w} \left(P_{\hat{\Theta}_{1s}}^s | x \right) \right\|_2^2 \right] + 2\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{1s}}^s | x \right) - \tilde{w} (P^* | x) \right\|_2^2 \right]. \end{aligned} \quad (24)$$

In what follows, we bound the two terms on the right-hand side separately.

Step 1: the first term in (24). Let us focus on the first term $\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{w2s}}^s | x \right) - \tilde{w} \left(P_{\hat{\Theta}_{1s}}^s | x \right) \right\|_2^2 \right]$. Inside the norm is the difference between the actions recommended by two different training procedures (cf. (6) and (17)), both of which are based

upon a given empirical weak teacher $P_{\Theta_w}^w$. It turns out that this quantity can be controlled as the number of unlabeled samples N increases, at a rate enjoyed by standard generalization bounds for empirical risk minimization.

The remainder of this step is devoted to proving the following bound.

$$\begin{aligned}
& \mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_{w2s}}^s \mid x \right) - \tilde{w} \left(P_{\Theta_{ls}}^s \mid x \right) \right\|_2^2 \right] \right] \\
& \leq \left(\frac{1}{2} \frac{LKG^2}{\mu^3} + \frac{1}{2} \frac{KG^2}{\mu^2} \right) \cdot \left(\mathbb{E}_{\tilde{\mathcal{B}}} \left[\inf_{\Theta \in \mathcal{H}_B} \mathbb{E}_x \left\| \Theta^\top \phi_s(x) - \eta_{\Theta_w}^w(x) \right\|_2^2 \right] \right. \\
& \quad \left. + \frac{2(B_\Theta^w)^2}{\sqrt{N}} \sqrt{\mathbb{E}_x \|\phi_w(x)\|_2^4} + \frac{8B^2}{\sqrt{N}} \left(\mathbb{E}_x \|\phi_s(x)\|_2^4 \right)^{\frac{1}{2}} + \frac{16BB_\Theta^w}{\sqrt{N}} \sqrt{\mathbb{E}_x \left[\|\phi_s(x)\|_2^2 \|\phi_w(x)\|_2^2 \right]} \right) \\
& \quad + 16\sqrt{2} \frac{KG^2B\sqrt{\text{tr}(\Sigma_s)}}{\mu^2} \frac{1}{\sqrt{N}} + \frac{8C_{\text{unif}}}{\mu} \frac{1}{\sqrt{N}}. \tag{25}
\end{aligned}$$

First, using the fact that $\|a+b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$, we have

$$\begin{aligned}
& \mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_{w2s}}^s \mid x \right) - \tilde{w} \left(P_{\Theta_{ls}}^s \mid x \right) \right\|_2^2 \right] \right] \\
& \leq 2\mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_{w2s}}^s \mid x \right) - \tilde{w} \left(P_{\Theta_w}^w \mid x \right) \right\|_2^2 \right] \right] + 2\mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_w}^w \mid x \right) - \tilde{w} \left(P_{\Theta_{ls}}^s \mid x \right) \right\|_2^2 \right] \right] \tag{26}
\end{aligned}$$

It turns out that first term in (26) can be controlled by logits difference between the least-squares strong model and the weak model, proved in Lemma 7 in Appendix B.3. Namely, we have

$$\begin{aligned}
& \mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_{w2s}}^s \mid x \right) - \tilde{w} \left(P_{\Theta_w}^w \mid x \right) \right\|_2^2 \right] \right] \\
& \leq \frac{1}{4} \frac{LKG^2}{\mu^3} \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \eta_{\Theta_{ls}}^s(x) - \eta_{\Theta_w}^w(x) \right\|_2^2 \right] \right] + 8\sqrt{2} \frac{KG^2B\sqrt{\text{tr}(\Sigma_s)}}{\mu^2} \frac{1}{\sqrt{N}} + \frac{4C_{\text{unif}}}{\mu} \frac{1}{\sqrt{N}}. \tag{27}
\end{aligned}$$

Building on (26) and (27), and invoking Lemma 2, we have

$$\begin{aligned}
& \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_{w2s}}^s \mid x \right) - \tilde{w} \left(P_{\Theta_{ls}}^s \mid x \right) \right\|_2^2 \right] \right] \\
& \leq \frac{1}{2} \frac{LKG^2}{\mu^3} \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \eta_{\Theta_{ls}}^s(x) - \eta_{\Theta_w}^w(x) \right\|_2^2 \right] \right] + 16\sqrt{2} \frac{KG^2B\sqrt{\text{tr}(\Sigma_s)}}{\mu^2} \frac{1}{\sqrt{N}} + \frac{8C_{\text{unif}}}{\mu} \frac{1}{\sqrt{N}} \\
& \quad + 2\mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_w}^w \mid x \right) - \tilde{w} \left(P_{\Theta_{ls}}^s \mid x \right) \right\|_2^2 \right] \right] \\
& \leq \left(\frac{1}{2} \frac{LKG^2}{\mu^3} + \frac{1}{2} \frac{KG^2}{\mu^2} \right) \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \eta_{\Theta_{ls}}^s(x) - \eta_{\Theta_w}^w(x) \right\|_2^2 \right] \right] + 16\sqrt{2} \frac{KG^2B\sqrt{\text{tr}(\Sigma_s)}}{\mu^2} \frac{1}{\sqrt{N}} + \frac{8C_{\text{unif}}}{\mu} \frac{1}{\sqrt{N}} \tag{28}
\end{aligned}$$

Now, it suffices to upper bound the quantity $\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \eta_{\Theta_{ls}}^s(x) - \eta_{\Theta_w}^w(x) \right\|_2^2 \right] \right]$.

Given $\eta_{\Theta_w}^w(x)$, we can write

$$\mathbb{E}_x \left[\left\| \eta_{\Theta_{ls}}^s(x) - \eta_{\Theta_w}^w(x) \right\|_2^2 \right]$$

$$= \inf_{\Theta \in \mathcal{H}_B} \mathbb{E}_x \left[\left\| \Theta^\top \phi_s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] + \mathbb{E}_x \left[\left\| \eta_{\hat{\Theta}_{ls}}^s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] - \inf_{\Theta \in \mathcal{H}_B} \mathbb{E}_x \left[\left\| \Theta^\top \phi_s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] \quad (29)$$

To proceed, let us consider the population and the empirical multivariate least-squares objectives:

$$L_{ls}(\Theta) \stackrel{\text{def}}{=} \mathbb{E}_x \left[\left\| \Theta^\top \phi_s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] \quad \text{and} \quad \hat{L}_{ls}(\Theta) \stackrel{\text{def}}{=} \frac{1}{N} \sum_{j=1}^N \left\| \Theta^\top \phi_s(x_j) - \eta_{\hat{\Theta}_w}^w(x_j) \right\|_2^2.$$

Recalling $\eta_{\hat{\Theta}_{ls}}^s(x) = (\hat{\Theta}_{ls})^\top \phi_s(x)$, we see that our goal is exactly to upper bound $\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \eta_{\hat{\Theta}_{ls}}^s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] \right] = \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[L_{ls}(\hat{\Theta}_{ls}) \right]$.

Let $\Theta_* \in \text{argmin}_{\Theta \in \mathcal{H}_B} L_{ls}(\Theta)$ be any population minimizer. Because $\hat{\Theta}_{ls}$ minimizes $\hat{L}_{ls}$, we have $\hat{L}_{ls}(\hat{\Theta}_{ls}) \leq \hat{L}_{ls}(\Theta_*)$. In addition, since we assume that $\hat{\Theta}_{ls} \in \mathcal{H}_B$ almost surely, we can write

$$\begin{aligned} L_{ls}(\hat{\Theta}_{ls}) - L_{ls}(\Theta_*) &= \left(L_{ls}(\hat{\Theta}_{ls}) - \hat{L}_{ls}(\hat{\Theta}_{ls}) \right) + \left(\hat{L}_{ls}(\hat{\Theta}_{ls}) - \hat{L}_{ls}(\Theta_*) \right) + \left(\hat{L}_{ls}(\Theta_*) - L_{ls}(\Theta_*) \right) \\ &\leq \sup_{\Theta \in \mathcal{H}_B} \left| L_{ls}(\Theta) - \hat{L}_{ls}(\Theta) \right| + 0 + \sup_{\Theta \in \mathcal{H}_B} \left| L_{ls}(\Theta) - \hat{L}_{ls}(\Theta) \right| \\ &= 2 \sup_{\Theta \in \mathcal{H}_B} \left| L_{ls}(\Theta) - \hat{L}_{ls}(\Theta) \right|. \end{aligned}$$

Taking conditional expectations on both sides yields that

$$\mathbb{E}_{\tilde{\mathcal{B}}|\tilde{\mathcal{B}}} \left[L_{ls}(\hat{\Theta}_{ls}) \right] \leq \inf_{\Theta \in \mathcal{H}_B} L_{ls}(\Theta) + 2 \mathbb{E}_{\tilde{\mathcal{B}}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| L_{ls}(\Theta) - \hat{L}_{ls}(\Theta) \right| \right]. \quad (30)$$

To proceed, we need the following lemma, which is a generalization bound for multivariate regression with squared loss.

LEMMA 3. *Conditioned on the labeled dataset $\tilde{\mathcal{B}}$, we have*

$$\begin{aligned} &\mathbb{E}_{\tilde{\mathcal{B}}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| L_{ls}(\Theta) - \hat{L}_{ls}(\Theta) \right| \right] \\ &\leq \frac{(B_{\Theta}^w)^2}{\sqrt{N}} \left(\mathbb{E}_x \left\| \phi_w(x) \right\|_2^4 \right)^{1/2} + \frac{4B^2}{\sqrt{N}} \left(\mathbb{E}_x \left\| \phi_s(x) \right\|_2^4 \right)^{1/2} + \frac{8BB_{\Theta}^w}{\sqrt{N}} \left(\mathbb{E}_x \left[\left\| \phi_w(x) \right\|_2^2 \left\| \phi_s(x) \right\|_2^2 \right] \right)^{1/2}. \end{aligned} \quad (31)$$

The proof is deferred to Appendix B.8.

Invoking Lemma 3 and taking expectation over $\tilde{\mathcal{B}}$ on both sides of (30), we have

$$\begin{aligned} &\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left\| \eta_{\hat{\Theta}_{ls}}^s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] \\ &\leq \mathbb{E}_{\tilde{\mathcal{B}}} \left[\inf_{\Theta \in \mathcal{H}_B} \mathbb{E}_x \left\| \Theta^\top \phi_s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] \\ &\quad + \frac{2(B_{\Theta}^w)^2}{\sqrt{N}} \sqrt{\mathbb{E}_x \left\| \phi_w(x) \right\|_2^4} + \frac{8B^2}{\sqrt{N}} \sqrt{\mathbb{E}_x \left\| \phi_s(x) \right\|_2^4} + \frac{16BB_{\Theta}^w}{\sqrt{N}} \sqrt{\mathbb{E}_x \left[\left\| \phi_s(x) \right\|_2^2 \left\| \phi_w(x) \right\|_2^2 \right]}. \end{aligned} \quad (32)$$

Step 1 now is complete.

Step 2: the second term in (24). Now, let us turn attention to the term $\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{\text{ls}}}^s \mid x \right) - \tilde{w} \left(P^* \mid x \right) \right\|_2^2 \right] \right]$. We proceed to show that whenever $N \gtrsim d_s$, we have

$$\begin{aligned} & \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{\text{ls}}}^s \mid x \right) - \tilde{w} \left(P^* \mid x \right) \right\|_2^2 \right] \right] \\ & \lesssim \frac{G^2 K}{\mu^2} \left(\lambda_{\max}(\Lambda_w) \mathbb{E}_{\tilde{\mathcal{B}}} \left[\left\| \hat{\Theta}_w - \Theta_w^* \right\|_F^2 \right] \left(d_{s \wedge w} + \frac{1}{N} d_s (d_w - d_{s \wedge w}) \right) + \rho_w + \rho_s \right) \\ & \quad + \frac{1}{\sqrt{N}} \frac{G^2 K}{\mu^2} \left(\left(\mathbb{E} \left[\|\eta^*(x)\|_2^4 \right] \right)^{\frac{1}{2}} + B^2 \left(\mathbb{E} \left[\|\phi_s(x)\|_2^4 \right] \right)^{\frac{1}{2}} + B \left(\mathbb{E} \left[\|\eta^*(x)\|_2^4 \right] \right)^{\frac{1}{4}} \left(\mathbb{E} \left[\|\phi_s(x)\|_2^4 \right] \right)^{\frac{1}{4}} \right) \end{aligned} \quad (33)$$

To proceed, invoking Lemma 2 yields that

$$\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\hat{\Theta}_{\text{ls}}}^s \mid x \right) - \tilde{w} \left(P^* \mid x \right) \right\|_2^2 \right] \leq \frac{G^2 K}{4\mu^2} \mathbb{E}_x \left[\left\| \eta_{\hat{\Theta}_{\text{ls}}}^s(x) - \eta^*(x) \right\|_2^2 \right]. \quad (34)$$

Let us consider the population and the empirical least-squares objectives:

$$L(\Theta) \stackrel{\text{def}}{=} \mathbb{E}_x \left[\left\| \Theta^\top \phi_s(x) - \eta^*(x) \right\|_2^2 \right] \quad \text{and} \quad \hat{L}(\Theta) \stackrel{\text{def}}{=} \frac{1}{N} \sum_{j=1}^N \left\| \Theta^\top \phi_s(x_j) - \eta^*(x_j) \right\|_2^2. \quad (35)$$

Recall $\eta_{\hat{\Theta}_{\text{ls}}}^s(x) = \hat{\Theta}_{\text{ls}}^\top \phi_s(x) \in \mathbb{R}^K$. Hence, we have

$$\mathbb{E}_x \left[\left\| \eta_{\hat{\Theta}_{\text{ls}}}^s(x) - \eta^*(x) \right\|_2^2 \right] \leq \frac{1}{N} \sum_{j=1}^N \left\| \eta_{\hat{\Theta}_{\text{ls}}}^s(x_j) - \eta^*(x_j) \right\|_2^2 + \sup_{\Theta \in \mathcal{H}_B} |L(\Theta) - \hat{L}(\Theta)|. \quad (36)$$

For the latter term, following the same reasoning as in the proof of Lemma 3 with $g(x) = \eta^*(x)$ yields that

$$\begin{aligned} & \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} |L(\Theta) - \hat{L}(\Theta)| \right] \\ & \leq \frac{1}{\sqrt{N}} \left(\left(\mathbb{E} \left[\|\eta^*(x)\|_2^4 \right] \right)^{\frac{1}{2}} + 4B^2 \left(\mathbb{E} \left[\|\phi_s(x)\|_2^4 \right] \right)^{\frac{1}{2}} + 8B \left(\mathbb{E} \left[\|\eta^*(x)\|_2^4 \right] \right)^{\frac{1}{4}} \left(\mathbb{E} \left[\|\phi_s(x)\|_2^4 \right] \right)^{\frac{1}{4}} \right). \end{aligned} \quad (37)$$

Hence, it remains to control the first term in (36).

To this end, we need to introduce some notation. For the labeled data set $\tilde{\mathcal{B}} = \{\tilde{x}_i, \tilde{\xi}_i\}_{i=1}^n$, we define the feature matrices $\tilde{\Phi}_w, \tilde{\Phi}_s \in \mathbb{R}^{n \times d_\phi}$ with $[\tilde{\Phi}_w]_{i,:} = \phi_w(\tilde{x}_i)^\top$ and $[\tilde{\Phi}_s]_{i,:} = \phi_s(\tilde{x}_i)^\top$. For the unlabeled data set $\mathcal{B} = \{x_j\}_{j=1}^N$, we define the feature matrices $\Phi_w, \Phi_s \in \mathbb{R}^{N \times d_\phi}$ with $[\Phi_w]_{j,:} = \phi_w(x_j)^\top$ and $[\Phi_s]_{j,:} = \phi_s(x_j)^\top$.

We denote $\eta^*(X) \in \mathbb{R}^{N \times K}$ as the matrix whose j -th row is $\eta^*(x_j)^\top$. Denote $P_s = \Phi_s \Phi_s^\dagger$, where $\dagger$ represents the Moore-Penrose pseudo-inverse. We denote $\eta_{\hat{\Theta}_{\text{ls}}}^s(X) \in \mathbb{R}^{N \times K}$ as the matrix whose j -th row is $\eta_{\hat{\Theta}_{\text{ls}}}^s(x_j)$. Then we can write $\sum_{j=1}^N \left\| \eta_{\hat{\Theta}_{\text{ls}}}^s(x_j) - \eta^*(x_j) \right\|_2^2 = \left\| \eta_{\hat{\Theta}_{\text{ls}}}^s(X) - \eta^*(X) \right\|_F^2$. Recalling the definition of $\hat{\Theta}_{\text{ls}}$ in (17), we have $\eta_{\hat{\Theta}_{\text{ls}}}^s(X) = \Phi_s \hat{\Theta}_{\text{ls}} = \Phi_s \Phi_s^\dagger \eta_{\hat{\Theta}_w}^w(X) = P_s \Phi_w \hat{\Theta}_w$. Denote the residual matrices by $R_w(X) = \eta^*(X) - \Phi_w \Theta_w^* \in \mathbb{R}^{N \times K}$ and $R_s(X) = \eta^*(X) - \Phi_s \Theta_s^* \in \mathbb{R}^{N \times K}$, where we recall that Θ_w^* and Θ_s^* are defined in (14) and (15). Then, clearly we have $\mathbb{E}_{\mathcal{B}} \left[\|R_w(X)\|_F^2 \right] = N \rho_w$.

Similarly, we have $\mathbb{E}_{\mathcal{B}} [\|R_s(X)\|_F^2] = N\rho_s$. By adding and subtracting terms, we can write the decomposition

$$\begin{aligned}\eta_{\widehat{\Theta}_{\text{ls}}}^s(X) - \eta^*(X) &= P_s \left(\Phi_w \widehat{\Theta}_w - \eta^*(X) \right) - (I - P_s) \eta^*(X) \\ &= P_s \left(\Phi_w \widehat{\Theta}_w - \Phi_w \Theta_w^* + \Phi_w \Theta_w^* - \eta^*(X) \right) - (I - P_s) (\Phi_s \Theta_s^* + R_s(X)) \\ &= P_s \Phi_w \left(\widehat{\Theta}_w - \Theta_w^* \right) - P_s R_w(X) - (I - P_s) R_s(X).\end{aligned}\quad (38)$$

Since P_s is an orthogonal projection, we have

$$\begin{aligned}\mathbb{E}_{\mathcal{B}} [\|P_s R_w(X)\|_F^2] &\leq \mathbb{E}_{\mathcal{B}} [\|R_w(X)\|_F^2] = N\rho_w, \\ \mathbb{E}_{\mathcal{B}} [\|(I - P_s) R_s(X)\|_F^2] &\leq \mathbb{E}_{\mathcal{B}} [\|R_s(X)\|_F^2] = N\rho_s.\end{aligned}\quad (39)$$

Combining (38) and (39), we have

$$\begin{aligned}&\mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\left\| \eta_{\widehat{\Theta}_{\text{ls}}}^s(X) - \eta^*(X) \right\|_F^2 \right] \\ &\leq 3\mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\left\| P_s \Phi_w \left(\widehat{\Theta}_w - \Theta_w^* \right) \right\|_F^2 \right] + 3\mathbb{E}_{\mathcal{B}} [\|P_s R_w(X)\|_F^2] + 3\mathbb{E}_{\mathcal{B}} [\|(I - P_s) R_s(X)\|_F^2] \\ &\leq 3\mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\left\| P_s \Phi_w \left(\widehat{\Theta}_w - \Theta_w^* \right) \right\|_F^2 \right] + 3N\rho_w + 3N\rho_s.\end{aligned}\quad (40)$$

We proceed to upper bound the first term in (40). Let $\Delta = \widehat{\Theta}_w - \Theta_w^* \in \mathbb{R}^{d_\phi \times K}$. Recall the spectral decomposition $\Sigma_w = V_w \Lambda_w V_w^\top$ and $\Sigma_s = V_s \Lambda_s V_s^\top$. We define $\gamma_s(x) = \Lambda_s^{-\frac{1}{2}} V_s^\top \phi_s(x) \in \mathbb{R}^{d_s}$ and $\gamma_w(x) = \Lambda_w^{-\frac{1}{2}} V_w^\top \phi_w(x) \in \mathbb{R}^{d_w}$. One can verify that $\gamma_s(x)$ and $\gamma_w(x)$ are random vectors with zero mean and identity covariance. Indeed, for instance, $\mathbb{E}[\gamma_s(x) \gamma_s(x)^\top] = \mathbb{E}[\Lambda_s^{-\frac{1}{2}} V_s^\top \phi_s(x) \phi_s(x)^\top V_s \Lambda_s^{-\frac{1}{2}}] = \Lambda_s^{-\frac{1}{2}} V_s^\top V_s \Lambda_s V_s^\top V_s \Lambda_s^{-\frac{1}{2}} = I_{d_s}$. We define the whitened design matrices to be

$$\Gamma_s = \Phi_s V_s \Lambda_s^{-\frac{1}{2}} \in \mathbb{R}^{N \times d_s}, \quad \Gamma_w = \Phi_w V_w \Lambda_w^{-\frac{1}{2}} \in \mathbb{R}^{N \times d_w}.$$

Hence, row i of Γ_s is exactly $\gamma_s(x_i)^\top$, and row i of Γ_w is $\gamma_w(x_i)^\top$. This way, we have the exact expression $\Phi_w = \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top$ and $\Phi_s = \Gamma_s \Lambda_s^{\frac{1}{2}} V_s^\top$. In addition, we denote $\Delta_w = \Lambda_w^{\frac{1}{2}} V_w^\top \Delta$. Therefore, we can write

$$P_s \Phi_w \left(\widehat{\Theta}_w - \Theta_w^* \right) = P_s \Phi_w \Delta = P_s \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top \Delta = P_s \Gamma_w \Delta_w. \quad (41)$$

Using the fact $\|AB\|_F^2 = \text{tr}(B^\top A^\top AB)$ and P_s is an orthogonal projection matrix, we further have

$$\left\| P_s \Phi_w \left(\widehat{\Theta}_w - \Theta_w^* \right) \right\|_F^2 = \|P_s \Gamma_w \Delta_w\|_F^2 = \text{tr}(\Delta_w^\top \Gamma_w^\top P_s^\top P_s \Gamma_w \Delta_w) = \text{tr}(\Delta_w^\top \Gamma_w^\top P_s \Gamma_w \Delta_w) = \text{tr}(\Gamma_w^\top P_s \Gamma_w \Delta_w \Delta_w^\top). \quad (42)$$

Next, we make the following derivation:

$$\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\left\| P_s \Phi_w \Delta \right\|_F^2 \right]$$

$$\begin{aligned}
&\stackrel{(a)}{=} \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w \Delta_w \Delta_w^\top)] \\
&= \mathbb{E}_{\mathcal{B}} [\mathbb{E}_{\tilde{\mathcal{B}}|\mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w \Delta_w \Delta_w^\top) | \mathcal{B}]] \\
&\stackrel{(b)}{=} \mathbb{E}_{\mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w \mathbb{E}_{\tilde{\mathcal{B}}|\mathcal{B}} [\Delta_w \Delta_w^\top | \mathcal{B}])] \\
&\stackrel{(c)}{=} \mathbb{E}_{\mathcal{B}} \left[\text{tr} \left(\Gamma_w^\top P_s \Gamma_w \mathbb{E}_{\tilde{\mathcal{B}}|\mathcal{B}} \left[\Lambda_w^{\frac{1}{2}} V_w^\top \Delta \Delta^\top V_w \Lambda_w^{\frac{1}{2}} | \mathcal{B} \right] \right) \right] \\
&\stackrel{(d)}{=} \mathbb{E}_{\mathcal{B}} \left[\text{tr} \left(\Gamma_w^\top P_s \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top \mathbb{E}_{\tilde{\mathcal{B}}|\mathcal{B}} [\Delta \Delta^\top | \mathcal{B}] V_w \Lambda_w^{\frac{1}{2}} \right) \right] \\
&\stackrel{(e)}{=} \mathbb{E}_{\mathcal{B}} \left[\text{tr} \left(\Gamma_w^\top P_s \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top \mathbb{E}_{\tilde{\mathcal{B}}} [\Delta \Delta^\top] V_w \Lambda_w^{\frac{1}{2}} \right) \right] \\
&\stackrel{(f)}{=} \mathbb{E}_{\mathcal{B}} \left[\text{tr} \left(V_w \Lambda_w^{\frac{1}{2}} \Gamma_w^\top P_s \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top \mathbb{E}_{\tilde{\mathcal{B}}} [\Delta \Delta^\top] \right) \right] \\
&\stackrel{(g)}{\leq} \mathbb{E}_{\mathcal{B}} \left[\text{tr} \left(V_w \Lambda_w^{\frac{1}{2}} \Gamma_w^\top P_s \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top \right) \lambda_{\max} (\mathbb{E}_{\tilde{\mathcal{B}}} [\Delta \Delta^\top]) \right] \\
&\stackrel{(h)}{=} \mathbb{E}_{\mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w \Lambda_w)] \lambda_{\max} (\mathbb{E}_{\tilde{\mathcal{B}}} [\Delta \Delta^\top]) \\
&\stackrel{(i)}{\leq} \mathbb{E}_{\mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w)] \lambda_{\max} (\Lambda_w) \lambda_{\max} (\mathbb{E}_{\tilde{\mathcal{B}}} [\Delta \Delta^\top]) \\
&= \mathbb{E}_{\mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w)] \lambda_{\max} (\Lambda_w) \lambda_{\max} (\mathbb{E}_{\tilde{\mathcal{B}}} [\Delta \Delta^\top]). \tag{43}
\end{aligned}$$

Equation (a) simply follows from (42). Equation (b) holds due to the linearity property of trace. Equation (c) is by the definition of Δ_w . Equation (d) holds since both $\Lambda_w^{\frac{1}{2}}$ and V_w are deterministic matrices. Equation (e) is due to the fact that $\mathcal{B}$ and $\tilde{\mathcal{B}}$ are independent. Equation (f) follows from the cyclic property of trace. The inequality (g) and (i) follow from the fact that for psd matrices A, B , we have $\text{tr}(AB) \leq \lambda_{\max}(A) \text{tr}(B)$. Equation (h) is due to $\text{tr} \left(V_w \Lambda_w^{\frac{1}{2}} \Gamma_w^\top P_s \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top \right) = \text{tr} \left(\Gamma_w^\top P_s \Gamma_w \Lambda_w^{\frac{1}{2}} V_w^\top V_w \Lambda_w^{\frac{1}{2}} \right) = \text{tr} (\Gamma_w^\top P_s \Gamma_w \Lambda_w)$, which follows from the cyclic property of trace and the fact that V_w is an orthogonal matrix.

In what follows, we proceed to upper bound (43). We proceed in two steps.

- First, we deal with the quantity $\mathbb{E}_{\mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w)]$. To this end, we need some more notation. We denote $M \stackrel{\text{def}}{=} \mathbb{E} [\gamma_s(x) \gamma_w(x)^\top] = \Sigma_s^{-\frac{1}{2}} \mathbb{E} [\phi_s(x) \phi_w(x)^\top] \Sigma_w^{-\frac{1}{2}} \in \mathbb{R}^{d_s \times d_w}$. We note that $d_{s \wedge w} = \|M\|_F^2$. We define $\varepsilon(x) \stackrel{\text{def}}{=} \gamma_w(x) - M^\top \gamma_s(x) \in \mathbb{R}^{d_w}$. Hence, we can write $\Gamma_w = \Gamma_s M + E$, where the i -th row of the matrix E is $\varepsilon(x_i)^\top$. We observe the following immediate properties of $\varepsilon(x)$:

$$\mathbb{E} [\gamma_s \varepsilon^\top] = \mathbb{E} [\gamma_s \gamma_w^\top] - \mathbb{E} [\gamma_s \gamma_s^\top] M = M - I_{d_s} M = 0, \tag{44}$$

and

$$\begin{aligned}
\mathbb{E} [\varepsilon \varepsilon^\top] &= \mathbb{E} \left[(\gamma_w - M^\top \gamma_s) (\gamma_w - M^\top \gamma_s)^\top \right] \\
&= \gamma_w \gamma_w^\top - \gamma_w (\gamma_s^\top M) - (M^\top \gamma_s) \gamma_w^\top + (M^\top \gamma_s) (M^\top \gamma_s)^\top = I_{d_w} - M^\top M. \tag{45}
\end{aligned}$$

Moreover, we have the following useful bounds on the fourth moments of $\gamma_s(x)$, $\varepsilon(x)$ and $\phi_s(x)$.

LEMMA 4. *Under Assumption 4, we have the following facts:*

$$\begin{aligned}
& -\mathbb{E} \left[\|\gamma_s(x)\|_2^4 \right] \lesssim d_s^2 \\
& -\mathbb{E} \left[\|\varepsilon(x)\|_2^4 \right] \lesssim (d_w - d_{s \wedge w})^2 \\
& -\mathbb{E} \left[\|\phi_s(x)\|_2^4 \right] \lesssim (\text{tr}(\Sigma_s))^2
\end{aligned}$$

To handle the quantity $\mathbb{E}_{\mathcal{B}} [\text{tr}(\Gamma_w^\top P_s \Gamma_w)]$, we need the following lemma.

LEMMA 5. *Under Assumption 4, there exist constants $c, C > 0$ depending on the sub-gaussian norm of $\phi_s(x)$ and $\phi_w(x)$, such that whenever $N \geq C d_s$, we have*

$$\mathbb{E}_{\mathcal{B}} [\text{tr}(\Gamma_w^\top P_s \Gamma_w)] \leq N d_{s \wedge w} + C d_s (d_w - d_{s \wedge w}) + C d_s (d_w - d_{s \wedge w}) N \exp(-cN). \quad (46)$$

The proof is deferred to Appendix B.6.

- To control the quantity $\lambda_{\max}(\mathbb{E}_{\mathcal{B}}[\Delta \Delta^\top])$ in (43), we simply note that

$$\lambda_{\max}(\mathbb{E}_{\mathcal{B}}[\Delta \Delta^\top]) \leq \text{tr}(\mathbb{E}_{\mathcal{B}}[\Delta \Delta^\top]) = \mathbb{E}_{\mathcal{B}}[\text{tr}(\Delta \Delta^\top)] = \mathbb{E}_{\mathcal{B}}[\|\Delta\|_F^2] = \mathbb{E}_{\mathcal{B}}\left[\left\|\hat{\Theta}_w - \Theta_w^*\right\|_F^2\right],$$

where the inequalities follow from the fact that $\mathbb{E}_{\mathcal{B}}[\Delta \Delta^\top]$ is a psd matrix.

When combining (43) with Lemma 5, we absorb the exponentially small remainder in Lemma 5 into the term $d_s(d_w - d_{s \wedge w})$ under $N \gtrsim d_s$. The proof of (33) is complete by combining all the pieces together.

4. Benchmark Bounds and W2S Performance

The main upper bound becomes most informative when it is compared with a direct strong-only benchmark and paired with a concrete control of the weak teacher. This section develops those auxiliary bounds and then combines them in a case study in Section 4.3 that identifies sufficient regimes in which W2S improves over direct strong-model training.

4.1. Strong-only lower bound

We recall the setup of training the strong model. We train the strong model by directly minimizing the empirical decision risk on the labeled data:

$$\hat{\Theta}_s \in \arg \min_{\Theta \in \mathbb{R}^{d_\phi \times K}, \Theta \mathbf{1} = 0} \frac{1}{n} \sum_{i=1}^n c\left(\tilde{w}(P_\Theta^s | \tilde{x}_i), \tilde{\xi}_i\right).$$

We assume that such a minimizer exists almost surely.

In this subsection, we study how to lower bound $\mathbb{E}_{\mathcal{B}}[R(\hat{\pi}_s) - R^*]$. To this end, we first introduce some notation. Define the single-sample gradient vector

$$\varphi \stackrel{\text{def}}{=} \text{vec}\left(\nabla_{\Theta} c\left(\tilde{w}\left(P_{\Theta_s^*}^s(x)\right), \xi\right)\right) \in \mathbb{R}^{d_\phi K}$$

with $(x, \xi) \sim \mathcal{D}$. We denote $\Sigma_\varphi \stackrel{\text{def}}{=} \text{Var}(\varphi) \in \mathbb{R}^{d_\phi K \times d_\phi K}$.

THEOREM 2. *In addition to Assumption 1 and 2, suppose the following assumptions hold:*

1. *Assume that there exists $\gamma > 0$ such that for all $x \in \mathcal{X}$, we have almost surely*

$$\left\| \tilde{w} \left(P_{\hat{\Theta}_s}^s(x) \right) - \tilde{w} \left(P^*(x) \right) \right\| \geq \gamma \left\| P_{\hat{\Theta}_s}^s(x) - P^*(x) \right\|. \quad (47)$$

Moreover, we assume that there exists $B_\eta > 0$ such that

$$\left\| \hat{\Theta}_s^\top \phi_s(x) \right\|_\infty \leq B_\eta, \quad \left\| \eta^*(x) \right\|_\infty \leq B_\eta. \quad (48)$$

2. *There exists $L_H > 0$ such that, almost surely over $\tilde{\mathcal{B}}$,*

$$-L_H I_{d_\phi K} \preceq \nabla_{\Theta}^2 \left[\frac{1}{n} \sum_{i=1}^n c \left(\tilde{w} \left(P_{\Theta}^s(\tilde{x}_i) \right), \tilde{\xi}_i \right) \right] \preceq L_H I_{d_\phi K} \quad (49)$$

for any Θ on the line segment connecting Θ_s^ and $\hat{\Theta}_s$, where ∇_{Θ}^2 is viewed as an operator on $\text{vec}(\Theta) \in \mathbb{R}^{d_\phi K}$.*

Under the above assumptions, for excess decision risk of the strong model, we have

$$\mathbb{E}_{\tilde{\mathcal{B}}} [R(\hat{\pi}_s) - R^*] \gtrsim \mu \gamma^2 \frac{e^{-4B_\eta}}{K^2} \left(\rho_s + \frac{\lambda_{\min}(\Lambda_s) \text{tr}(\Sigma_\varphi)}{L_H^2} \frac{1}{n} \right). \quad (50)$$

This theorem shows that the excess decision risk of the strong-only benchmark admits two unavoidable sources of error. The first is the approximation term ρ_s , which reflects the representational mismatch between the ground-truth logit map and the strong feature class. The second term is the estimation error term, which scales as $\frac{\lambda_{\min}(\Lambda_s) \text{tr}(\Sigma_\varphi)}{L_H^2} \frac{1}{n}$. We stress that this is an algorithm-specific lower bound for our strong-only benchmark training procedure. The proof of Theorem 2 is relatively standard, and is deferred to Appendix C.1.

The W2S upper bound also depends on the estimation quality of the weak teacher through the term $\mathbb{E}_{\tilde{\mathcal{B}}} \left[\left\| \hat{\Theta}_w - \Theta_w^* \right\|_F^2 \right]$. The next result gives one concrete control of this term under maximum likelihood training.

4.2. Weak-model upper bound

For the sake of completeness and concreteness, this subsection gives a maximum likelihood estimator as an example of a weak teacher training procedure and quantifies its performance. But the general W2S framework is not confined to this specific weak teacher training procedure.

We define $\mathcal{C}_B = \{ \Theta \in \mathbb{R}^{d_\phi \times K} \mid \Theta \mathbf{1} = 0, \|\Theta\|_F \leq B_\Theta \}$ as the constrained parameter space for some $B_\Theta > 0$. We consider the following maximum likelihood estimator based on the weak features:

$$\hat{\Theta}_w = \arg \min_{\Theta \in \mathcal{C}_B} - \frac{1}{n} \sum_{i=1}^n \ln \left(p_{\Theta}^w \left(\tilde{\xi}_i \mid \tilde{x}_i \right) \right), \quad (51)$$

where we denote $p_{\Theta}^w \left(\tilde{\xi}_i \mid \tilde{x}_i \right)$ to be the likelihood of observing $\tilde{\xi}_i$ given input $\tilde{x}_i$ under the weak feature model parameterized by Θ . Let $\mu_w \stackrel{\text{def}}{=} \frac{e^{-2B_\Theta B_\phi}}{K} \lambda_{\min}(\Sigma_w)$.

ASSUMPTION 5 (identifiability of Θ_w^*). *The ground-truth weak model parameter Θ_w^* is identifiable in the sense that $\rho_w = 0$ and the minimizer Θ_w^* is unique. Moreover, we assume that Θ_w^* is attained in the interior of $\mathcal{C}_B$, i.e., $\|\Theta_w^*\|_F < B_\Theta$.*

ASSUMPTION 6 (bounded weak features). *There exists a constant $B_\phi > 0$ such that for all $x \in \mathcal{X}$, we have $\|\phi_w(x)\|_2 \leq B_\phi$.*

THEOREM 3. *Under Assumption 1, Assumption 5 and Assumption 6, the MLE estimator $\hat{\Theta}_w$ defined in (51) enjoys that*

$$\mathbb{E}_{\tilde{\mathcal{B}}} \left[\left\| \hat{\Theta}_w - \Theta_w^* \right\|_F^2 \right] \leq \frac{4}{\mu_w^2} \frac{\text{tr}(\Sigma_w)}{n} + 4B_\Theta^2 d_w \exp \left(-\frac{n\lambda_{\min}(\Sigma_w)}{8B_\phi^2} \right), \quad (52)$$

The proof of this theorem is deferred to Appendix C.2.

4.3. When does W2S outperform?

We now turn the preceding bounds into a sufficient condition under which W2S improves over direct strong-model training. Combining the W2S upper bound in Theorem 1 with the strong-only lower bound in Theorem 2, it suffices, up to universal constants, that

$$\mathcal{E}_{\text{imit}} + \mathcal{E}_N + \mathcal{E}_{\text{teacher}} + \mathcal{E}_{\text{approx}} \lesssim \mu\gamma^2 \frac{e^{-4B_\eta}}{K^2} \left(\rho_s + \frac{\lambda_{\min}(\Lambda_s) \text{tr}(\Sigma_\varphi)}{L_H^2 n} \right). \quad (53)$$

The left-hand side collects the costs paid by W2S: imitation, unlabeled-sample fluctuation, teacher estimation, and model misspecification. The right-hand side is the strong model's approximation and labeled-sample statistical error. Thus, the comparison asks whether the weak teacher and the unlabeled sample can offset the label scarcity faced by direct strong-model training.

We next plug the weak-model estimate from Theorem 3 into the weak-to-strong term $\mathcal{E}_{\text{teacher}}$ defined after Theorem 1. To obtain a transparent sufficient condition, we work in the large- n regime made explicit in Corollary 1. Also define $C_t \stackrel{\text{def}}{=} \frac{4LK^2}{\mu^2\mu_w^2} \lambda_{\max}(\Lambda_w) \text{tr}(\Sigma_w)$. Then $\mathcal{E}_{\text{teacher}}$ is bounded by

$$\mathcal{E}_{\text{teacher}} \lesssim \frac{C_t}{n} \left(d_{s \wedge w} + \frac{d_s(d_w - d_{s \wedge w})}{N} \right). \quad (54)$$

In the decomposition following Theorem 1, all terms in the unlabeled-sample statistical error $\mathcal{E}_N$ scale as $N^{-1/2}$. We therefore write $\mathcal{E}_N = C_N/\sqrt{N}$, where C_N collects the coefficients in $\mathcal{E}_N$ that do not depend on N . The imitation term $\mathcal{E}_{\text{imit}}$, which contains $\mathbb{E}_{\tilde{\mathcal{B}}} \left[\rho_{s \rightarrow w}(\hat{\Theta}_w) \right]$, is also independent of N .

Define the residual margin to be

$$\mathcal{M}_n \stackrel{\text{def}}{=} \mu\gamma^2 \frac{e^{-4B_\eta}}{K^2} \left(\rho_s + \frac{\lambda_{\min}(\Lambda_s) \text{tr}(\Sigma_\varphi)}{L_H^2 n} \right) - \mathcal{E}_{\text{imit}} - \mathcal{E}_{\text{approx}} - \frac{C_t d_{s \wedge w}}{n}.$$

With this definition, the condition (53) reduces to

$$\frac{C_N}{\sqrt{N}} + \frac{C_t d_s(d_w - d_{s \wedge w})}{nN} \lesssim \mathcal{M}_n. \quad (55)$$

COROLLARY 1 (A sufficient condition for W2S outperformance). *Suppose the assumptions of Theorems 1, 2, and 3 hold. Assume the following three conditions hold.*

- (i) *The labeled sample size n is large enough that the exponentially small term in Theorem 3 is dominated by its $1/n$ term: $B_{\Theta}^2 d_w \exp\left(-\frac{n\lambda_{\min}(\Sigma_w)}{8B_{\phi}^2}\right) \lesssim \frac{\text{tr}(\Sigma_w)}{\mu_w^2 n}$.*
- (ii) *The imitation error $\mathcal{E}_{\text{imit}}$, the model-approximation error $\mathcal{E}_{\text{approx}}$, and the overlap cost $C_t d_{s \wedge w}/n$ are small enough relative to the strong-only benchmark term to leave a positive margin:*

$$\mathcal{E}_{\text{imit}} + \mathcal{E}_{\text{approx}} + \frac{C_t d_{s \wedge w}}{n} < \mu \gamma^2 \frac{e^{-4B_{\eta}}}{K^2} \left(\rho_s + \frac{\lambda_{\min}(\Lambda_s) \text{tr}(\Sigma_{\varphi})}{L_H^2 n} \right). \quad (56)$$

Equivalently, $\mathcal{M}_n > 0$.

- (iii) *The unlabeled sample size N is large enough:*

$$N \gtrsim \max \left\{ d_s, \left(\frac{C_N + \sqrt{C_N^2 + 4\mathcal{M}_n C_t d_s (d_w - d_{s \wedge w})/n}}{2\mathcal{M}_n} \right)^2 \right\}. \quad (57)$$

Under these conditions, the preceding bounds certify, up to universal constants, that

$$\mathbb{E}_{\tilde{B}, B} [R(\hat{\pi}_{w2s}) - R^*] \leq \mathbb{E}_{\tilde{B}} [R(\hat{\pi}_s) - R^*]. \quad (58)$$

Therefore, W2S can outperform only if the N -independent costs $\mathcal{E}_{\text{imit}}$, $\mathcal{E}_{\text{approx}}$, and $C_t d_{s \wedge w}/n$ are small enough to leave a positive residual margin $\mathcal{M}_n > 0$. This condition combines three requirements: the strong student must imitate the weak teacher accurately, the two feature classes must have small approximation error, and the overlap dimension $d_{s \wedge w}$ must be sufficiently small relative to the labeled sample size n . Intuitively, a smaller overlap dimension means that the ways of thinking of the teacher and student are different enough, so the teacher's mistakes are more likely to be averaged out rather than directly inherited. In (55), once $\mathcal{M}_n > 0$, the finite- N terms $C_N/\sqrt{N}$ and $C_t d_s (d_w - d_{s \wedge w})/(nN)$ decrease as N grows; with fixed problem-dependent constants, (57) scales qualitatively as $N \gtrsim \max\{d_s, n^2\}$.

5. Numerical Experiments

In this section, we present numerical experiments to validate our theoretical findings. In Section 5.1, we evaluate the empirical performance of W2S in a contextual newsvendor problem, where the data-generating process can be precisely controlled to isolate and validate the theoretical mechanisms identified in the previous sections. Furthermore, Section 5.2 evaluates W2S on a real-world comment moderation dataset to test whether the predicted gains persist in a more realistic operational setting.

5.1. Synthetic Experiments

We consider a synthetic contextual single-item newsvendor problem.

Setup. We take a finite discrete support $\Xi = \{z_1, \dots, z_K\} \subseteq \mathbb{R}$ where $z_1 < z_2 < \dots < z_K$, and interpret $\xi \in \Xi$ as the random demand for a product. The decision is an order quantity $w \in \mathbb{R}$ before the demand ξ is realized. We use the standard newsvendor cost function $c(w, \xi) = c_u(\xi - w)_+ + c_o(w - \xi)_+$, where $c_u > 0$ is the underage cost, $c_o > 0$ is the overage cost, and $(a)_+ = \max\{a, 0\}$. We note that such a cost function is neither strongly convex nor smooth in w , and hence does not satisfy Assumption 2. It is well-known that the optimal order quantity given the context vector x and the demand distribution $P(\cdot | x)$ is given by the quantile function:

$$\tilde{w}(P | x) = \inf \left\{ z \in \mathbb{R} : \mathbf{Pr}_{\xi \sim P(\cdot | x)} [\xi \leq z] \geq \frac{c_u}{c_u + c_o} \right\}. \quad (59)$$

One can verify that $\tilde{w}(P | x) \in \Xi$. We work with a context vector x sampled from $\mathcal{N}(0, I_{d_x})$. We consider the strong and weak feature models $\phi_s(x) = V_s^\top x$ and $\phi_w(x) = V_w^\top x$, where $V_s \in \mathbb{R}^{d_x \times d_\phi}$ and $V_w \in \mathbb{R}^{d_x \times d_\phi}$ are fixed matrices with orthonormal columns. We explicitly construct V_s and V_w such that $\mathbb{E}[\phi_s(x)\phi_w(x)^\top] = \mathbb{E}[V_s^\top x x^\top V_w] = V_s^\top V_w = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$. First, we need $0 \leq r \leq \min\{d_s, d_w\}$, $d_\phi \geq \max\{d_s, d_w\}$ and $d_x \geq d_s + d_w - r$. We choose an orthonormal basis $Q = [q_1, \dots, q_{d_x}] \in \mathbb{R}^{d_x \times d_x}$. Then we set $U_s = [q_1, \dots, q_r, q_{r+1}, \dots, q_{d_s}] \in \mathbb{R}^{d_x \times d_s}$ and $U_w = [q_1, \dots, q_r, q_{d_s+1}, \dots, q_{d_s+(d_w-r)}] \in \mathbb{R}^{d_x \times d_w}$. This way, we have $U_s^\top U_w = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$. To construct V_s and V_w , it suffices to pad zeroes: $V_s = \begin{bmatrix} U_s & 0_{d_x \times (d_\phi - d_s)} \end{bmatrix}$, $V_w = \begin{bmatrix} U_w & 0_{d_x \times (d_\phi - d_w)} \end{bmatrix}$. Hence, we have $\mathbb{E}[\phi_s(x)\phi_s(x)^\top] = V_s^\top V_s = \begin{bmatrix} I_{d_s} & 0 \\ 0 & 0 \end{bmatrix}$, and similarly for Σ_w . One can easily verify that $d_{s \wedge w} = \left\| \Sigma_s^{-\frac{1}{2}} \mathbb{E}[\phi_s(x)\phi_w(x)^\top] \Sigma_w^{-\frac{1}{2}} \right\|_F^2 = \|V_s^\top V_w\|_F^2 = r$.

We make the ground-truth environment realizable by the strong model (hence there is no approximation error for the strong model, i.e., $\rho_s = 0$), by defining the ground-truth logits to be linear in the strong features: $\eta^*(x) = \Theta_*^\top \phi_s(x) \in \mathbb{R}^K$ and $P^*(\xi = z_k | x) = \text{softmax}_k(\eta^*(x))$.

As noted in Qi et al. (2025), the W2S training objective

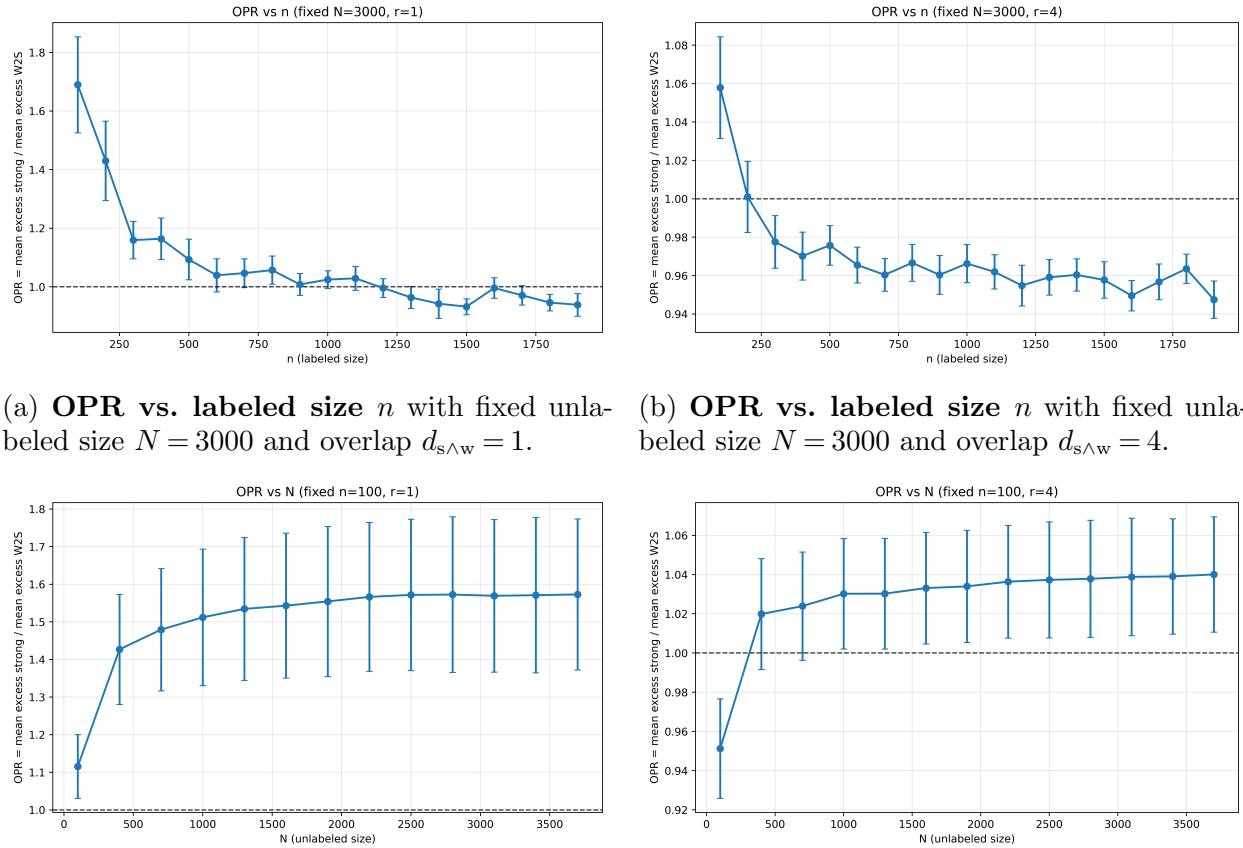
$$\hat{\Theta}_{\text{w2s}} \in \arg \min_{\Theta \in \mathbb{R}^{d_\phi \times K}, \|\Theta\|_F \leq B} \frac{1}{N} \sum_{j=1}^N \mathbb{E}_{\xi \sim P_{\Theta_w}^w(\cdot | x_j)} [c(\tilde{w}(P_\Theta^s | x_j), \xi_j)]$$

is in general neither convex nor differentiable, which is also the case even in this synthetic newsvendor problem. Indeed, the objective function is piecewise constant with respect to Θ because the optimal order quantity $\tilde{w}(P_\Theta^s | x_j)$ only changes when the quantile of the distribution $P_\Theta^s(\cdot | x_j)$ crosses one of the demand values in Ξ . To this end, we work with a smoothed surrogate optimal action function instead of the exact optimal action function in (59). To be specific, the induced conditional pmf on Ξ by the strong model is given by $P_\Theta^s(\xi = z_k | x) = \frac{\exp(\Theta_k^\top V_s^\top x)}{\sum_{j=1}^K \exp(\Theta_j^\top V_s^\top x)}$. For $\beta > 0$, define $a_{\Theta, k}^\beta(x) = -\beta \left(\sum_{\ell=1}^k P_\Theta^s(\xi = z_\ell | x) - \frac{c_u}{c_u + c_o} \right)^2$. We then use $\hat{w}_\beta(P_\Theta^s | x) = \sum_{k=1}^K \text{softmax}_k(a_{\Theta, k}^\beta(x)) z_k$ as the surrogate of $\tilde{w}(P_\Theta^s | x)$, where β controls the sharpness of the approximation. Moreover, we use $\tau \log(1 + e^{t/\tau})$ to approximate $(t)_+$ for some small smoothing parameter $\tau > 0$. With all these

in hand, we use a first-order method to approximately solve for $\hat{\Theta}_{w2s}$. Again following the spirit of Qi et al. (2025), in the numerical implementation reported below, we use the following regularized version of the surrogate objective:

$$\hat{\Theta}_{w2s} \in \arg \min_{\Theta \in \mathbb{R}^{d_\phi \times K}} \frac{1}{N} \sum_{j=1}^N \mathbb{E}_{\xi \sim P_{\Theta}^w(\cdot | x_j)} [c(\hat{w}_\beta(P_\Theta^s | x_j), \xi)] + \lambda_{w2s} \|\Theta - \Theta_s^0\|_F^2, \quad (60)$$

where Θ_s^0 denotes the initial strong-model parameter used to initialize W2S and $\lambda_{w2s} = 5 \times 10^{-4}$. We note that we only use the smoothed surrogate objective for numerical optimization; the evaluation of decision risk is still based on the original newsvendor cost function without smoothing.



(c) OPR vs. unlabeled size N with fixed labeled size $n = 100$ and overlap $d_{s \wedge w} = 1$. (d) OPR vs. unlabeled size N with fixed labeled size $n = 100$ and overlap $d_{s \wedge w} = 4$.

Figure 1 Sensitivity of the optimization performance ratio (OPR) to labeled and unlabeled sample sizes and to the overlap dimension. OPR is defined as $\text{OPR} := \frac{\text{ExcessRisk}(\text{Strong Benchmark})}{\text{ExcessRisk}(\text{W2S})} = \frac{\mathbb{E}_{\tilde{\mathcal{B}}} [R(\tilde{\pi}_s) - R^*]}{\mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} [R(\tilde{\pi}_{w2s}) - R^*]}$, where excess risk is measured relative to the oracle using the ground-truth conditional distribution. Each plotted value is the ratio of mean excess risks across 60 independent trials; error bars show approximate 95% normal confidence bands computed as 1.96 times a bootstrap standard error for this ratio-of-means estimator. Parameters used for the experiment: $d_x = 80, \Xi = \{0, 0.5, 1, \dots, 10\}, d_s = 10, d_w = 48, c_o = 1, c_u = 3, \beta = 35, \tau = 0.5$.

Results. We can make a few observations on Figure 1.

1. W2S provides the largest gains when labeled data are scarce, especially under low overlap. In the top-left panel ($d_{s \wedge w} = 1$, fixed $N = 3000$), the OPR starts well above 1 and then decreases toward 1 as n grows. This suggests that when the strong model is data-limited, leveraging abundant unlabeled data through the weak teacher can materially reduce excess decision risk compared to training the strong model on labeled data alone.
2. The benefit of W2S diminishes as overlap increases. Comparing $d_{s \wedge w} = 1$ vs $d_{s \wedge w} = 4$, the curves are systematically closer to 1 when $d_{s \wedge w} = 4$. In the top-right panel, OPR is only slightly above (or near) 1 for small n and drifts below (or around) 1 as n increases. Namely, when the weak and strong representations share more directions, the extra information conveyed by the weak teacher is less distinctive, so W2S yields only modest improvement.
3. More unlabeled data improves W2S up to a plateau, with stronger effects at low overlap. For example, in the bottom-left panel ($d_{s \wedge w} = 1$, fixed $n = 100$), OPR increases as N grows and then levels off, indicating that additional unlabeled samples help W2S learn a better policy until the benefit saturates.

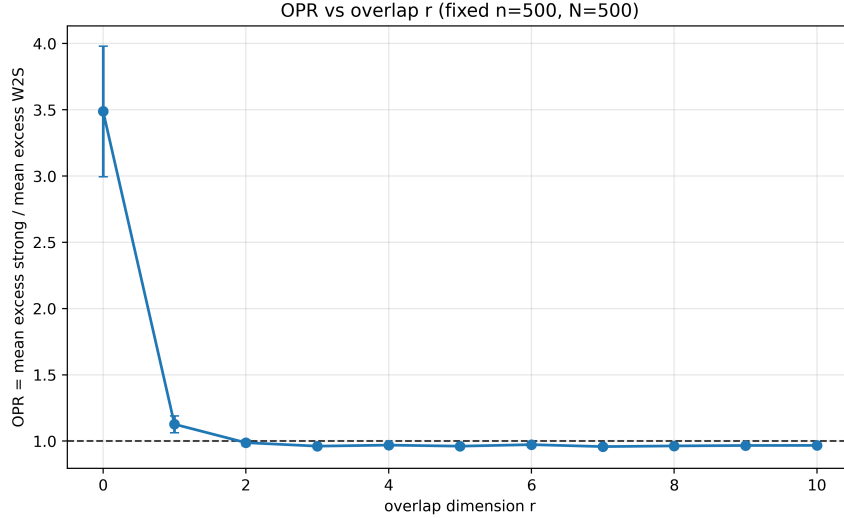


Figure 2 Effect of the strong-weak feature overlap on the OPR. We fix the labeled and unlabeled sample sizes to $n = 500$ and $N = 500$, and vary the overlap dimension $d_{s \wedge w}$. The large OPR at small $d_{s \wedge w}$ indicates that W2S substantially reduces excess decision risk when the weak and strong representations share little (or no) subspace, whereas the curve approaching $\text{OPR} \approx 1$ for larger $d_{s \wedge w}$ indicates that the benefit of W2S diminishes as the two feature spaces become more aligned.

In Figure 2, we further examine the effect of the overlap dimension $d_{s \wedge w}$ on the OPR, fixing $n = 500$ and $N = 500$. We observe that the OPR decreases as $d_{s \wedge w}$ increases, confirming that W2S is most beneficial when the weak and strong feature representations share little subspace. When

$d_{s\wedge w}$ is small, W2S can leverage information from directions that are inaccessible to the strong model alone, leading to significant reductions in excess decision risk.

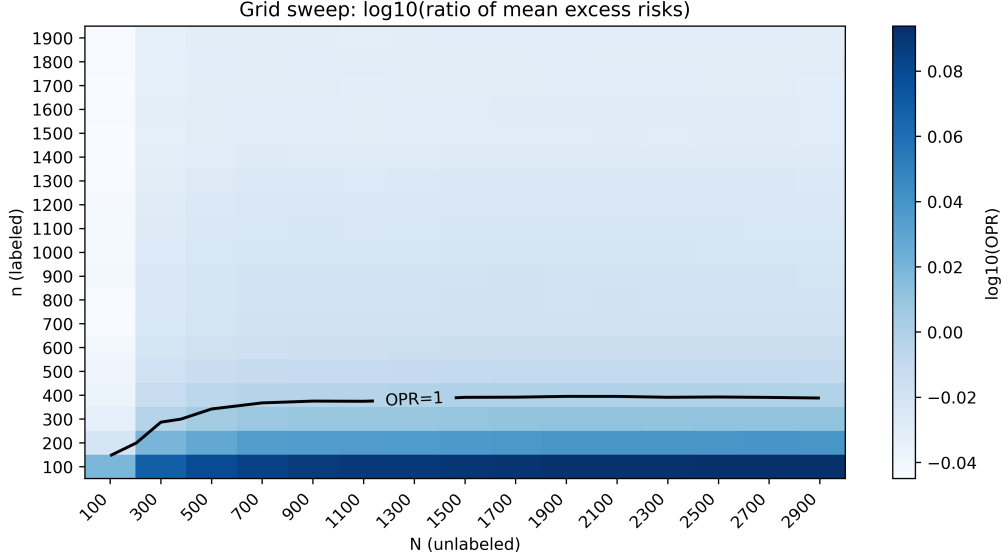


Figure 3 Grid sweep of the OPR over labeled and unlabeled sample sizes. Each cell reports $\log(\text{OPR})$, where OPR is computed as the ratio of mean excess risks across 60 independent trials, at overlap $d_{s\wedge w} = 2$. Darker (more positive) values indicate $\text{OPR} > 1$, i.e., W2S achieves lower excess decision risk than the strong-only baseline; lighter (more negative) values indicate $\text{OPR} < 1$. The overlaid contour highlights the level set $\text{OPR} = 1$, separating the two regimes.

Figure 3 presents an exhaustive grid sweep of the OPR over labeled and unlabeled sample sizes at fixed overlap $d_{s\wedge w} = 2$. It reinforces the earlier observations that W2S excels when labeled data are limited and unlabeled data are abundant, e.g., for the region $n \leq 300$, we see the largest gains. At moderate labeled sizes $n \approx 500$, the improvement is smaller but still visible. The solid contour marks the boundary $\text{OPR} = 1$, indicating that beyond roughly $n \sim 10^3$ the advantage of W2S becomes marginal.

5.2. Comment Moderation Experiments

In this section, we evaluate W2S on a more realistic comment moderation task, a canonical application in AI-assisted service systems (Lee et al. 2024).

A platform receives a stream of comments: most are harmless, some are toxic, and a small fraction are severely toxic. The operational task is to route each comment to automatic approval, automatic removal, or one of several human-review routes. The central tradeoff is that allowing severely toxic content creates exposure harm, while removing harmless content creates user-friction and over-moderation cost. Human review can reduce both types of mistakes, but reviewer capacity

is limited: sending more comments to review increases delay, and delay is more costly for comments that are likely to be harmful. Thus, the value of a predictor is measured not only by classification accuracy, but by the downstream cost of the moderation route it induces. This setting also naturally creates a labeled/unlabeled split: raw comments are abundant as platform traffic, whereas reliable severity labels require crowd or expert moderation judgments.

Setup. The raw comments come from the Jigsaw Unintended Bias/Civil Comments corpus, which contains online comments with crowd-sourced toxicity and identity annotations (Borkan et al. 2019, Jigsaw 2019). The context x is the comment text. The raw data reports a toxicity score for each comment, which we discretize into three severity classes $\Xi = \{0, 1, 2\}$ using thresholds 0.5 and 0.8: non-toxic, toxic, and severely toxic. The feasible action set is

$$\mathcal{A} = \{\text{allow, remove, standard review, hold review, priority review}\}.$$

Here, *allow* means that the comment is automatically approved and remains visible on the platform. *Remove* means that the comment is automatically suppressed without human review. *Standard review* sends the comment to a regular human-review queue while the comment remains visible during the review process. *Hold review* also sends the comment to human review, but the comment is held off the platform until the review is completed. *Priority review* is a faster human-review route for comments that may be more urgent; it has higher review cost but shorter delay. These actions trade off different types of operational cost: exposure cost from leaving harmful comments visible, user-friction cost from suppressing benign comments, reviewer cost, and delay cost from human review. The original data set is highly imbalanced, with approximately 14% toxic comments and only 2.6% severely toxic comments. To address this imbalance, we augment the data by prompting Claude Sonnet, using real severe-toxicity comments as style anchors, to generate additional plausible CivilComments-style comments intended to belong to the severe class. In the augmented data set, the severely toxic class is increased to about 10% of the total comments. Then, we further partition the augmented comment data into separate splits for (1) task pretraining, (2) downstream W2S training with labeled and unlabeled samples, and (3) held-out evaluation.

In the notation of Section 2, the random outcome is the severity class $\xi \in \Xi$, and the action is the moderation route $a \in \mathcal{A}$. We encode the cost of assigning route a to a comment with severity ξ by a route-severity cost matrix C , so that $c(a, \xi) = C_{a, \xi}$. The exact entries of C are reported in Appendix D; they are chosen to reflect the operational tradeoffs described above. Given a conditional model $P : \mathcal{X} \rightarrow \Delta(\Xi)$, the induced plug-in route follows the decision rule used throughout the paper: $\tilde{w}(P(x)) \in \arg \min_{a \in \mathcal{A}} \sum_{\xi \in \Xi} C_{a, \xi} P(\xi | x)$. We refer to this as the *hard-route* decision.

Pretraining. To obtain the two feature maps ϕ_w and ϕ_s , we start from two generic language-model backbones with different capacities: a compact two-layer BERT encoder for the weak model and a DistilBERT encoder for the strong model (Devlin et al. 2019, Sanh et al. 2019). To map each encoder representation to the three severity classes, we attach a newly initialized classification head to each model and task-pretrain the resulting models end-to-end on labeled comments from the pretraining split using cross-entropy loss; sample-size and optimizer-level details are reported in Appendix D. To align the experiment with the weak-to-strong regime (Burns et al. 2024), we restrict the task-pretraining compute allocated to the larger DistilBERT model so that it does not already dominate the smaller model outright.

Training protocol. To validate the theoretical findings in Section 4, we compare W2S with the strong-only baseline across a range of labeled and unlabeled sample sizes. For each experimental setting, we draw a labeled sample $\tilde{\mathcal{B}}$ and an unlabeled sample $\mathcal{B}$ uniformly without replacement from their respective pools. The strong-only benchmark is initialized from the strong pre-trained model and trained on $\tilde{\mathcal{B}}$ using the supervised decision-aware objective in (9), with the optimal action oracle $\tilde{w}$ replaced by a differentiable soft-route surrogate to enable gradient-based optimization. It is optimized using AdamW with weight decay. For W2S, we initialize the weak teacher from the weak pretrained model and fine-tune it on $\tilde{\mathcal{B}}$ by cross-entropy. We then implement the W2S training in the same way as in (60), with the route-severity cost matrix replacing the synthetic newsvendor cost. It is optimized with AdamW with a cosine learning-rate scheduler.

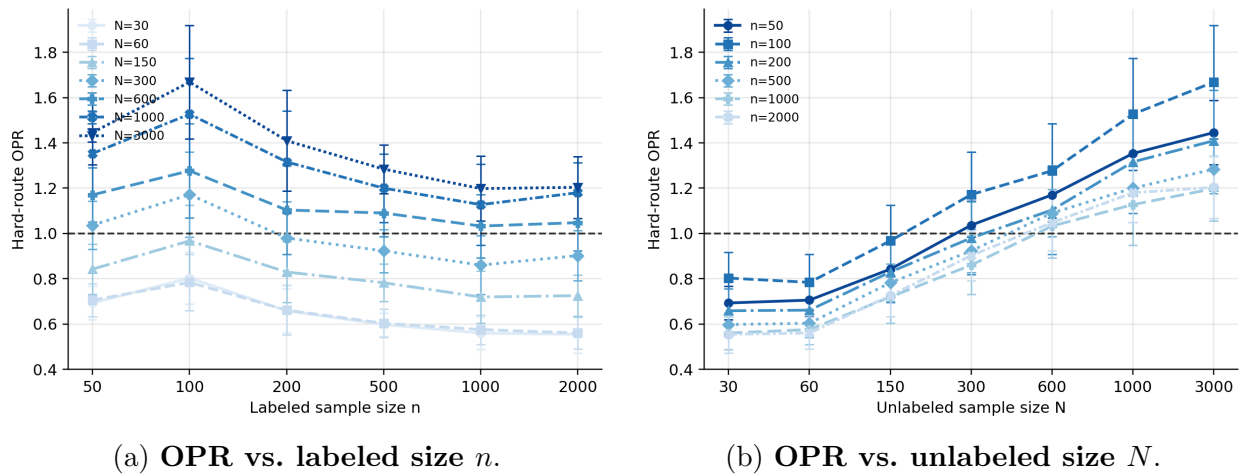


Figure 4 **Hard-route OPR for the augmented queue-aware comment moderation experiment. Panels (a)–(b) plot empirical OPR across 30 trials with 1.96 bootstrap standard-error bars.**

Results. Since we do not have access to the ground-truth distribution for computing OPR (11), we compute the empirical OPR using the held-out split of the data set. Figure 4 shows how the labeled and unlabeled sample sizes affect the OPR in this real-world operational setting. We observe that, consistent with the qualitative implication of Corollary 1, W2S does not improve over the strong-only baseline when the unlabeled sample size N is small, but achieves OPR above one once N is greater than 600. Another finding consistent with the synthetic experiment is that the advantage of W2S is most pronounced when the labeled sample size n is small, and the gain narrows as n increases.

6. Concluding Remarks

This paper takes a first step toward understanding weak-to-strong learning as a decision-making problem. In the contextual stochastic optimization setting, we show that weak supervision can improve downstream decision performance when labeled data are scarce and unlabeled contexts are abundant, with the overlap between weak and strong representations determining how much teacher error is inherited rather than averaged out. The numerical experiments support this mechanism in both controlled simulations and a comment moderation routing task. A natural next step is to move beyond one-shot decisions to sequential decision making, including reinforcement learning and online control. In such settings, weak supervision may affect not only the final decision rule, but also exploration, state distribution shift, and the accumulation of decision errors over time.

Appendix A: Omitted Proofs in Section 2

LEMMA 6. We always have $0 \leq d_{s \wedge w} \leq \min\{d_s, d_w\}$.

Proof: We define $\gamma_s(x) = \Lambda_s^{-\frac{1}{2}} V_s^\top \phi_s(x)$ and $\gamma_w(x) = \Lambda_w^{-\frac{1}{2}} V_w^\top \phi_w(x)$. One can verify that they are zero-mean random vectors with identity covariance matrix. Then by definition $M \stackrel{\text{def}}{=} \mathbb{E}[\gamma_s(x) \gamma_w(x)^\top] = \mathbb{E}\left[\Lambda_s^{-\frac{1}{2}} V_s^\top \phi_s(x) \phi_w(x)^\top V_w \Lambda_w^{-\frac{1}{2}}\right] = \Lambda_s^{-\frac{1}{2}} V_s^\top \mathbb{E}[\phi_s(x) \phi_w(x)^\top] V_w \Lambda_w^{-\frac{1}{2}}$.

We next show that $\|M\|_2 \leq 1$. First of all, by definition of the 2-norm and by invoking Cauchy-Schwarz inequality, we have $\|M\|_2 = \sup_{\|v\|_2=1} \|Mv\|_2 = \sup_{\|u\|_2=\|v\|_2=1} |u^\top Mv|$. For any vectors $u \in \mathbb{R}^{d_s}$ and $v \in \mathbb{R}^{d_w}$, we have

$$u^\top Mv = u^\top \mathbb{E}[\gamma_s \gamma_w^\top] v = \mathbb{E}[u^\top \gamma_s \gamma_w^\top v].$$

By Cauchy-Schwarz inequality, we have $\mathbb{E}[u^\top \gamma_s \gamma_w^\top v] \leq \sqrt{\mathbb{E}[(u^\top \gamma_s)^2]} \sqrt{\mathbb{E}[(\gamma_w^\top v)^2]}$. Because both γ_s and γ_w are isotropic, we have $\sqrt{\mathbb{E}[(u^\top \gamma_s)^2]} = \|u\|_2$ and $\sqrt{\mathbb{E}[(\gamma_w^\top v)^2]} = \|v\|_2$. Hence, we conclude that $\|M\|_2 = \sup_{\|u\|_2=\|v\|_2=1} \mathbb{E}[u^\top \gamma_s \gamma_w^\top v] \leq 1$.

Now, we proceed to upper bound the overlap dimension. We have

$$\begin{aligned} d_{s \wedge w} &= \left\| \Sigma_s^{-\frac{1}{2}} \mathbb{E}[\phi_s(x) \phi_w(x)^\top] \Sigma_w^{-\frac{1}{2}} \right\|_F^2 \\ &= \left\| V_s \Lambda_s^{-\frac{1}{2}} V_s^\top \mathbb{E}[\phi_s(x) \phi_w(x)^\top] V_w \Lambda_w^{-\frac{1}{2}} V_w^\top \right\|_F^2 \end{aligned} \quad (61)$$

$$\begin{aligned} &= \|V_s M V_w^\top\|_F^2 \\ &= \text{tr}(V_s M V_w^\top V_w M^\top V_s^\top) \end{aligned} \quad (62)$$

$$\begin{aligned} &= \text{tr}(M M^\top V_s^\top V_s) \\ &= \text{tr}(M M^\top). \end{aligned} \quad (63)$$

Equation (61) follows from the definition of $\Sigma_s^{-\frac{1}{2}}$ and $\Sigma_w^{-\frac{1}{2}}$. Equation (62) is due to the fact that $\|A\|_F^2 = \text{tr}(A A^\top)$ for any matrix A . The last equation follows from the fact that $V_s^\top V_s = I$.

Finally, we note that $\text{tr}(M M^\top) = \|M\|_F^2 = \sum_{i=1}^{\min\{d_s, d_w\}} \sigma_i^2$, where σ_i is the i -th singular value of M . Because $\|M\|_2 \leq 1$, we have $\sigma_i \leq 1$ for all i . Hence, the proof is complete. $\square$

Appendix B: Omitted Proofs in Section 3

B.1. Proof of Lemma 1

Proof: First of all, since c is μ -strongly convex and L -smooth in its first argument, we have

$$\frac{\mu}{2} \|a' - a\|_2^2 \leq c(a', z) - c(a, z) - \langle \nabla_a c(a, z), a' - a \rangle \leq \frac{L}{2} \|a' - a\|_2^2 \quad (64)$$

for every $z \in \Xi$ and $a, a' \in \mathcal{A}$. Multiplying by (64) by $P(\xi = z_k | x)$ and sum over $k = 1, 2, \dots, K$, we have

$$\frac{\mu}{2} \|a' - a\|_2^2 \leq \sum_{k=1}^K P(\xi = z_k | x) [c(a', z_k) - c(a, z_k) - \langle \nabla_a c(a, z_k), a' - a \rangle] \leq \frac{L}{2} \|a' - a\|_2^2. \quad (65)$$

The middle term can be written as

$$\sum_{k=1}^K P(\xi = z_k | x) [c(a', z_k) - c(a, z_k)] - \left\langle \sum_{k=1}^K P(\xi = z_k | x) \nabla_a c(a, z_k), a' - a \right\rangle$$

$$= \mathbb{E}_{\xi \sim P(\cdot|x)} [c(a', \xi) - c(a, \xi)] - \left\langle \sum_{k=1}^K P(\xi = z_k | x) \nabla_a c(a, z_k), a' - a \right\rangle.$$

Taking $a' = \tilde{w}(P' | x)$ and $a = \tilde{w}(P | x)$, we have

$$\begin{aligned} & \frac{\mu}{2} \|\tilde{w}(P' | x) - \tilde{w}(P | x)\|_2^2 \\ & \leq \mathbb{E}_{\xi \sim P(\cdot|x)} [c(\tilde{w}(P' | x), \xi) - c(\tilde{w}(P | x), \xi)] \\ & \quad - \left\langle \sum_{k=1}^K P(\xi = z_k | x) \nabla_w c(\tilde{w}(P | x), z_k), \tilde{w}(P' | x) - \tilde{w}(P | x) \right\rangle \\ & \leq \frac{L}{2} \|\tilde{w}(P' | x) - \tilde{w}(P | x)\|_2^2. \end{aligned} \tag{66}$$

To conclude the proof, we recall the definition of $\tilde{w}(P | x)$, which is given by

$$\tilde{w}(P | x) = \operatorname{argmin}_{w \in \mathcal{A}} \sum_{k=1}^K P(\xi = z_k | x) c(w, z_k).$$

The first-order optimality condition implies that

$$\sum_{k=1}^K P(\xi = z_k | x) \nabla_w c(\tilde{w}(P | x), z_k) = 0. \tag{67}$$

□

B.2. Proof of Lemma 2

Proof: For a logit η , let P_η be the corresponding softmax distribution, namely, $P_\eta(\cdot | x) = \operatorname{softmax}(\eta(x))$. Let $\eta_t = \eta' + t(\eta - \eta')$ for $t \in [0, 1]$. For the distribution P_{η_t} ,

$$\tilde{w}(P_{\eta_t} | x) = \operatorname{argmin}_{w \in \mathcal{A}} \sum_{k=1}^K P_{\eta_t}(\xi = z_k | x) c(w, z_k).$$

By the first-order optimality condition, we have

$$0 = \sum_{k=1}^K P_{\eta_t}(\xi = z_k | x) \nabla_w c(\tilde{w}(P_{\eta_t} | x), z_k).$$

Differentiating both sides with respect to t , we have

$$0 = \sum_{k=1}^K P_{\eta_t}(\xi = z_k | x) \nabla_w^2 c(\tilde{w}(P_{\eta_t} | x), z_k) \frac{d\tilde{w}(P_{\eta_t} | x)}{dt} + \sum_{k=1}^K \frac{dP_{\eta_t}(\xi = z_k | x)}{dt} \nabla_w c(\tilde{w}(P_{\eta_t} | x), z_k). \tag{68}$$

Since $\nabla_w^2 c(\tilde{w}(P_{\eta_t} | x), z_k) \succeq \mu I$ for any x and each k , we have

$$\frac{d\tilde{w}(P_{\eta_t} | x)}{dt} = - \left(\sum_{k=1}^K P_{\eta_t}(\xi = z_k | x) \nabla_w^2 c(\tilde{w}(P_{\eta_t} | x), z_k) \right)^{-1} \sum_{k=1}^K \frac{dP_{\eta_t}(\xi = z_k | x)}{dt} \nabla_w c(\tilde{w}(P_{\eta_t} | x), z_k) \tag{69}$$

and $\left\| \left(\sum_{k=1}^K P_{\eta_t}(\xi = z_k | x) \nabla_w^2 c(\tilde{w}(P_{\eta_t} | x), z_k) \right)^{-1} \right\|_2 \leq \frac{1}{\mu}$. Therefore, we have

$$\begin{aligned} \left\| \frac{d\tilde{w}(P_{\eta_t} | x)}{dt} \right\|_2 & \leq \frac{1}{\mu} \left\| \sum_{k=1}^K \frac{dP_{\eta_t}(\xi = z_k | x)}{dt} \nabla_w c(\tilde{w}(P_{\eta_t} | x), z_k) \right\|_2 \\ & \leq \frac{1}{\mu} \left(\sum_{k=1}^K \left(\frac{dP_{\eta_t}(\xi = z_k | x)}{dt} \right)^2 \sum_{k=1}^K \|\nabla_w c(\tilde{w}(P_{\eta_t} | x), z_k)\|_2^2 \right)^{\frac{1}{2}} \end{aligned} \tag{70}$$

$$\leq \frac{G\sqrt{K}}{\mu} \left(\sum_{k=1}^K \left(\frac{dP_{\eta_t}(\xi = z_k | x)}{dt} \right)^2 \right)^{\frac{1}{2}} \quad (71)$$

$$= \frac{G\sqrt{K}}{\mu} \left\| \frac{d}{dt} p(\eta_t) \right\|_2 \quad (72)$$

where $p(\eta_t) = \begin{bmatrix} P_{\eta_t}(\xi = z_1 | x) \\ \vdots \\ P_{\eta_t}(\xi = z_K | x) \end{bmatrix}$ and (70) follows from Cauchy-Schwarz inequality. To finish the proof, we note the fact that $\left\| \frac{d}{dt} p(\eta_t) \right\|_2 = \|(\text{Diag}(p(\eta_t)) - p(\eta_t)p(\eta_t)^\top)(\eta - \eta')\|_2 \leq \frac{1}{2} \|\eta - \eta'\|_2$, see e.g. Nair (2025). $\square$

B.3. Statement and proof of Lemma 7

LEMMA 7. *We have*

$$\begin{aligned} & \mathbb{E}_{\mathcal{B}, \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w} \left(P_{\Theta_{w2s}}^s | x \right) - \tilde{w} \left(P_{\hat{\Theta}_w}^w | x \right) \right\|_2^2 \right] \right] \\ & \leq \frac{1}{4} \frac{LKG^2}{\mu^3} \mathbb{E}_{\tilde{\mathcal{B}}, \mathcal{B}} \left[\mathbb{E}_x \left[\left\| \eta_{\Theta_{1s}}^s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] \right] + 8\sqrt{2} \frac{KG^2 B \sqrt{\text{tr}(\Sigma_s)}}{\mu^2} \frac{1}{\sqrt{N}} + \frac{4C_{\text{unif}}}{\mu} \frac{1}{\sqrt{N}}. \end{aligned} \quad (73)$$

Proof: To proceed, we define the following population and empirical estimated costs associated with the W2S procedure:

$$\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta) \stackrel{\text{def}}{=} \mathbb{E}_x \mathbb{E}_{\xi \sim P_{\hat{\Theta}_w}^w(\cdot | x)} [c(\tilde{w}(P_{\Theta}^s | x), \xi)], \quad (74)$$

and

$$\hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\Theta) \stackrel{\text{def}}{=} \frac{1}{N} \sum_{j=1}^N \mathbb{E}_{\xi \sim P_{\hat{\Theta}_w}^w(\cdot | x_j)} [c(\tilde{w}(P_{\Theta}^s | x_j), \xi)]. \quad (75)$$

Let

$$\Theta_{w2s} \in \arg \min_{\Theta \in \mathcal{H}_B} \mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta)$$

denote a population minimizer of the W2S objective.

Conditioned on the labeled data $\tilde{\mathcal{B}}$ used to train the weak teacher $\hat{\Theta}_w$, we first note that by Lemma 1, we have

$$\begin{aligned} & \mathbb{E}_{\mathcal{B} | \tilde{\mathcal{B}}} \left[\frac{\mu}{2} \mathbb{E}_x \left[\left\| \tilde{w}(P_{\Theta_{w2s}}^s | x) - \tilde{w}(P_{\hat{\Theta}_w}^w | x) \right\|_2^2 \right] \middle| \tilde{\mathcal{B}} \right] \\ & \leq \mathbb{E}_{\mathcal{B} | \tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\mathbb{E}_{\xi \sim P_{\hat{\Theta}_w}^w(\cdot | x)} \left[c(\tilde{w}(P_{\Theta_{w2s}}^s | x), \xi) - c(\tilde{w}(P_{\hat{\Theta}_w}^w | x), \xi) \right] \right] \middle| \tilde{\mathcal{B}} \right] \end{aligned} \quad (76)$$

$$= \mathbb{E}_{\mathcal{B} | \tilde{\mathcal{B}}} \left[\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\hat{\Theta}_{w2s}) - \mathbb{E}_x \left[\mathbb{E}_{\xi \sim P_{\hat{\Theta}_w}^w(\cdot | x)} \left[c(\tilde{w}(P_{\hat{\Theta}_w}^w | x), \xi) \right] \right] \middle| \tilde{\mathcal{B}} \right]. \quad (77)$$

Moreover, in view of Lemma 8, we can upper bound the quantity $\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\hat{\Theta}_{w2s}) - \mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta_{w2s})$. Using the uniform absolute-deviation bound in Lemma 8, we obtain

$$\begin{aligned} & \mathbb{E}_{\mathcal{B} | \tilde{\mathcal{B}}} \left[\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\hat{\Theta}_{w2s}) - \mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta_{w2s}) \right] \\ & = \mathbb{E}_{\mathcal{B} | \tilde{\mathcal{B}}} \left[\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\hat{\Theta}_{w2s}) - \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\hat{\Theta}_{w2s}) + \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\hat{\Theta}_{w2s}) - \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\Theta_{w2s}) + \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\Theta_{w2s}) - \mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta_{w2s}) \right] \\ & \leq 2\mathbb{E}_{\mathcal{B} | \tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta) - \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\Theta) \right| \right] \\ & \leq 2\sqrt{\frac{V_0(\hat{\Theta}_w)}{N}} + 4\sqrt{2} \frac{G^2 KB}{\mu} \frac{1}{\sqrt{N}} \sqrt{\text{tr}(\Sigma_s)} \end{aligned}$$

$$\leq \frac{2C_{\text{unif}}}{\sqrt{N}} + 4\sqrt{2} \frac{G^2 KB}{\mu} \frac{1}{\sqrt{N}} \sqrt{\text{tr}(\Sigma_s)}. \quad (78)$$

Here, the first inequality uses the empirical optimality of $\hat{\Theta}_{w2s}$, and the final inequality uses $V_0(\hat{\Theta}_w) \leq C_{\text{unif}}^2$.

On the other hand, in view of the optimality of Θ_{w2s} with respect to the population objective $\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\cdot)$, we have

$$\begin{aligned} & \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta_{w2s}) - \mathbb{E}_x \left[\mathbb{E}_{\xi \sim P_{\hat{\Theta}_w}^w(\cdot|x)} \left[c\left(\tilde{w}\left(P_{\hat{\Theta}_w}^w|x\right), \xi\right) \right] \right] \middle| \tilde{\mathcal{B}} \right] \\ & \leq \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\hat{\Theta}_{1s}) - \mathbb{E}_x \left[\mathbb{E}_{\xi \sim P_{\hat{\Theta}_w}^w(\cdot|x)} \left[c\left(\tilde{w}\left(P_{\hat{\Theta}_w}^w|x\right), \xi\right) \right] \right] \middle| \tilde{\mathcal{B}} \right] \end{aligned} \quad (79)$$

$$\leq \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\frac{L}{2} \mathbb{E}_x \left[\left\| \tilde{w}(P_{\hat{\Theta}_{1s}}^s|x) - \tilde{w}(P_{\hat{\Theta}_w}^w|x) \right\|_2^2 \right] \middle| \tilde{\mathcal{B}} \right] \quad (80)$$

where (80) again follows from Lemma 1.

Combining all above and invoking Lemma 2, we get

$$\begin{aligned} & \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \tilde{w}(P_{\hat{\Theta}_{w2s}}^s|x) - \tilde{w}(P_{\hat{\Theta}_w}^w|x) \right\|_2^2 \right] \middle| \tilde{\mathcal{B}} \right] \\ & \leq \frac{1}{4} \frac{LKG^2}{\mu^3} \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\mathbb{E}_x \left[\left\| \eta_{\hat{\Theta}_{1s}}^s(x) - \eta_{\hat{\Theta}_w}^w(x) \right\|_2^2 \right] \middle| \tilde{\mathcal{B}} \right] + 8\sqrt{2} \frac{KG^2 B \sqrt{\text{tr}(\Sigma_s)}}{\mu^2} \frac{1}{\sqrt{N}} + \frac{4C_{\text{unif}}}{\mu} \frac{1}{\sqrt{N}}. \end{aligned} \quad (81)$$

Finally, taking expectation over $\tilde{\mathcal{B}}$ on both sides concludes the proof. $\square$

B.4. Statement and proof of Lemma 8

LEMMA 8. Define $C_{\text{unif}} \stackrel{\text{def}}{=} \max_{k \in [K]} |c(\tilde{w}(\text{softmax}(0)), z_k)|$. For any fixed $\hat{\Theta}_w$, we have

$$\mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta) - \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\Theta) \right| \right] \leq \frac{C_{\text{unif}}}{\sqrt{N}} + 2\sqrt{2} \frac{G^2 KB}{\mu} \frac{1}{\sqrt{N}} \sqrt{\text{tr}(\Sigma_s)}. \quad (82)$$

Proof: For brevity, we let $\phi(x) = \phi_s(x)$ be the strong feature mapping. Given the weak teacher $P_{\hat{\Theta}_w}^w$, denote $Q_k(x) = P_{\hat{\Theta}_w}^w(\xi = z_k | x)$. Given Θ , the function $g_\Theta : \mathcal{X} \rightarrow \mathbb{R}$ defined by

$$\begin{aligned} g_\Theta(x) & \stackrel{\text{def}}{=} \sum_{k=1}^K Q_k(x) c(\tilde{w}(\text{softmax}(\Theta^\top \phi(x))), z_k) \\ & = \mathbb{E}_{\xi \sim P_{\hat{\Theta}_w}^w(\cdot|x)} [c(\tilde{w}(\text{softmax}(\Theta^\top \phi(x))), \xi)] \end{aligned}$$

is $\frac{G^2 \sqrt{K}}{2\mu} \|\phi(x)\|_2$ -Lipschitz with respect to Θ . Indeed, for any Θ, Θ' , denote $w = \tilde{w}(\text{softmax}(\Theta^\top \phi(x)))$, $w' = \tilde{w}(\text{softmax}((\Theta')^\top \phi(x)))$. Then we have

$$\begin{aligned} |g_\Theta(x) - g_{\Theta'}(x)| & = \left| \sum_{k=1}^K Q_k(x) (c(w, z_k) - c(w', z_k)) \right| \\ & \leq \sum_{k=1}^K Q_k(x) |c(w, z_k) - c(w', z_k)| \\ & \leq G \sum_{k=1}^K Q_k(x) \|w - w'\|_2 \end{aligned} \quad (83)$$

$$\leq \frac{G^2 \sqrt{K}}{2\mu} \|(\Theta - \Theta')^\top \phi(x)\|_2 \quad (84)$$

$$\leq \frac{G^2 \sqrt{K}}{2\mu} \|\phi(x)\|_2 \|\Theta - \Theta'\|_2. \quad (85)$$

Inequality (83) follows since $\|\nabla_w c(w, z)\|_2 \leq G$, and Inequality (84) is due to Lemma 2.

By definition, we have $\mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta) - \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\Theta) = \mathbb{E}_x[g_\Theta(x)] - \frac{1}{N} \sum_{j=1}^N g_\Theta(x_j)$. For the proof, define

$$g_0(x) \stackrel{\text{def}}{=} g_{\Theta=0}(x) = \sum_{k=1}^K P_{\Theta_w}^w(\xi = z_k | x) c(\tilde{w}(\text{softmax}(0)), z_k), \quad V_0 \stackrel{\text{def}}{=} \text{Var}_x(g_0(x)),$$

and set $h_\Theta(x) = g_\Theta(x) - g_0(x)$. Then

$$\begin{aligned} & \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathbb{E}_x[g_\Theta(x)] - \frac{1}{N} \sum_{j=1}^N g_\Theta(x_j) \right| \right] \\ & \leq \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\left| \mathbb{E}_x[g_0(x)] - \frac{1}{N} \sum_{j=1}^N g_0(x_j) \right| \right] + \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathbb{E}_x[h_\Theta(x)] - \frac{1}{N} \sum_{j=1}^N h_\Theta(x_j) \right| \right] \\ & \leq \sqrt{\frac{V_0}{N}} + \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathbb{E}_x[h_\Theta(x)] - \frac{1}{N} \sum_{j=1}^N h_\Theta(x_j) \right| \right]. \end{aligned} \quad (86)$$

The second inequality follows from Cauchy's inequality, since

$$\mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\left| \mathbb{E}_x[g_0(x)] - \frac{1}{N} \sum_{j=1}^N g_0(x_j) \right| \right] \leq \sqrt{\mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\left(\mathbb{E}_x[g_0(x)] - \frac{1}{N} \sum_{j=1}^N g_0(x_j) \right)^2 \right]} = \sqrt{\frac{V_0}{N}}.$$

By the standard symmetrization inequality (Wainwright 2019, Chapter 4), applied to the centered class $\{h_\Theta : \Theta \in \mathcal{H}_B\}$, we have

$$\mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathbb{E}_x[h_\Theta(x)] - \frac{1}{N} \sum_{j=1}^N h_\Theta(x_j) \right| \right] \leq 2\mathbb{E}_{\mathcal{B}, \sigma|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \frac{1}{N} \sum_{j=1}^N \sigma_j h_\Theta(x_j) \right| \right]. \quad (87)$$

The class $\{h_\Theta : \Theta \in \mathcal{H}_B\}$ contains the zero function, because $h_0 \equiv 0$. Thus

$$\mathbb{E}_{\mathcal{B}, \sigma|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \frac{1}{N} \sum_{j=1}^N \sigma_j h_\Theta(x_j) \right| \right] \leq 2\mathbb{E}_{\mathcal{B}, \sigma|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sigma_j h_\Theta(x_j) \right]. \quad (88)$$

For each fixed x_j , the inner map $u \mapsto \sum_{k=1}^K Q_k(x_j) c(\tilde{w}(\text{softmax}(u)), z_k) - g_0(x_j)$ is $\frac{G^2\sqrt{K}}{2\mu}$ -Lipschitz in u and vanishes at $u = 0$. By virtue of the vector contraction inequality (Lemma 15), we conclude that

$$\begin{aligned} \mathbb{E}_{\mathcal{B}, \sigma|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sigma_j h_\Theta(x_j) \right] & \leq \sqrt{2} \frac{G^2\sqrt{K}}{2\mu} \mathbb{E}_{\mathcal{B}, \sigma|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sum_{k=1}^K \sigma_{j,k} [\Theta^\top \phi(x_j)]_k \right] \\ & \leq \sqrt{2} \frac{G^2\sqrt{K}}{2\mu} \frac{B\sqrt{K}}{\sqrt{N}} \sqrt{\text{tr}(\mathbb{E}[\phi(x)\phi(x)^\top])}, \end{aligned} \quad (89)$$

where the second inequality follows from Lemma 9. Combining (86), (87), (88), and (89) gives

$$\begin{aligned} & \mathbb{E}_{\mathcal{B}|\tilde{\mathcal{B}}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathcal{L}_{w2s}^{\hat{\Theta}_w}(\Theta) - \hat{\mathcal{L}}_{w2s}^{\hat{\Theta}_w}(\Theta) \right| \right] \\ & \leq \sqrt{\frac{V_0}{N}} + 2\sqrt{2} \frac{G^2KB}{\mu} \sqrt{\frac{\text{tr}(\Sigma_s)}{N}}. \end{aligned} \quad (90)$$

Since $Q_k(x) \geq 0$, $\sum_{k=1}^K Q_k(x) = 1$, and $|c(\tilde{w}(\text{softmax}(0)), z_k)| \leq C_{\text{unif}}$ for every k ,

$$|g_0(x)| \leq \sum_{k=1}^K Q_k(x) C_{\text{unif}} = C_{\text{unif}}.$$

Therefore $V_0 = \text{Var}_x(g_0(x)) \leq \mathbb{E}_x[g_0(x)^2] \leq C_{\text{unif}}^2$. Substituting $V_0 \leq C_{\text{unif}}^2$ into (90) yields the claimed inequality. $\square$

B.5. Proof of Lemma 4

Proof:

- We note that $\gamma_s(x)$ is a linear map of $\phi(x) = \begin{bmatrix} \phi_s(x) \\ \phi_w(x) \end{bmatrix}$, with $\gamma_s = B_s \phi$, where $B_s = \begin{bmatrix} \Lambda_s^{-1/2} V_s^\top & 0 \end{bmatrix}$. Hence, for any $u \in \mathbb{R}^{d_s}$, we have

$$\mathbb{E} \left[e^{u^\top \gamma_s(x)} \right] = \mathbb{E} \left[e^{(B_s^\top u)^\top \phi} \right] \leq \exp \left(\frac{1}{2} c^2 u^\top (B_s \Sigma_\phi B_s^\top) u \right). \quad (91)$$

But $B_s \Sigma_\phi B_s^\top = \text{Cov}(\gamma_s)$ and we know that $\text{Cov}(\gamma_s) = I_{d_s}$. Lemma 10 then implies that $\mathbb{E} \left[\|\gamma_s(x)\|_2^4 \right] \lesssim (\text{tr}(I_{d_s}))^2 = d_s^2$.

- We note that $\varepsilon(x)$ is a linear map of $\phi(x)$. To be precise, we have $\varepsilon = \gamma_w - M^\top \gamma_s = \left(\begin{bmatrix} 0 & \Lambda_w^{-\frac{1}{2}} V_w^\top \end{bmatrix} - M^\top \begin{bmatrix} \Lambda_s^{-1/2} V_s^\top & 0 \end{bmatrix} \right) \phi \stackrel{\text{def}}{=} B_\varepsilon \phi$. Hence, for any $u \in \mathbb{R}^{d_w}$, we have

$$\mathbb{E} \left[e^{u^\top \varepsilon} \right] = \mathbb{E} \left[\mathbb{E} e^{u^\top B_\varepsilon \phi} \right] = \mathbb{E} \left[e^{(B_\varepsilon^\top u)^\top \phi} \right] \leq \exp \left(\frac{1}{2} c^2 u^\top (B_\varepsilon \Sigma_\phi B_\varepsilon^\top) u \right). \quad (92)$$

We notice that $B_\varepsilon \Sigma_\phi B_\varepsilon^\top = \text{Cov}(\varepsilon) = I_{d_w} - M^\top M \preceq I_{d_w}$, which implies that $\mathbb{E} \left[\|\varepsilon(x)\|_2^4 \right] \lesssim (d_w - \|M\|_F^2)^2$.

- We note that $\phi_s(x)$ is a linear map of $\phi(x)$, with $\phi_s = B_{\phi_s} \phi$, where $B_{\phi_s} = \begin{bmatrix} I_{d_s} & 0 \end{bmatrix}$. Following the same argument as above suffices.

□

B.6. Proof of Lemma 5

Proof: To start off the analysis, by algebraic calculations, we obtain that

$$\begin{aligned} \text{tr}(\Gamma_w^\top P_s \Gamma_w) &= \text{tr}(\Gamma_w^\top P_s^\top P_s \Gamma_w) \\ &= \|P_s \Gamma_w\|_F^2 \end{aligned} \quad (93)$$

$$\begin{aligned} &= \|P_s \Gamma_s M + P_s E\|_F^2 \\ &\leq 2 \|\Gamma_s M\|_F^2 + 2 \|P_s E\|_F^2. \end{aligned} \quad (94)$$

Equation (93) holds since P_s is an orthogonal projection matrix. The last inequality follows from the fact that $\|A + B\|_F^2 \leq 2\|A\|_F^2 + 2\|B\|_F^2$ for any matrices A, B of the same dimension, and the fact that P_s is an orthogonal projection matrix, which implies $\|P_s A\|_F \leq \|A\|_F$ for any matrix A .

For the first term in (94), taking expectation over $\mathcal{B}$ yields that

$$\begin{aligned} \mathbb{E} \left[\|\Gamma_s M\|_F^2 \right] &= \mathbb{E} \left[\text{tr}(M^\top \Gamma_s^\top \Gamma_s M) \right] \\ &= \text{tr}(M^\top \mathbb{E} [\Gamma_s^\top \Gamma_s] M) \\ &= N \text{tr}(M^\top M) \end{aligned} \quad (95)$$

where the last step follows from the fact that $\mathbb{E} [\Gamma_s^\top \Gamma_s] = \sum_{j=1}^N \mathbb{E} [\gamma_s(x_j) \gamma_s(x_j)^\top] = N I_{d_s}$.

To deal with the second term in (94), we need to handle the randomness of P_s . To this end, we define the good event to be

$$\mathcal{G} = \left\{ \Gamma_s^\top \Gamma_s \succeq \frac{N}{4} I_{d_s} \right\}. \quad (96)$$

The good event $\mathcal{G}$ ensures that the matrix $\Gamma_s^\top \Gamma_s$ is invertible. It also happens with high probability. To be precise, there exist constants $c_{\text{NG}}, C_{\text{NG}} > 0$ depending on the sub-gaussian norm of $\phi_s(x)$, such that whenever $N \geq C_{\text{NG}} d_s$, we have

$$\mathbf{Pr} \left[\mathcal{G}^c \right] \leq 2 \exp(-c_{\text{NG}} N). \quad (97)$$

To see this, we first note that the rows of Γ_s are i.i.d. sub-Gaussian random vectors with identity covariance. Then, the i -th row of Γ_s is given by $\gamma_i^\top V_s$, where $\gamma_i^\top$ is the i -th row of Γ . Its covariance matrix is $\mathbb{E} \left[(\gamma_i^\top V_s)^\top \gamma_i^\top V_s \right] = V_s^\top \mathbb{E} [\gamma_i \gamma_i^\top] V_s = I_{d_s}$. By applying Theorem 5.39 in Vershynin (2010) (cf. Lemma 16) to the matrix $\Gamma_s^\top \Gamma_s$, we know that there exist absolute constants $c_{\text{NG}}, C'_{\text{NG}}$ depending on the sub-gaussian norm of the rows of Γ_s such that for any $t > 0$, with probability at least $1 - 2 \exp(-c_{\text{NG}} t^2)$, we have

$$\left\| \frac{1}{N} \Gamma_s^\top \Gamma_s - I_{d_s} \right\|_2 \leq \max(\delta, \delta^2), \quad \delta = C'_{\text{NG}} \sqrt{\frac{d_s}{N}} + \frac{t}{\sqrt{N}}. \quad (98)$$

This implies that if $\max(\delta, \delta^2) \leq \frac{3}{4}$, then

$$\frac{1}{N} \Gamma_s^\top \Gamma_s \succeq \frac{1}{4} I_{d_s} \quad (99)$$

holds, which means the event $\mathcal{G}$ happens. In order to ensure $\max(\delta, \delta^2) \leq \frac{3}{4}$, it suffices to let $\delta \leq \frac{3}{4}$. This can be achieved by choosing $t = \sqrt{N}/4$ and $N \geq 4C'_{\text{NG}} d_s$. The proof of (97) is now complete.

Under event $\mathcal{G}$, the matrix $\Gamma_s^\top \Gamma_s$ is invertible and hence we have

$$\|P_s E\|_F^2 = \text{tr}(E^\top P_s^\top P_s E) = \text{tr}\left(E^\top \Gamma_s (\Gamma_s^\top \Gamma_s)^{-1} \Gamma_s^\top E\right) \leq \frac{4}{N} \|\Gamma_s^\top E\|_F^2. \quad (100)$$

For the case when the event $\mathcal{G}$ does not happen, we have

$$\mathbb{E} \left[\|P_s E\|_F^2 \mathbb{1}[\mathcal{G}^c] \right] \leq \mathbb{E} \left[\|E\|_F^2 \mathbb{1}[\mathcal{G}^c] \right] \leq \sqrt{\mathbb{E} [\|E\|_F^4] \mathbf{Pr}[\mathcal{G}^c]}. \quad (101)$$

The first inequality follows from the fact that $\|P_s A\|_F \leq \|A\|_F$ for any matrix A as P_s is a projection matrix, and the second inequality is by Cauchy-Schwarz inequality. Combining (100) and (101), we obtain that

$$\begin{aligned} \mathbb{E}_{\mathcal{B}} \left[\|P_s E\|_F^2 \right] &= \mathbb{E}_{\mathcal{B}} \left[\|P_s E\|_F^2 \mathbb{1}[\mathcal{G}] \right] + \mathbb{E}_{\mathcal{B}} \left[\|P_s E\|_F^2 \mathbb{1}[\mathcal{G}^c] \right] \\ &\leq \frac{4}{N} \mathbb{E}_{\mathcal{B}} \left[\|\Gamma_s^\top E\|_F^2 \right] + \sqrt{\mathbb{E}_{\mathcal{B}} [\|E\|_F^4] \mathbf{Pr}[\mathcal{G}^c]}. \end{aligned} \quad (102)$$

We proceed to control the two terms in (102) respectively.

1. For the first term, by algebraic calculations, we have

$$\|\Gamma_s^\top E\|_F^2 = \left\| \sum_{j=1}^N \gamma_s(x_j) \varepsilon(x_j)^\top \right\|_F^2 = \sum_{i=1}^N \sum_{j=1}^N \gamma_s(x_i)^\top \gamma_s(x_j) \varepsilon(x_j)^\top \varepsilon(x_i). \quad (103)$$

For terms in the sum with $i \neq j$, taking expectation, we have

$$\begin{aligned} &\mathbb{E} [\gamma_s(x_i)^\top \gamma_s(x_j) \varepsilon(x_j)^\top \varepsilon(x_i)] \\ &= \mathbb{E} \left[\sum_{a=1}^{d_s} [\gamma_s(x_i)]_a [\gamma_s(x_j)]_a \sum_{b=1}^{d_w} [\varepsilon(x_j)]_b [\varepsilon(x_i)]_b \right] \\ &= \sum_{a=1}^{d_s} \sum_{b=1}^{d_w} \mathbb{E} [[\gamma_s(x_i)]_a [\varepsilon(x_i)]_b] \mathbb{E} [[\gamma_s(x_j)]_a [\varepsilon(x_j)]_b] \end{aligned} \quad (104)$$

$$= \sum_{a=1}^{d_s} \sum_{b=1}^{d_w} (\mathbb{E} [\gamma_s(x)_a \varepsilon(x)_b])^2 \quad (105)$$

$$= \|\mathbb{E} [\gamma_s \varepsilon^\top]\|_F^2 \\ = 0. \quad (106)$$

Equation (104) holds since x_i and x_j are independent for $i \neq j$. Equation (105) is due to the fact that x_i and x_j are identically distributed for any $i, j \in [N]$. The last step follows from (44). Therefore, we conclude that

$$\mathbb{E} [\|\Gamma_s^\top E\|_F^2] = \sum_{i=1}^N \mathbb{E} [\gamma_s(x_i)^\top \gamma_s(x_i) \varepsilon(x_i)^\top \varepsilon(x_i)] = N \mathbb{E} [\|\gamma_s(x)\|_2^2 \|\varepsilon(x)\|_2^2]. \quad (107)$$

After applying Cauchy-Schwarz inequality, $\mathbb{E} [\|\gamma_s(x)\|_2^2 \|\varepsilon(x)\|_2^2] \leq \sqrt{\mathbb{E} [\|\gamma_s(x)\|_2^4]} \sqrt{\mathbb{E} [\|\varepsilon(x)\|_2^4]}$, we can further use Lemma 4.

2. For the second term in (102), we observe that since $\|E\|_F^2 = \sum_{i=1}^N \|\varepsilon(x_i)\|_2^2$, it follows that $\|E\|_F^4 = \left(\sum_{i=1}^N \|\varepsilon(x_i)\|_2^2\right)^2 = \sum_{i=1}^N \|\varepsilon(x_i)\|_2^4 + 2 \sum_{i < j} \|\varepsilon(x_i)\|_2^2 \|\varepsilon(x_j)\|_2^2$. Hence, taking expectation yields that

$$\mathbb{E} [\|E\|_F^4] = N \mathbb{E} [\|\varepsilon(x)\|_2^4] + 2N(N-1) \left(\mathbb{E} [\|\varepsilon(x)\|_2^2]\right)^2. \quad (108)$$

Furthermore, we can calculate $\mathbb{E} [\|\varepsilon(x)\|_2^2]$ as follows

$$\mathbb{E} [\|\varepsilon(x)\|_2^2] = \text{tr} (\mathbb{E} [\varepsilon(x) \varepsilon(x)^\top]) = \text{tr} (I_{d_w} - M^\top M) = d_w - d_{s \wedge w}. \quad (109)$$

Then, it suffices to note that the event $\mathcal{G}^c$ happens with probability at most $2 \exp(-c_{\text{NG}} N)$ as shown in (97).

Now, combining (95) and (102) yields that

$$\begin{aligned} & \mathbb{E}_{\mathcal{B}} [\text{tr} (\Gamma_w^\top P_s \Gamma_w)] \\ & \leq 4N \text{tr} (M^\top M) + \frac{2}{N} N \sqrt{\mathbb{E} [\|\gamma_s(x)\|_2^4]} \sqrt{\mathbb{E} [\|\varepsilon(x)\|_2^4]} \\ & \quad + 2 \sqrt{N \mathbb{E} [\|\varepsilon(x)\|_2^4] + 2N(N-1) \left(\mathbb{E} [\|\varepsilon(x)\|_2^2]\right)^2} \sqrt{2 \exp(-c_{\text{NG}} N)} \end{aligned} \quad (110)$$

$$\lesssim N d_{s \wedge w} + d_s (d_w - d_{s \wedge w}) + N d_s (d_w - d_{s \wedge w}) \exp\left(-\frac{1}{2} c_{\text{NG}} N\right), \quad (111)$$

where we use Lemma 4 to conclude Inequality (110). □

B.7. Statement and proof of Lemma 9

LEMMA 9. Given a set $\mathcal{X}$, let $\phi: \mathcal{X} \rightarrow \mathbb{R}^d$. Let $\mathcal{H}_B = \{\Theta \in \mathbb{R}^{d \times K} : \|\Theta\|_F \leq B\}$ for some constant $B > 0$. Consider the vector-valued function class $\mathcal{F} = \{\Theta^\top \phi(x) : \Theta \in \mathcal{H}_B\}$. Let $\mathcal{B} = \{x_j\}_{j=1}^N$ be N i.i.d. samples from $\mathcal{D}_x$. Then we have

$$\mathbb{E}_{\mathcal{B}, \sigma} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sum_{k=1}^K \sigma_{j,k} [\Theta^\top \phi(x_j)]_k \right] \leq \frac{B\sqrt{K}}{\sqrt{N}} \sqrt{\text{tr} (\mathbb{E}_{\mathcal{D}_x} [\phi(x) \phi(x)^\top])}. \quad (112)$$

Proof: Recalling the algebraic fact that $\langle A, uv^\top \rangle_F = \text{tr}(A^\top uv^\top)$ for any matrix A and vectors u, v , we have

$$\sum_{j=1}^N \sum_{k=1}^K \sigma_{j,k} [\Theta^\top \phi(x_j)]_k = \sum_{j=1}^N \sum_{k=1}^K \sigma_{j,k} e_k^\top \Theta^\top \phi(x_j) = \sum_{j=1}^N \phi(x_j)^\top \Theta \sigma_j = \sum_{j=1}^N \langle \Theta, \phi(x_j) \sigma_j^\top \rangle_F.$$

Hence, we can rewrite the empirical Rademacher complexity of $\mathcal{F}$ as

$$\widehat{\mathfrak{R}}_N(\mathcal{F} | \mathcal{B}) = \mathbb{E}_\sigma \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \langle \Theta, \phi(x_j) \sigma_j^\top \rangle_F \right] \quad (113)$$

$$\begin{aligned} &= \frac{1}{N} \mathbb{E}_\sigma \left[\sup_{\Theta \in \mathcal{H}_B} \left\langle \Theta, \sum_{j=1}^N \phi(x_j) \sigma_j^\top \right\rangle_F \right] \\ &= \frac{B}{N} \mathbb{E}_\sigma \left[\left\| \sum_{j=1}^N \phi(x_j) \sigma_j^\top \right\|_F \right]. \end{aligned} \quad (114)$$

Indeed, the last equality is due to the following fact. Let $(\mathbb{H}, \langle \cdot, \cdot \rangle)$ be a real inner-product space with induced norm $\|h\| = \sqrt{\langle h, h \rangle}$. Let $a \in \mathbb{H}$ and $B \geq 0$. Then we have $\sup_{\|h\| \leq B} \langle h, a \rangle = B\|a\|$. Next, by Jensen's inequality, we have

$$\begin{aligned} \mathbb{E}_\sigma \left[\left\| \sum_{j=1}^N \phi(x_j) \sigma_j^\top \right\|_F \right] &\leq \left(\mathbb{E}_\sigma \left[\left\| \sum_{j=1}^N \phi(x_j) \sigma_j^\top \right\|_F^2 \right] \right)^{\frac{1}{2}} \\ &= \left(\sum_{j=1}^N \|\phi(x_j)\|_2^2 \mathbb{E}_\sigma [\|\sigma_j\|_2^2] \right)^{\frac{1}{2}} \end{aligned} \quad (115)$$

$$= \left(\sum_{j=1}^N \|\phi(x_j)\|_2^2 K \right)^{\frac{1}{2}}. \quad (116)$$

Specifically, Equation 115 is due to the following calculation:

$$\mathbb{E}_\sigma \left[\left\| \sum_{j=1}^N \phi(x_j) \sigma_j^\top \right\|_F^2 \right] = \sum_{a=1}^d \sum_{k=1}^K \sum_{i=1}^N \sum_{j=1}^N \mathbb{E}_\sigma [\phi(x_j)_a \sigma_{j,k} \phi(x_i)_a \sigma_{i,k}] = \sum_{j=1}^N \sum_{k=1}^K \sum_{a=1}^d \phi(x_j)_a^2 = \sum_{j=1}^N \|\phi(x_j)\|_2^2.$$

Equation 116 is due to the independence of Rademacher random variables $\sigma_{j,k}$ across different j and k , and the fact that $\mathbb{E}_\sigma [\|\sigma_j\|_2^2] = \sum_{k=1}^K \mathbb{E} [\sigma_{j,k}^2] = K$.

Therefore, combining (114) and (116), we have

$$\widehat{\mathfrak{R}}_N(\mathcal{F} | \mathcal{B}) \leq \frac{B}{N} \left(\sum_{j=1}^N \|\phi(x_j)\|_2^2 K \right)^{\frac{1}{2}}. \quad (117)$$

Invoking Jensen's inequality, we have

$$\mathbb{E}_\mathcal{B} \left[\left(\sum_{j=1}^N \|\phi(x_j)\|_2^2 \right)^{\frac{1}{2}} \right] \leq \sqrt{\mathbb{E}_\mathcal{B} \left[\sum_{j=1}^N \|\phi(x_j)\|_2^2 \right]} = \sqrt{N \text{tr}(\mathbb{E}_{\mathcal{D}_x} [\phi(x) \phi(x)^\top])}. \quad (118)$$

Taking expectation over $\mathcal{B}$ on both sides of (117) concludes the proof. $\square$

B.8. Proof of Lemma 3

Proof: Conditioned on the labeled data $\tilde{\mathcal{B}}$, the teacher logit map $x \mapsto \eta_{\tilde{\Theta}_w}^w(x)$ is fixed. For notational simplicity, in this proof, we denote $\phi(x) = \phi_s(x)$, $g(x) = \eta_{\tilde{\Theta}_w}^w(x)$, $L(\Theta) = L_{\text{ls}}(\Theta)$, $\hat{L}_N(\Theta) = \hat{L}_{\text{ls}}(\Theta)$,

$$\ell(x, \Theta) = \|\Theta^\top \phi(x) - g(x)\|_2^2, \quad \ell_0(x) = \ell(x, 0) = \|g(x)\|_2^2,$$

and $\bar{\ell}(x, \Theta) = \ell(x, \Theta) - \ell_0(x)$. Thus

$$\bar{\ell}(x, \Theta) = \|\Theta^\top \phi(x)\|_2^2 - 2g(x)^\top \Theta^\top \phi(x).$$

All expectations in this proof are conditioned on $\tilde{\mathcal{B}}$.

The decomposition below follows directly from the identity $\ell(x, \Theta) = \ell_0(x) + \bar{\ell}(x, \Theta)$. Indeed, using $L(\Theta) = \mathbb{E}_x[\ell(x, \Theta)]$ and $\hat{L}_N(\Theta) = N^{-1} \sum_{j=1}^N \ell(x_j, \Theta)$, subtracting the empirical average from the population average yields, for every $\Theta \in \mathcal{H}_B$,

$$\begin{aligned} L(\Theta) - \hat{L}_N(\Theta) &= \mathbb{E}_x[\ell_0(x) + \bar{\ell}(x, \Theta)] - \frac{1}{N} \sum_{j=1}^N (\ell_0(x_j) + \bar{\ell}(x_j, \Theta)) \\ &= \left(\mathbb{E}_x[\ell_0(x)] - \frac{1}{N} \sum_{j=1}^N \ell_0(x_j) \right) + \left(\mathbb{E}_x[\bar{\ell}(x, \Theta)] - \frac{1}{N} \sum_{j=1}^N \bar{\ell}(x_j, \Theta) \right). \end{aligned} \quad (119)$$

Therefore

$$\begin{aligned} &\mathbb{E}_{\mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} |L(\Theta) - \hat{L}_N(\Theta)| \right] \\ &\leq \mathbb{E}_{\mathcal{B}} \left[\left| \mathbb{E}_x[\ell_0(x)] - \frac{1}{N} \sum_{j=1}^N \ell_0(x_j) \right| \right] + \mathbb{E}_{\mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathbb{E}_x[\bar{\ell}(x, \Theta)] - \frac{1}{N} \sum_{j=1}^N \bar{\ell}(x_j, \Theta) \right| \right]. \end{aligned} \quad (120)$$

The baseline term is bounded by Cauchy-Schwarz as

$$\mathbb{E}_{\mathcal{B}} \left[\left| \mathbb{E}_x[\ell_0(x)] - \frac{1}{N} \sum_{j=1}^N \ell_0(x_j) \right| \right] \leq \sqrt{\frac{\text{Var}_x(\ell_0(x))}{N}} \leq \frac{1}{\sqrt{N}} \left(\mathbb{E}_x[\|g(x)\|_2^4] \right)^{1/2}. \quad (121)$$

Applying the standard symmetrization inequality to the function class $\{x \mapsto \bar{\ell}(x, \Theta) : \Theta \in \mathcal{H}_B\}$ gives

$$\mathbb{E}_{\mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \mathbb{E}_x[\bar{\ell}(x, \Theta)] - \frac{1}{N} \sum_{j=1}^N \bar{\ell}(x_j, \Theta) \right| \right] \leq 2\mathbb{E}_{\mathcal{B}, \sigma} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \frac{1}{N} \sum_{j=1}^N \sigma_j \bar{\ell}(x_j, \Theta) \right| \right]. \quad (122)$$

Since $\bar{\ell}(x, 0) \equiv 0$, the function class $\{x \mapsto \bar{\ell}(x, \Theta) : \Theta \in \mathcal{H}_B\}$ contains the zero function. Hence, using the symmetry of the Rademacher variables,

$$\mathbb{E}_{\mathcal{B}, \sigma} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \frac{1}{N} \sum_{j=1}^N \sigma_j \bar{\ell}(x_j, \Theta) \right| \right] \leq 2\mathbb{E}_{\mathcal{B}, \sigma} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sigma_j \bar{\ell}(x_j, \Theta) \right]. \quad (123)$$

Moreover, expanding $\bar{\ell}$ gives

$$\begin{aligned} &\mathbb{E}_{\mathcal{B}, \sigma} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sigma_j \bar{\ell}(x_j, \Theta) \right] \\ &\leq \mathbb{E}_{\sigma, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sigma_j \|\Theta^\top \phi(x_j)\|_2^2 \right] + 2\mathbb{E}_{\sigma, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \frac{1}{N} \sum_{j=1}^N \sigma_j g(x_j)^\top \Theta^\top \phi(x_j) \right| \right]. \end{aligned} \quad (124)$$

For the first term in (124), keeping the normalization from (124), we have

$$\begin{aligned} & \mathbb{E}_{\sigma, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \frac{1}{N} \sum_{j=1}^N \sigma_j \|\Theta^\top \phi(x_j)\|_2^2 \right] \\ &= \frac{1}{N} \mathbb{E}_{\sigma, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \text{tr} \left(\Theta^\top \left(\sum_{j=1}^N \sigma_j \phi(x_j) \phi(x_j)^\top \right) \Theta \right) \right] \\ &\leq \frac{1}{N} \mathbb{E}_{\sigma, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \|\Theta\|_F^2 \left\| \sum_{j=1}^N \sigma_j \phi(x_j) \phi(x_j)^\top \right\|_F \right] \end{aligned} \quad (125)$$

$$\begin{aligned} &= \frac{B^2}{N} \mathbb{E}_{\sigma, \mathcal{B}} \left[\left\| \sum_{j=1}^N \sigma_j \phi(x_j) \phi(x_j)^\top \right\|_F \right] \\ &\leq \frac{B^2}{N} \sqrt{\mathbb{E}_{\sigma, \mathcal{B}} \left[\left\| \sum_{j=1}^N \sigma_j \phi(x_j) \phi(x_j)^\top \right\|_F^2 \right]} \end{aligned} \quad (126)$$

$$= \frac{B^2}{N} \sqrt{\mathbb{E}_{\mathcal{B}} \left[\sum_{j=1}^N \|\phi(x_j) \phi(x_j)^\top\|_F^2 \right]} \quad (127)$$

$$= \frac{B^2}{\sqrt{N}} \left(\mathbb{E}_{x \sim \mathcal{D}_x} [\|\phi(x)\|_2^4] \right)^{\frac{1}{2}}. \quad (128)$$

To arrive at (125), we recall the fact that for any symmetric matrix A compatible with matrix Θ , $\text{tr}(\Theta^\top A \Theta) \leq \|A\|_2 \|\Theta\|_F^2 \leq \|A\|_F \|\Theta\|_F^2$. Inequality (126) holds due to Cauchy-Schwarz inequality. Equality (127) follows from the zero-mean property and the independence of Rademacher variables σ_j across different j .

For the second term in (124), the Frobenius identity $g(x_j)^\top \Theta^\top \phi(x_j) = \langle \Theta, \phi(x_j) g(x_j)^\top \rangle_F$ gives the normalized bound

$$\begin{aligned} \mathbb{E}_{\sigma, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \frac{1}{N} \sum_{j=1}^N \sigma_j g(x_j)^\top \Theta^\top \phi(x_j) \right| \right] &= \frac{1}{N} \mathbb{E}_{\sigma, \mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| \left\langle \Theta, \sum_{j=1}^N \sigma_j \phi(x_j) g(x_j)^\top \right\rangle_F \right| \right] \\ &\leq \frac{B}{N} \mathbb{E}_{\sigma, \mathcal{B}} \left[\left\| \sum_{j=1}^N \sigma_j \phi(x_j) g(x_j)^\top \right\|_F \right] \end{aligned} \quad (129)$$

$$\begin{aligned} &\leq \frac{B}{N} \mathbb{E}_{\mathcal{B}} \left[\sqrt{\mathbb{E}_{\sigma} \left[\left\| \sum_{j=1}^N \sigma_j \phi(x_j) g(x_j)^\top \right\|_F^2 \mid \mathcal{B} \right]} \right] \\ &= \frac{B}{N} \mathbb{E}_{\mathcal{B}} \left[\sqrt{\sum_{j=1}^N \|\phi(x_j) g(x_j)^\top\|_F^2} \right] \end{aligned} \quad (130)$$

$$= \frac{B}{N} \mathbb{E}_{\mathcal{B}} \left[\sqrt{\sum_{j=1}^N \|g(x_j)\|_2^2 \|\phi(x_j)\|_2^2} \right] \quad (131)$$

$$\leq \frac{B}{\sqrt{N}} \left(\mathbb{E}_x [\|g(x)\|_2^2 \|\phi(x)\|_2^2] \right)^{\frac{1}{2}} \quad (132)$$

$$\leq \frac{B}{\sqrt{N}} \left(\mathbb{E} [\|g(x)\|_2^4] \right)^{\frac{1}{4}} \left(\mathbb{E} [\|\phi(x)\|_2^4] \right)^{\frac{1}{4}}. \quad (133)$$

In (130), the cross terms vanish by the independence and zero-mean property of the Rademacher variables. Equality (131) uses $\|uv^\top\|_F = \|u\|_2 \|v\|_2$, and (132) follows from Cauchy-Schwarz inequality.

The first term in (120) is bounded by (121). For the second term in (120), the symmetrization step and the one-sided reduction reduce the problem to the two Rademacher terms in (124): the quadratic term is bounded by (128), and the bilinear term is bounded by (132). Therefore

$$\mathbb{E}_{\mathcal{B}} \left[\sup_{\Theta \in \mathcal{H}_B} \left| L(\Theta) - \widehat{L}_N(\Theta) \right| \right] \leq \frac{1}{\sqrt{N}} \left(\mathbb{E}_x \left[\|g(x)\|_2^4 \right] \right)^{1/2} + \frac{4B^2}{\sqrt{N}} \left(\mathbb{E}_x \left[\|\phi(x)\|_2^4 \right] \right)^{1/2} + \frac{8B}{\sqrt{N}} \left(\mathbb{E}_x \left[\|g(x)\|_2^2 \|\phi(x)\|_2^2 \right] \right)^{1/2}. \quad (134)$$

Finally, under the boundedness condition $\|\widehat{\Theta}_w\|_F \leq B_{\Theta}^w$ a.s., for every x ,

$$\left\| \eta_{\widehat{\Theta}_w}^w(x) \right\|_2 = \left\| \widehat{\Theta}_w^\top \phi_w(x) \right\|_2 \leq B_{\Theta}^w \|\phi_w(x)\|_2.$$

Hence, with $g(x) = \eta_{\widehat{\Theta}_w}^w(x)$ and $\phi(x) = \phi_s(x)$, the two terms involving g in (134) satisfy

$$\left(\mathbb{E}_x \|g(x)\|_2^4 \right)^{1/2} \leq (B_{\Theta}^w)^2 \left(\mathbb{E}_x \|\phi_w(x)\|_2^4 \right)^{1/2}, \quad \left(\mathbb{E}_x \|g(x)\|_2^2 \|\phi_s(x)\|_2^2 \right)^{1/2} \leq B_{\Theta}^w \left(\mathbb{E}_x \|\phi_w(x)\|_2^2 \|\phi_s(x)\|_2^2 \right)^{1/2}.$$

Substituting these two inequalities into (134) proves Lemma 3. $\square$

B.9. Statement and proof of Lemma 10

LEMMA 10. *Let $X \in \mathbb{R}^d$ be mean-zero with covariance $\Sigma = \text{Cov}(X) \succeq 0$. Assume there exists $c > 0$ such that for all $u \in \mathbb{R}^d$,*

$$\mathbb{E} \left[\exp(u^\top X) \right] \leq \exp \left(\frac{1}{2} c^2 u^\top \Sigma u \right). \quad (135)$$

Then

$$\mathbb{E} \|X\|_2^4 \leq 16c^4 (\text{tr}(\Sigma))^2.$$

Proof: We start from the following algebraic identity

$$\|X\|_2^4 = \left(\sum_{i=1}^d X_i^2 \right)^2 = \sum_{i=1}^d \sum_{j=1}^d X_i^2 X_j^2.$$

Taking expectation on both sides and applying Cauchy-Schwarz inequality, we have

$$\mathbb{E} \left[\|X\|_2^4 \right] = \sum_{i=1}^d \sum_{j=1}^d \mathbb{E} [X_i^2 X_j^2] \leq \sum_{i=1}^d \sum_{j=1}^d \sqrt{\mathbb{E} [X_i^4] \mathbb{E} [X_j^4]} = \left(\sum_{i=1}^d \sqrt{\mathbb{E} [X_i^4]} \right)^2. \quad (136)$$

It suffices to upper bound $\mathbb{E} [X_i^4]$ for each $i \in [d]$. To this end, we recall the following well-known tail moment identity for any non-negative random variable Z : for any $p > 0$, we have

$$\mathbb{E} [Z^p] = p \int_0^\infty s^{p-1} \mathbf{Pr} [Z > s] ds.$$

Applying the above identity to $Z = |X_i|$ and $p = 4$, we have

$$\mathbb{E} [X_i^4] = 4 \int_0^\infty s^3 \mathbf{Pr} [|X_i| > s] ds. \quad (137)$$

To proceed, we use the Chernoff method. We take $u = te_i$ in (135), where e_i is the i -th standard basis vector in $\mathbb{R}^d$ and $t > 0$ is a free parameter to be determined later. Then

$$\mathbb{E} [e^{tX_i}] \leq \exp \left(\frac{1}{2} c^2 t^2 \Sigma_{ii} \right).$$

For any $t > 0$ and $s \geq 0$, we have

$$\Pr[X_i \geq s] = \Pr[e^{tX_i} \geq e^{ts}] \leq e^{-ts} \mathbb{E}[e^{tX_i}] \leq \exp\left(\frac{1}{2}c^2t^2\Sigma_{ii} - ts\right),$$

where the first inequality is by Markov's inequality. Taking $t = \frac{s}{c^2\Sigma_{ii}}$, we have $\Pr[X_i \geq s] \leq \exp\left(-\frac{s^2}{2c^2\Sigma_{ii}}\right)$.

Hence, we conclude that

$$\Pr[|X_i| \geq s] \leq 2\exp\left(-\frac{s^2}{2c^2\Sigma_{ii}}\right). \quad (138)$$

Therefore, combining (136), (137) and (138) yields that

$$\begin{aligned} \mathbb{E}[\|X\|_2^4] &\leq \left(\sum_{i=1}^d \sqrt{\mathbb{E}[X_i^4]}\right)^2 \\ &= 4 \left(\sum_{i=1}^d \sqrt{\int_0^\infty s^3 \Pr[|X_i| > s] ds}\right)^2 \\ &\leq 4 \left(\sum_{i=1}^d \sqrt{\int_0^\infty s^3 2\exp\left(-\frac{s^2}{2c^2\Sigma_{ii}}\right) ds}\right)^2 \\ &= 4 \left(\sum_{i=1}^d 2c^2\Sigma_{ii}\right)^2 \\ &= 16c^4(\text{tr}(\Sigma))^2, \end{aligned} \quad (139)$$

$$= 16c^4(\text{tr}(\Sigma))^2, \quad (140)$$

where (139) follows from the fact that $\int_0^\infty s^3 \exp\left(-\frac{s^2}{a}\right) ds = \frac{1}{2}a^2$ for $a > 0$.

□

B.10. Statement and proof of Lemma 11

LEMMA 11. *Let $A \succeq 0$ be symmetric, Π an orthogonal projection matrix, and $C \succeq 0$ symmetric such that $C = \Pi C \Pi$. Assume that $A - \alpha \Pi \succeq 0$ for some $\alpha > 0$. Then we have*

$$\text{tr}(ACA) \geq \alpha^2 \text{tr}(C). \quad (141)$$

Proof: We denote $H = A - \alpha \Pi \succeq 0$. We can write

$$\begin{aligned} ACA &= (\alpha \Pi + H) C (\alpha \Pi + H) \\ &= \alpha^2 \Pi C \Pi + \alpha (\Pi C H + H C \Pi) + H C H. \end{aligned}$$

Since $C = \Pi C \Pi$ and Π is an orthogonal projection, we have $\Pi C = C$ and $C \Pi = C$. Therefore,

$$ACA = \alpha^2 C + \alpha (CH + HC) + HCH. \quad (142)$$

Taking trace on both sides, we have

$$\text{tr}(ACA) = \alpha^2 \text{tr}(C) + 2\alpha \text{tr}(CH) + \text{tr}(HCH).$$

Since $H \succeq 0$ and $C \succeq 0$, we have $\text{tr}(HC) = \text{tr}\left(C^{\frac{1}{2}} H C^{\frac{1}{2}}\right) \geq 0$ and $\text{tr}(HCH) = \text{tr}\left(C^{\frac{1}{2}} H^2 C^{\frac{1}{2}}\right) \geq 0$. Hence, we have $\text{tr}(ACA) \geq \alpha^2 \text{tr}(C)$, which concludes the proof.

□

B.11. A no-free-lunch bound for zero overlap

The following lemma shows that, under exact realizability, the overlap dimension cannot be arbitrarily small.

LEMMA 12. *When $\rho_s = \rho_w = 0$, we have*

$$d_{s \wedge w} \geq \frac{1}{\lambda_{\max}(\Sigma_w)} \frac{\left\| \Sigma_s^{\frac{1}{2}} \Theta_s^* \right\|_F^2}{\left\| \Theta_w^* \right\|_F^2}. \quad (143)$$

Proof: Since $\rho_s = \rho_w = 0$, we have $\eta^*(x) = (\Theta_s^*)^\top \phi_s(x) = (\Theta_w^*)^\top \phi_w(x) = 0$ almost surely with respect to x . We denote $\mathbb{E}[\phi_s \phi_w^\top] = \Sigma_{sw}$. Hence, we can compute $\mathbb{E}[\phi_s(\eta^*)^\top]$ in two ways:

$$\mathbb{E}[\phi_s(\eta^*)^\top] = \mathbb{E}[\phi_s \phi_s^\top] \Theta_s^* = \Sigma_s \Theta_s^* \quad \text{and} \quad \mathbb{E}[\phi_s(\eta^*)^\top] = \mathbb{E}[\phi_s \phi_w^\top] \Theta_w^* = \Sigma_{sw} \Theta_w^*.$$

Hence, we have $\Sigma_s \Theta_s^* = \Sigma_{sw} \Theta_w^*$. Left multiplying both sides by $\Sigma_s^{-\frac{1}{2}}$, we have $\Sigma_s^{\frac{1}{2}} \Theta_s^* = \Sigma_s^{-\frac{1}{2}} \Sigma_{sw} \Theta_w^*$. Denote $\Pi_w = V_w V_w^\top$. Since $\phi_w(x) = \Pi_w \phi_w(x)$ almost surely, replacing Θ_w^* by $\Pi_w \Theta_w^*$ leaves the weak logits unchanged, so we can assume without loss of generality that $\Theta_w^* = \Pi_w \Theta_w^*$. Then we have $\Sigma_s^{\frac{1}{2}} \Theta_s^* = \Sigma_s^{-\frac{1}{2}} \Sigma_{sw} \Pi_w \Theta_w^*$. Denote $B = \Sigma_s^{-\frac{1}{2}} \Sigma_{sw} \Pi_w$. Then the fact that Frobenius norm admits submultiplicativity implies that

$$\left\| \Sigma_s^{\frac{1}{2}} \Theta_s^* \right\|_F \leq \|B\|_F \|\Theta_w^*\|_F. \quad (144)$$

On the other hand, we can compute the overlap dimension $d_{s \wedge w}$ as follows:

$$d_{s \wedge w} = \left\| \Sigma_s^{-\frac{1}{2}} \Sigma_{sw} \Sigma_w^{-\frac{1}{2}} \right\|_F^2 = \left\| \Sigma_s^{-\frac{1}{2}} \Sigma_{sw} \Pi_w \Sigma_w^{-\frac{1}{2}} \right\|_F^2 = \left\| B \Sigma_w^{-\frac{1}{2}} \right\|_F^2 = \text{tr} \left(\Sigma_w^{-\frac{1}{2}} B^\top B \Sigma_w^{-\frac{1}{2}} \right), \quad (145)$$

where the second equality follows from the fact that

$$\Pi_w \Sigma_w^{-\frac{1}{2}} = V_w V_w^\top V_w \Lambda_w^{-\frac{1}{2}} V_w^\top = V_w \Lambda_w^{-\frac{1}{2}} V_w^\top = \Sigma_w^{-\frac{1}{2}},$$

by simply noting that $V_w^\top V_w = I_{d_w}$. Next, we can verify that $\Sigma_w^{-\frac{1}{2}} \succeq \frac{1}{\sqrt{\lambda_{\max}(\Sigma_w)}} \Pi_w$. Indeed, by writing $\Sigma_w^{-\frac{1}{2}} - \frac{1}{\sqrt{\lambda_{\max}(\Sigma_w)}} \Pi_w = V_w \Lambda_w^{-\frac{1}{2}} V_w^\top - \frac{1}{\sqrt{\lambda_{\max}(\Sigma_w)}} V_w V_w^\top = V_w \left(\Lambda_w^{-\frac{1}{2}} - \frac{1}{\sqrt{\lambda_{\max}(\Sigma_w)}} I_{d_w} \right) V_w^\top$, we can verify that $\Lambda_w^{-\frac{1}{2}} - \frac{1}{\sqrt{\lambda_{\max}(\Sigma_w)}} I_{d_w}$ is a diagonal matrix with non-negative entries, and hence is positive semidefinite.

Therefore, invoking Lemma 11 on (145), we have

$$d_{s \wedge w} \geq \frac{1}{\lambda_{\max}(\Sigma_w)} \|B\|_F^2 \geq \frac{1}{\lambda_{\max}(\Sigma_w)} \frac{\left\| \Sigma_s^{\frac{1}{2}} \Theta_s^* \right\|_F^2}{\left\| \Theta_w^* \right\|_F^2}. \quad (146)$$

The last inequality follows from (144). $\square$

Appendix C: Omitted Proofs in Section 4

C.1. Proof of Theorem 2

Proof: Step 1: lower bound the decision risk by $O\left(\rho_s + \lambda_{\min}(\Lambda_s) \|\Delta\theta\|_2^2\right)$. By Lemma 1, we have for any x ,

$$\mathbb{E}_{\xi \sim P^*(\cdot|x)} \left[c\left(\tilde{w}\left(P_{\hat{\Theta}_s}^s | x\right), \xi\right) - c\left(\tilde{w}\left(P^* | x\right), \xi\right) \right] \geq \frac{\mu}{2} \left\| \tilde{w}\left(P_{\hat{\Theta}_s}^s | x\right) - \tilde{w}\left(P^* | x\right) \right\|_2^2. \quad (147)$$

We claim that under Assumption (1), we have

$$\left\| \tilde{w}\left(P_{\hat{\Theta}_s}^s | x\right) - \tilde{w}\left(P^* | x\right) \right\|_2 \geq \gamma \frac{e^{-2B\eta}}{K} \left\| \eta_{\hat{\Theta}_s}^s(x) - \eta^*(x) \right\|_2. \quad (148)$$

To see this, for brevity, for any logit vector $\eta \in \mathbb{R}^K$, we denote

$$s(\eta)_k \stackrel{\text{def}}{=} \frac{e^{\eta_k}}{\sum_{j=1}^K e^{\eta_j}} \text{ for } k \in [K]$$

to be the k th coordinate of the softmax function evaluated at η . If $\|\eta\|_\infty \leq B_\eta$, then for any k , $s(\eta)_k = \frac{e^{\eta_k}}{\sum_j e^{\eta_j}} \geq \frac{e^{-B_\eta}}{K e^{B_\eta}} = \frac{e^{-2B_\eta}}{K}$. If $\|\eta_0\|_\infty \leq B_\eta$ and $\|\eta_1\|_\infty \leq B_\eta$, then the entire segment $\eta(t) = (1-t)\eta_0 + t\eta_1$ also satisfies $\|\eta(t)\|_\infty \leq B_\eta$ for $t \in [0, 1]$, hence $\min_{t \in [0, 1]} \min_{k \in [K]} s(\eta(t))_k \geq \frac{e^{-2B_\eta}}{K}$. Then (48) implies that

$$\min_{t \in [0, 1]} \min_{k \in [K]} s(\eta^*(x) + t(\eta_{\widehat{\Theta}_s}(x) - \eta^*(x)))_k \geq \frac{e^{-2B_\eta}}{K}. \quad (149)$$

To proceed, we invoke the following lemma.

LEMMA 13. *Given $\eta_0, \eta_1 \in \mathbb{R}^K$ satisfying $\mathbf{1}^\top \eta_0 = 0$ and $\mathbf{1}^\top \eta_1 = 0$. Along the segment $\eta(t) = (1-t)\eta_0 + t\eta_1$ for $t \in [0, 1]$, the induced probability distributions satisfy*

$$\min_{k \in [K]} s(\eta(t))_k \geq \alpha, \quad \forall t \in [0, 1] \quad (150)$$

for some constant $\alpha > 0$. Then we have

$$\|s(\eta_1) - s(\eta_0)\|_2 \geq \alpha \|\eta_1 - \eta_0\|_2. \quad (151)$$

Proof: Let $v = \eta_1 - \eta_0$, so $\mathbf{1}^\top v = 0$. Define $p(t) = s(\eta(t))$. By the fundamental theorem of calculus, we have

$$s(\eta_1) - s(\eta_0) = \int_0^1 \frac{d}{dt} s(\eta(t)) dt = \int_0^1 J(\eta(t)) v dt, \quad (152)$$

where $J(\eta) = \text{Diag}(p) - pp^\top$ is the Jacobian matrix of the softmax function at η .

Recall $\Pi = I_K - \frac{1}{K} \mathbf{1} \mathbf{1}^\top$ is the projection onto the subspace orthogonal to $\mathbf{1}$. Next, we claim that for every $t \in [0, 1]$,

$$J(\eta(t)) \succeq \alpha \Pi. \quad (153)$$

Suppose the claim holds for now. Then immediately implies that $M \stackrel{\text{def}}{=} \int_0^1 J(\eta(t)) dt \succeq \alpha \Pi$. Since $v^\top \mathbf{1} = 0$, we have $\Pi v = v$. Hence, $v^\top \Pi v = v^\top v = \|v\|_2^2$. Then $M \succeq \alpha \Pi$ implies that $v^\top M v \geq \alpha v^\top \Pi v = \alpha \|v\|_2^2$. By Cauchy-Schwarz inequality, we have

$$\|Mv\|_2 \geq \frac{v^\top M v}{\|v\|_2} \geq \alpha \|v\|_2. \quad (154)$$

Noting that $s(\eta_1) - s(\eta_0) = Mv$, the claim then follows.

Now, we return to prove (153). In fact, for any arbitrary $x \in \mathbb{R}^K$, we decompose it as $x = \Pi x + \frac{1}{K}(\mathbf{1}^\top x)\mathbf{1}$. Then, we notice that

$$x^\top J(\eta(t)) x = (\Pi x + \frac{1}{K}(\mathbf{1}^\top x)\mathbf{1})^\top J(\eta(t)) (\Pi x + \frac{1}{K}(\mathbf{1}^\top x)\mathbf{1}) = (\Pi x)^\top J(\eta(t)) (\Pi x). \quad (155)$$

The last step holds since $J(\eta(t))\mathbf{1} = \text{diag}(p)\mathbf{1} - pp^\top \mathbf{1} = p - p = 0$. Let $m \stackrel{\text{def}}{=} \min_{k \in [K]} s(\eta(t))_k \geq \alpha$. Using Lemma 14 and the fact that $\mathbf{1}^\top \Pi x = 0$, we have

$$x^\top J(\eta(t)) x = (\Pi x)^\top J(\eta(t)) (\Pi x) \geq m \|\Pi x\|_2^2 \geq \alpha x^\top \Pi x, \quad (156)$$

which is exactly (153).

□

In view of Lemma 13 and the condition given in (47), the proof of (148) is now complete. Combining (147) and (148) yields that

$$\mathbb{E}_{\xi \sim P^*(\cdot|x)} \left[c \left(\tilde{w} \left(P_{\hat{\Theta}_s}^s | x \right), \xi \right) - c \left(\tilde{w} \left(P^* | x \right), \xi \right) \right] \geq \frac{1}{2} \mu \left(\gamma \frac{e^{-2B\eta}}{K} \right)^2 \left\| \eta_{\hat{\Theta}_s}^s(x) - \eta^*(x) \right\|_2^2. \quad (157)$$

Next, we proceed to show that

$$\mathbb{E}_{\tilde{\mathcal{B}}, x} \left[\left\| \eta_{\hat{\Theta}_s}^s(x) - \eta^*(x) \right\|_2^2 \right] \geq \rho_s + \mathbb{E}_{\tilde{\mathcal{B}}, x} \left[\left\| \left(\hat{\Theta}_s - \Theta_s^* \right)^\top \phi_s(x) \right\|_2^2 \right]. \quad (158)$$

Indeed, we note that

$$\begin{aligned} \left\| \eta_{\hat{\Theta}_s}^s(x) - \eta^*(x) \right\|_2^2 &= \left\| \left(\hat{\Theta}_s - \Theta_s^* \right)^\top \phi_s(x) \right\|_2^2 + \left\| (\Theta_s^*)^\top \phi_s(x) - \eta^*(x) \right\|_2^2 \\ &\quad + 2 \left\langle \left(\hat{\Theta}_s - \Theta_s^* \right)^\top \phi_s(x), (\Theta_s^*)^\top \phi_s(x) - \eta^*(x) \right\rangle. \end{aligned} \quad (159)$$

For the cross term, taking expectation over x and $\tilde{\mathcal{B}}$, we further have

$$\begin{aligned} &\mathbb{E}_{\tilde{\mathcal{B}}} \left[\mathbb{E}_{x|\tilde{\mathcal{B}}} \left[\left\langle \left(\hat{\Theta}_s - \Theta_s^* \right)^\top \phi_s(x), (\Theta_s^*)^\top \phi_s(x) - \eta^*(x) \right\rangle \middle| \tilde{\mathcal{B}} \right] \right] \\ &= \mathbb{E}_{\tilde{\mathcal{B}}} \left[\mathbb{E}_{x|\tilde{\mathcal{B}}} \left[\left\langle \hat{\Theta}_s - \Theta_s^*, \phi_s(x) \left((\Theta_s^*)^\top \phi_s(x) - \eta^*(x) \right)^\top \right\rangle_F \middle| \tilde{\mathcal{B}} \right] \right] \\ &= \mathbb{E}_{\tilde{\mathcal{B}}} \left[\left\langle \hat{\Theta}_s - \Theta_s^*, \mathbb{E}_{x|\tilde{\mathcal{B}}} \left[\phi_s(x) \left((\Theta_s^*)^\top \phi_s(x) - \eta^*(x) \right)^\top \middle| \tilde{\mathcal{B}} \right] \right\rangle_F \right] \\ &= \mathbb{E}_{\tilde{\mathcal{B}}} \left[\left\langle \hat{\Theta}_s - \Theta_s^*, \mathbb{E}_x \left[\phi_s(x) \left((\Theta_s^*)^\top \phi_s(x) - \eta^*(x) \right)^\top \right] \right\rangle_F \right]. \end{aligned}$$

Now we recall that Θ_s^* is defined as the minimizer of

$$\Theta_s^* = \arg \min_{\Theta \in \mathbb{R}^{d_\phi \times K}, \Theta \mathbf{1} = 0} L(\Theta), \quad (160)$$

where $L(\Theta) = \mathbb{E}_x \left[\left\| \Theta^\top \phi_s(x) - \eta^*(x) \right\|_2^2 \right]$ is a convex function of Θ . We denote $\mathcal{C} = \{\Theta \in \mathbb{R}^{d_\phi \times K} : \Theta \mathbf{1} = 0\}$. By the first-order optimality condition of convex optimization, we have

$$\langle \nabla L(\Theta_s^*), \Theta - \Theta_s^* \rangle_F \geq 0, \quad \forall \Theta \in \mathcal{C}. \quad (161)$$

By noting that

$$\nabla L(\Theta) = 2 \mathbb{E}_x \left[\phi_s(x) \left(\Theta^\top \phi_s(x) - \eta^*(x) \right)^\top \right],$$

we have thus proved (158).

Denote $\Delta\Theta = \hat{\Theta}_s - \Theta_s^* \in \mathbb{R}^{d_\phi \times K}$ and $\Delta\theta = \text{vec}(\hat{\Theta}_s - \Theta_s^*) \in \mathbb{R}^{d_\phi K}$.

We note that by the cyclic property and linearity property of trace, we have

$$\begin{aligned} \mathbb{E} \left[\left\| \Delta\Theta^\top \phi_s(x) \right\|_2^2 \middle| \tilde{\mathcal{B}} \right] &= \mathbb{E}_x \left[\phi_s(x)^\top \Delta\Theta \Delta\Theta^\top \phi_s(x) \middle| \tilde{\mathcal{B}} \right] \\ &= \mathbb{E}_x \left[\text{tr} \left(\phi_s(x)^\top \Delta\Theta \Delta\Theta^\top \phi_s(x) \right) \middle| \tilde{\mathcal{B}} \right] \\ &= \mathbb{E}_x \left[\text{tr} \left(\Delta\Theta \Delta\Theta^\top \phi_s(x) \phi_s(x)^\top \right) \middle| \tilde{\mathcal{B}} \right] \\ &= \text{tr} \left(\Delta\Theta \Delta\Theta^\top \mathbb{E}_x \left[\phi_s(x) \phi_s(x)^\top \middle| \tilde{\mathcal{B}} \right] \right) \\ &= \text{tr} \left(\Delta\Theta \Delta\Theta^\top \Sigma_s \right). \end{aligned} \quad (162)$$

The last equation holds since x is sampled independent of $\tilde{\mathcal{B}}$. Recall that for any compatible matrices A, B, X , we have $\text{tr}(AXBX^\top) = \text{vec}(X)^\top (B^\top \otimes A) \text{vec}(X)$. Taking $X = \Delta\Theta$, $A = \Sigma_s$, $B = I_K$, we have

$$\begin{aligned} \mathbb{E} \left[\|\Delta\Theta^\top \phi_s(x)\|_2^2 \mid \tilde{\mathcal{B}} \right] &= \Delta\theta^\top (I_K \otimes \Sigma_s) \Delta\theta \\ &= \Delta\theta^\top (I_K \otimes V_s) (I_K \otimes \Lambda_s) (I_K \otimes V_s^\top) \Delta\theta \end{aligned} \quad (163)$$

$$\begin{aligned} &= \left\| \left(I_K \otimes \Lambda_s^{1/2} \right) (I_K \otimes V_s^\top) \Delta\theta \right\|_2^2 \\ &\geq \lambda_{\min}(\Lambda_s) \left\| (I_K \otimes V_s^\top) \Delta\theta \right\|_2^2 \end{aligned} \quad (164)$$

$$= \lambda_{\min}(\Lambda_s) \Delta\theta^\top (I_K \otimes V_s V_s^\top) \Delta\theta. \quad (165)$$

Equation (163) follows from the property of Kronecker product that $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ valid for shape-compatible matrices A, B, C, D . Equation (164) is due to the simple fact that $(I_K \otimes \Lambda_s^{1/2})^\top = I_K \otimes (\Lambda_s^{1/2})^\top$. For the inequality, we note that the eigenvalues of $I_K \otimes \Lambda_s^{1/2}$ are given by $\lambda_j(I_K) \cdot \lambda_i(\Lambda_s^{1/2}) = \lambda_i(\Lambda_s^{1/2})$ for $i = 1, 2, \dots, d_s$ and $j = 1, 2, \dots, K$. Hence, the minimum eigenvalue of $\Lambda_s^{1/2} \otimes I_K$ is equal to $\lambda_{\min}(\Lambda_s^{1/2}) = \sqrt{\lambda_{\min}(\Lambda_s)}$.

It remains to relate the projected quadratic form in (165) to $\|\Delta\theta\|_2^2$. Since $\Sigma_s = \mathbb{E}[\phi_s(x)\phi_s(x)^\top] = V_s \Lambda_s V_s^\top$,

$$\begin{aligned} \mathbb{E} \left\| (I - V_s V_s^\top) \phi_s(x) \right\|_2^2 &= \text{tr} \left([(I - V_s V_s^\top) \mathbb{E}[\phi_s(x)\phi_s(x)^\top]] \right) \\ &= \text{tr} \left([(I - V_s V_s^\top) V_s \Lambda_s V_s^\top] \right) = 0, \end{aligned}$$

where $(I - V_s V_s^\top) V_s = 0$. Since the left-hand side is nonnegative, $\phi_s(x) = V_s V_s^\top \phi_s(x)$ almost surely. Hence, for any Θ , $(V_s V_s^\top \Theta)^\top \phi_s(x) = \Theta^\top V_s V_s^\top \phi_s(x) = \Theta^\top \phi_s(x)$ almost surely. Moreover, for any $\Theta \in \mathcal{C}$, $(V_s V_s^\top \Theta) \mathbf{1} = V_s V_s^\top (\Theta \mathbf{1}) = 0$. Therefore, it is without loss of generality to assume that $\Theta_s^* = V_s V_s^\top \Theta_s^*$ and $\hat{\Theta}_s = V_s V_s^\top \hat{\Theta}_s$. Subtracting the two projected identities and using the algebraic identity $\text{vec}(ABC) = (C^\top \otimes A) \text{vec}(B)$, we have

$$\Delta\Theta = \hat{\Theta}_s - \Theta_s^* = V_s V_s^\top \Delta\Theta, \quad \Delta\theta = \text{vec}(V_s V_s^\top \Delta\Theta) = (I_K \otimes V_s V_s^\top) \Delta\theta.$$

Hence, $\Delta\theta^\top (I_K \otimes V_s V_s^\top) \Delta\theta = \Delta\theta^\top \Delta\theta = \|\Delta\theta\|_2^2$. We hence have shown that $\mathbb{E} \left[\|\Delta\Theta^\top \phi_s(x)\|_2^2 \mid \tilde{\mathcal{B}} \right] \geq \lambda_{\min}(\Lambda_s) \|\Delta\theta\|_2^2$.

Step 2: lower bound $\|\Delta\theta\|_2$.

Let

$$F_n(\Theta) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n c \left(\tilde{w}(P_\Theta^s(\tilde{x}_i)), \tilde{\xi}_i \right).$$

Let $\Pi = I_K - \frac{1}{K} \mathbf{1}\mathbf{1}^\top \in \mathbb{R}^{K \times K}$ be the projection matrix to the subspace $\{v \in \mathbb{R}^K : v^\top \mathbf{1} = 0\}$. We denote $\Theta(t) \stackrel{\text{def}}{=} \Theta_s^* + t(\hat{\Theta}_s - \Theta_s^*) \in \mathbb{R}^{d_\phi \times K}$ for $t \in [0, 1]$. Since $\hat{\Theta}_s = \arg \min_{\Theta \in \mathcal{C}} F_n(\Theta)$, the KKT conditions imply that there exists a multiplier $\nu \in \mathbb{R}^{d_\phi}$ such that

$$\nabla_\Theta F_n(\hat{\Theta}_s) + \nu \mathbf{1}^\top = 0, \quad \text{and} \quad \hat{\Theta}_s \mathbf{1} = 0. \quad (166)$$

Right-multiplying the above condition by Π and using the fact that $\mathbf{1}^\top \Pi = 0$, we have

$$\nabla_\Theta F_n(\hat{\Theta}_s) \Pi = 0. \quad (167)$$

Applying the fundamental theorem of calculus to $g(\Theta(t)) \stackrel{\text{def}}{=} \text{vec}(\nabla_{\Theta} F_n(\Theta(t))\Pi)$, we have

$$g(\hat{\Theta}_s) - g(\Theta_s^*) = \int_0^1 \frac{d}{dt} g(\Theta(t)) dt = \int_0^1 (\Pi \otimes I_{d_\phi}) \nabla_{\Theta}^2 F_n(\Theta(t)) \Delta\theta dt. \quad (168)$$

We denote $H_n \stackrel{\text{def}}{=} \int_0^1 (\Pi \otimes I_{d_\phi}) \nabla_{\Theta}^2 F_n(\Theta(t)) dt \in \mathbb{R}^{d_\phi K \times d_\phi K}$ and $\varphi_n \stackrel{\text{def}}{=} g(\Theta_s^*) = \text{vec}(\nabla_{\Theta} F_n(\Theta_s^*)\Pi)$. Equation (167) and (168) hence imply that

$$0 - \varphi_n = H_n \Delta\theta. \quad (169)$$

By Assumption (49), we have

$$\|H_n\|_2 \leq \int_0^1 \|(\Pi \otimes I_{d_\phi}) \nabla_{\Theta}^2 F_n(\Theta(t))\|_2 dt \leq \int_0^1 \|(\Pi \otimes I_{d_\phi})\|_2 \|\nabla_{\Theta}^2 F_n(\Theta(t))\|_2 dt \leq L_H. \quad (170)$$

The last inequality follows because $\|\Pi \otimes I\|_2 = \|\Pi\|_2 = 1$. Hence, we obtain

$$\|\varphi_n\|_2 = \|H_n \Delta\theta\|_2 \leq \|H_n\|_2 \|\Delta\theta\|_2 \leq L_H \|\Delta\theta\|_2. \quad (171)$$

Continuing from (165) and taking expectation with respect to $\tilde{\mathcal{B}}$, we have

$$\mathbb{E} \left[\|\Delta\Theta^\top \phi_s(x)\|_2^2 \right] = \mathbb{E}_{\tilde{\mathcal{B}}} \left[\mathbb{E} \left[\|\Delta\Theta^\top \phi_s(x)\|_2^2 \mid \tilde{\mathcal{B}} \right] \right] \geq \frac{\lambda_{\min}(\Lambda_s)}{L_H^2} \mathbb{E}_{\tilde{\mathcal{B}}} \left[\|\varphi_n\|_2^2 \right]. \quad (172)$$

Step 3: combine everything.

First, we claim that

$$\nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right) \Pi = \nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right). \quad (173)$$

To see this, we first note that $\nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right) \Pi = \nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right) (I_K - \frac{1}{K} \mathbf{1}\mathbf{1}^\top)$. Hence, it suffices to show that $\nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right) \mathbf{1} = 0$.

Given any $u \in \mathbb{R}^{d_\phi}$, we let $U = u\mathbf{1}^\top \in \mathbb{R}^{d_\phi \times K}$. Define the scalar function of one variable t by $\psi(t) \stackrel{\text{def}}{=} c \left(\tilde{w} \left(P_{\Theta_s^* + tU}^s(x) \right), \xi \right)$. Then for every t ,

$$(\Theta_s^* + tu\mathbf{1}^\top)^\top \phi(x) = \Theta_s^{*\top} \phi(x) + t\mathbf{1}u^\top \phi(x)$$

so the logits are shifted by the same constant $tu^\top \phi(x)$ across all classes. Since softmax is invariant to such shifts, we have $P_{\Theta_s^* + tU}^s(x) = P_{\Theta_s^*}^s(x)$. Therefore, $\psi(t)$ is a constant function in t . So we have $\psi'(0) = 0$. By the chain rule, one can verify that

$$0 = \psi'(0) = \left\langle \nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right), U \right\rangle_F = \left\langle \nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right) \mathbf{1}, u \right\rangle.$$

Since u is arbitrary, it follows that

$$\nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_s^*}^s(x) \right), \xi \right) \mathbf{1} = 0.$$

Hence, we have proved (173).

Now, we proceed to lower bound $\mathbb{E}_{\tilde{\mathcal{B}}} \left[\|\varphi_n\|_2^2 \right]$. Recalling the definition $\varphi_n \stackrel{\text{def}}{=} \text{vec} \left(\nabla_{\Theta} \frac{1}{n} \sum_{i=1}^n c \left(\tilde{w} \left(P_{\Theta}^s(\tilde{x}_i) \right), \tilde{\xi}_i \right) \Pi \right)$, we have

$$\begin{aligned} \mathbb{E}_{\tilde{\mathcal{B}}} \left[\|\varphi_n\|_2^2 \right] &= \text{tr}(\text{Var}_{\tilde{\mathcal{B}}}(\varphi_n)) + \|\mathbb{E}_{\tilde{\mathcal{B}}}[\varphi_n]\|_2^2 \\ &\geq \text{tr}(\text{Var}_{\tilde{\mathcal{B}}}(\varphi_n)) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n} \text{tr} \left(\text{Var}_{\tilde{B}} \left(\text{vec} \left(\nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_*}^s(\tilde{x}_i) \right), \tilde{\xi}_i \right) \Pi \right) \right) \right) \\
&= \frac{1}{n} \text{tr} \left(\text{Var}_{\tilde{B}} \left(\text{vec} \left(\nabla_{\Theta} c \left(\tilde{w} \left(P_{\Theta_*}^s(\tilde{x}_i) \right), \tilde{\xi}_i \right) \right) \right) \right) \tag{174} \\
&= \frac{1}{n} \text{tr} (\Sigma_{\varphi}). \tag{175}
\end{aligned}$$

The first equality follows from the fact that for any psd matrix M and random vector Z , we have $\mathbb{E}[Z^\top M Z] = \text{tr}(M \text{Var}(Z)) + \mathbb{E}[Z]^\top M \mathbb{E}[Z] \geq \text{tr}(M \text{Var}(Z))$. Equation (174) follows from (173). $\square$

C.2. Proof of Theorem 3

Proof: We denote $P_{\Theta}^w(x) = \text{softmax}(\Theta^\top \phi_w(x)) \in \mathbb{R}^K$ and p_{Θ}^w to be the corresponding probability mass function. We let

$$\ell(\Theta; \tilde{x}_i, \tilde{\xi}_i) \stackrel{\text{def}}{=} -\ln p_{\Theta}^w(\tilde{\xi}_i | \tilde{x}_i) = -\eta_{\Theta, \tilde{\xi}_i}^w(\tilde{x}_i) + \ln \left(\sum_{k=1}^K e^{\eta_{\Theta, k}^w(\tilde{x}_i)} \right). \tag{176}$$

With slight abuse of notation, we denote $\theta = \text{vec}(\Theta) \in \mathbb{R}^{d_{\phi} K}$ to be the vectorized version of Θ . We also write

$$\tilde{\mathcal{C}}_B \stackrel{\text{def}}{=} \{\text{vec}(\Theta) : \Theta \in \mathcal{C}_B\}, \quad \tilde{\mathcal{C}} \stackrel{\text{def}}{=} \{\text{vec}(\Delta) : \Delta \in \mathbb{R}^{d_{\phi} \times K}, \Delta \mathbf{1} = 0\}.$$

In this proof, we denote $F(\Theta) = \mathbb{E}[\ell(\Theta; \tilde{x}_i, \tilde{\xi}_i)]$ and $\hat{F}_n(\Theta) = \frac{1}{n} \sum_{i=1}^n \ell(\Theta; \tilde{x}_i, \tilde{\xi}_i)$. We recall that Θ_w^* is the minimizer of (15). Under Assumption 5, Θ_w^* is also the minimizer of $\arg \min_{\Theta \in \mathcal{C}_B} F(\Theta)$, by noting that

$$\arg \min_{\Theta \in \mathcal{C}_B} F(\Theta) = \arg \min_{\Theta \in \mathcal{C}_B} \mathbb{E}[-\ln p_{\Theta}^w(\tilde{\xi}_i | \tilde{x}_i)] = \arg \min_{\Theta \in \mathcal{C}_B} \mathbb{E}[\text{KL}(P^*(\tilde{\xi}_i | \tilde{x}_i) \parallel P_{\Theta}^w(\tilde{\xi}_i | \tilde{x}_i))]. \tag{177}$$

Assumption 5 also implies Σ_w is nonsingular. Suppose instead that $v \neq 0$ and $v^\top \Sigma_w v = 0$; since $\Sigma_w = \mathbb{E}[\phi_w(x) \phi_w(x)^\top]$, this gives $v^\top \phi_w(x) = 0$ almost surely. Choose $a \neq 0$ with $a^\top \mathbf{1} = 0$, possible since $K \geq 2$. Then $va^\top \mathbf{1} = 0$, so the centering constraint in $\mathcal{C}_B$ is preserved, and $(va^\top)^\top \phi_w(x) = a v^\top \phi_w(x) = 0$ almost surely. Hence $\Theta_w^* + \epsilon va^\top$ leaves $\Theta^\top \phi_w(x)$ unchanged and induces the same conditional model as Θ_w^* . Since Θ_w^* is interior, $\Theta_w^* + \epsilon va^\top \in \mathcal{C}_B$ for all sufficiently small $\epsilon \neq 0$, giving a distinct point with the same value of F and contradicting uniqueness; hence $\lambda_{\min}(\Sigma_w) > 0$ and $d_w = d_{\phi}$.

Under the assumption that $\|\Theta_w^*\|_F < B_{\Theta}$, there exists $\nu \in \mathbb{R}^{d_{\phi}}$ such that the KKT conditions hold:

$$\nabla_{\Theta} F(\Theta_w^*) + \nu \mathbf{1}^\top = 0, \quad \text{and} \quad \Theta_w^* \mathbf{1} = 0. \tag{178}$$

Hence, right multiplying $\mathbf{1}$ on both sides yields $\nabla_{\Theta} F(\Theta_w^*) \mathbf{1} + \nu \mathbf{1}^\top \mathbf{1} = \nabla_{\Theta} F(\Theta_w^*) \mathbf{1} + \nu K = 0$. One can show that $\nabla_{\Theta} \ell(\Theta; \tilde{x}_i, \tilde{\xi}_i) = \phi_w(\tilde{x}_i) (\text{softmax}(\Theta^\top \phi_w(\tilde{x}_i)) - e_{\tilde{\xi}_i})^\top$. Hence, $\nabla_{\Theta} F(\Theta_w^*) \mathbf{1} = \mathbb{E}[\nabla_{\Theta} \ell(\Theta_w^*; \tilde{x}_i, \tilde{\xi}_i) \mathbf{1}] = 0$. Therefore, it has to be that $\nu = 0$ in the KKT condition (178). Hence, the KKT condition (178) reduces to

$$\nabla_{\Theta} F(\Theta_w^*) = 0, \quad \text{and} \quad \Theta_w^* \mathbf{1} = 0. \tag{179}$$

Recalling the formula $\text{vec}(uv^\top) = v \otimes u$, we get $\nabla_{\theta} \ell(\theta; \tilde{x}_i, \tilde{\xi}_i) \stackrel{\text{def}}{=} \text{vec}(\nabla_{\Theta} \ell(\Theta; \tilde{x}_i, \tilde{\xi}_i)) = (\text{softmax}(\Theta^\top \phi_w(\tilde{x}_i)) - e_{\tilde{\xi}_i}) \otimes \phi_w(\tilde{x}_i)$. Denote $J(p) = \text{Diag}(p) - pp^\top \in \mathbb{R}^{K \times K}$ for probability vector $p \in \Delta_K$. One can verify that the Hessian in vector θ is given by

$$\nabla_{\theta}^2 \ell(\theta; \tilde{x}_i, \tilde{\xi}_i) = J(p_{\Theta}^w(\tilde{x}_i)) \otimes (\phi_w(\tilde{x}_i) \phi_w(\tilde{x}_i)^\top). \tag{180}$$

We denote $\theta^* = \text{vec}(\Theta_w^*)$ and $\hat{\theta} = \text{vec}(\hat{\Theta}_w)$. We let $\delta = \hat{\theta} - \theta^*$ and $\Delta = \hat{\Theta}_w - \Theta_w^*$. By convexity of the objective $\hat{F}_n$ and the feasible region $\tilde{\mathcal{C}}_B$, the optimality of $\hat{\Theta}_w$ implies that

$$(\theta - \hat{\theta})^\top \nabla_\theta \hat{F}_n(\hat{\theta}) \geq 0, \quad \forall \theta \in \tilde{\mathcal{C}}_B.$$

In particular, taking $\theta = \theta^* \in \tilde{\mathcal{C}}_B$, we have

$$\delta^\top \nabla_\theta \hat{F}_n(\hat{\theta}) \leq 0. \quad (181)$$

By the fundamental theorem of calculus, we have

$$\nabla_\theta \hat{F}_n(\hat{\theta}) - \nabla_\theta \hat{F}_n(\theta^*) = \int_0^1 \nabla_\theta^2 \hat{F}_n(\theta^* + t\delta) dt \cdot \delta. \quad (182)$$

For brevity, we denote $H_n = \int_0^1 \nabla_\theta^2 \hat{F}_n(\theta^* + t\delta) dt$.

Combining (181) and (182), we have

$$\delta^\top H_n \delta + \delta^\top \nabla_\theta \hat{F}_n(\theta^*) = \delta^\top \nabla_\theta \hat{F}_n(\hat{\theta}) \leq 0. \quad (183)$$

Let the good event $\mathcal{G}$ be defined as

$$\mathcal{G} = \left\{ \inf_{\theta \in \tilde{\mathcal{C}}_B} \inf_{u \in \tilde{\mathcal{C}}, \|u\|_2=1} u^\top \nabla_\theta^2 \hat{F}_n(\theta) u \geq \frac{1}{2} \mu_w \right\} \quad (184)$$

where μ_w is defined after (51). On the good event $\mathcal{G}$, we have $\delta^\top H_n \delta \geq \frac{1}{2} \mu_w \|\delta\|_2^2$, which in turn implies that $\frac{1}{2} \mu_w \|\delta\|_2^2 \leq -\delta^\top \nabla_\theta \hat{F}_n(\theta^*) \leq \|\delta\|_2 \left\| \nabla_\theta \hat{F}_n(\theta^*) \right\|_2$ and hence

$$\|\delta\|_2^2 \leq \frac{4}{\mu_w^2} \left\| \nabla_\theta \hat{F}_n(\theta^*) \right\|_2^2. \quad (185)$$

Therefore, by conditioning on the good event $\mathcal{G}$, we conclude that

$$\begin{aligned} \mathbb{E} \left[\|\delta\|_2^2 \right] &= \mathbb{E} \left[\|\delta\|_2^2 \mathbb{1}[\mathcal{G}] + \|\delta\|_2^2 \mathbb{1}[\mathcal{G}^c] \right] \\ &\leq \frac{4}{\mu_w^2} \mathbb{E} \left[\left\| \nabla_\theta \hat{F}_n(\theta^*) \right\|_2^2 \mathbb{1}[\mathcal{G}] \right] + \mathbb{E} \left[\|\delta\|_2^2 \mathbb{1}[\mathcal{G}^c] \right] \end{aligned} \quad (186)$$

$$\leq \frac{4}{\mu_w^2} \mathbb{E} \left[\left\| \nabla_\theta \hat{F}_n(\theta^*) \right\|_2^2 \right] + 4B_\Theta^2 \mathbf{Pr}[\mathcal{G}^c]. \quad (187)$$

Inequality (186) follows from (185). Inequality (187) follows from the fact that both $\hat{\Theta}_w$ and Θ_w^* lie in $\mathcal{C}_B$.

To control $\mathbb{E} \left[\left\| \nabla_\theta \hat{F}_n(\theta^*) \right\|_2^2 \right]$, we observe that

$$\begin{aligned} \mathbb{E} \left[\left\| \nabla_\theta \hat{F}_n(\theta^*) \right\|_2^2 \right] &= \mathbb{E} \left[\left\| \frac{1}{n} \sum_{i=1}^n \nabla_\theta \ell(\theta^*; \tilde{x}_i, \tilde{\xi}_i) \right\|_2^2 \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{E} \left[\left\| \nabla_\theta \ell(\theta^*; \tilde{x}_i, \tilde{\xi}_i) \right\|_2^2 \right] + \frac{2}{n^2} \mathbb{E} \left[\sum_{i < j} \left\langle \nabla_\theta \ell(\theta^*; \tilde{x}_i, \tilde{\xi}_i), \nabla_\theta \ell(\theta^*; \tilde{x}_j, \tilde{\xi}_j) \right\rangle \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{E} \left[\left\| \nabla_\theta \ell(\theta^*; \tilde{x}_i, \tilde{\xi}_i) \right\|_2^2 \right], \end{aligned} \quad (188)$$

where the last equality follows from the independence of the samples and the fact that $\mathbb{E} \left[\nabla_\theta \ell(\theta^*; \tilde{x}_i, \tilde{\xi}_i) \right] = \nabla_\theta F(\theta^*) = 0$ by (179). Now, using the algebraic identity $\|a \otimes b\|_2 = \|a\|_2 \|b\|_2$, we have

$$\mathbb{E} \left[\left\| \nabla_\theta \ell(\theta^*; \tilde{x}_i, \tilde{\xi}_i) \right\|_2^2 \right] = \mathbb{E} \left[\left\| \left(\text{softmax} \left((\Theta_w^*)^\top \phi_w(\tilde{x}_i) \right) - e_{\tilde{\xi}_i} \right) \otimes \phi_w(\tilde{x}_i) \right\|_2^2 \right]$$

$$= \mathbb{E} \left[\left\| \text{softmax} \left((\Theta_w^*)^\top \phi_w(\tilde{x}_i) \right) - e_{\tilde{\xi}_i} \right\|_2^2 \right]. \quad (189)$$

To proceed, for brevity, we denote $p \stackrel{\text{def}}{=} \text{softmax} \left((\Theta_w^*)^\top \phi_w(\tilde{x}_i) \right) \in \mathbb{R}^K$ and p_k to be its k th coordinate. We denote $e_{\tilde{\xi}}$ to be the one-hot encoding in $\mathbb{R}^K$ with 1 in the $\tilde{\xi}$ -th coordinate and 0 elsewhere. We note that conditional on $\tilde{x}_i$,

$$\mathbb{E} \left[\|p - e_{\tilde{\xi}_i}\|_2^2 \mid \tilde{x}_i \right] = \sum_{k=1}^K p_k \|p - e_k\|_2^2 = \sum_{k=1}^K p_k (\|p\|_2^2 + 1 - 2p_k) = 1 - \|p\|_2^2 \leq 1, \quad (190)$$

and hence

$$\mathbb{E} \left[\left\| \nabla_{\theta} \ell \left(\theta^*; \tilde{x}_i, \tilde{\xi}_i \right) \right\|_2^2 \right] = \mathbb{E} \left[\|\phi_w(\tilde{x}_i)\|_2^2 \mathbb{E} \left[\|p - e_{\tilde{\xi}_i}\|_2^2 \mid \tilde{x}_i \right] \right] \leq \mathbb{E} \left[\|\phi_w(\tilde{x}_i)\|_2^2 \right] = \text{tr}(\Sigma_w). \quad (191)$$

It remains to bound $\mathbf{Pr}[\mathcal{G}^c]$ in (187). To this end, we introduce the notation $\hat{\Sigma}_n \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \phi_w(\tilde{x}_i) \phi_w(\tilde{x}_i)^\top$.

We proceed to show that whenever $\lambda_{\min}(\hat{\Sigma}_n) \geq \frac{1}{2} \lambda_{\min}(\Sigma_w)$, then for any $\theta \in \tilde{\mathcal{C}}_B$ and every $\delta \in \tilde{\mathcal{C}}$,

$$\delta^\top \nabla_{\theta}^2 \hat{F}_n(\theta) \delta \stackrel{(*)}{\geq} \frac{e^{-2B_{\Theta} B_{\phi}}}{K} \lambda_{\min}(\hat{\Sigma}_n) \|\delta\|_2^2 \geq \frac{1}{2} \frac{e^{-2B_{\Theta} B_{\phi}}}{K} \lambda_{\min}(\Sigma_w) \|\delta\|_2^2, \quad (192)$$

which implies that the good event $\mathcal{G}$ holds. A standard matrix Chernoff lower tail bound ensures that for $\varepsilon \in (0, 1)$,

$$\mathbf{Pr} \left[\lambda_{\min}(\hat{\Sigma}_n) \leq (1 - \varepsilon) \lambda_{\min}(\Sigma_w) \right] \leq d_w \exp \left(-\frac{n \lambda_{\min}(\Sigma_w) \varepsilon^2}{2B_{\phi}^2} \right). \quad (193)$$

Taking $\varepsilon = \frac{1}{2}$ yields that

$$\mathbf{Pr}[\mathcal{G}^c] \leq d_w \exp \left(-\frac{n \lambda_{\min}(\Sigma_w)}{8B_{\phi}^2} \right). \quad (194)$$

Now, we turn to prove the inequality marked as $(*)$ above in (192). For any $\delta \in \tilde{\mathcal{C}}$,

$$\begin{aligned} \delta^\top \nabla_{\theta}^2 \hat{F}_n(\theta) \delta &= \frac{1}{n} \sum_{i=1}^n \delta^\top \nabla_{\theta}^2 \ell(\theta; \tilde{x}_i, \tilde{\xi}_i) \delta \\ &\stackrel{(a)}{=} \frac{1}{n} \sum_{i=1}^n \delta^\top \left(J(p_{\Theta}^w(\tilde{x}_i)) \otimes (\phi_w(\tilde{x}_i) \phi_w(\tilde{x}_i)^\top) \right) \delta \\ &\stackrel{(b)}{=} \frac{1}{n} \sum_{i=1}^n \text{tr}(\phi_w(\tilde{x}_i) \phi_w(\tilde{x}_i)^\top \Delta J(p_{\Theta}^w(\tilde{x}_i)) \Delta^\top) \\ &\stackrel{(c)}{=} \frac{1}{n} \sum_{i=1}^n (\Delta^\top \phi_w(\tilde{x}_i))^\top J(p_{\Theta}^w(\tilde{x}_i)) (\Delta^\top \phi_w(\tilde{x}_i)) \\ &\stackrel{(d)}{\geq} \frac{e^{-2B_{\Theta} B_{\phi}}}{K} \frac{1}{n} \sum_{i=1}^n \|\Delta^\top \phi_w(\tilde{x}_i)\|_2^2. \end{aligned} \quad (195)$$

In Equation (a), we plug in the expression of the Hessian (180). Equation (b) follows from the algebraic identity $\text{vec}(X)^\top (A \otimes B) \text{vec}(X) = \text{tr}(X^\top B X A)$ for compatible matrices A, B, X . Equation (c) is due to the cyclic property of trace. To conclude Inequality (d), we need the following Lemma.

LEMMA 14. *Let $p \in \Delta_K = \{u \in \mathbb{R}^K : u_k \geq 0, \sum_{k=1}^K u_k = 1\}$. Let $m = \min_{k \in [K]} p_k$. We define $J(p) = \text{diag}(p) - pp^\top \in \mathbb{R}^{K \times K}$. Then for every $u \in \mathbb{R}^K$ satisfying $\mathbf{1}^\top u = 0$, we have*

$$u^\top J(p) u \geq m \|u\|_2^2. \quad (196)$$

Proof: Denote $q = p - m\mathbf{1} \in \mathbb{R}^K$, so $q_k \geq 0$ and $\sum_{k=1}^K q_k = 1 - Km$. For any u with $\mathbf{1}^\top u = 0$, we have $p^\top u = (m\mathbf{1} + q)^\top u = q^\top u$. Then

$$\begin{aligned} u^\top (\text{diag}(p) - pp^\top) u &= \sum_{k=1}^K p_k u_k^2 - (p^\top u)^2 \\ &= m \sum_{k=1}^K u_k^2 + \sum_{k=1}^K q_k u_k^2 - (q^\top u)^2 \\ &\geq m \|u\|_2^2 + \sum_{k=1}^K q_k u_k^2 - (1 - Km) \sum_{k=1}^K q_k u_k^2 \end{aligned} \quad (197)$$

$$\begin{aligned} &= m \|u\|_2^2 + Km \sum_{k=1}^K q_k u_k^2 \\ &\geq m \|u\|_2^2. \end{aligned} \quad (198)$$

Inequality (197) follows from the Cauchy-Schwarz inequality. $\square$

We note that given $\|\Theta\|_F \leq B_\Theta$ and $\|\phi_w(x)\|_2 \leq B_\phi$, it is straightforward to verify that for every $x \in \mathcal{X}$,

$$\min_{k \in [K]} P_\Theta^w(\xi = z_k | x) \geq \frac{e^{-2B_\Theta B_\phi}}{K}, \quad (199)$$

as $P_\Theta^w(\xi = z_k | x) = \frac{e^{e_k^\top \Theta^\top \phi_w(x)}}{\sum_{j=1}^K e^{e_j^\top \Theta^\top \phi_w(x)}} \geq \frac{e^{-B_\Theta B_\phi}}{K e^{B_\Theta B_\phi}}$. Moreover, by noting that $(\Delta^\top \phi_w(\tilde{x}_i))^\top \mathbf{1} = \phi_w(\tilde{x}_i)^\top \Delta \mathbf{1} = 0$, Lemma 14 implies that Inequality (d) holds.

Then, continuing from (195), by cyclic and linear properties of trace, we have

$$\frac{1}{n} \sum_{i=1}^n \|\Delta^\top \phi_w(\tilde{x}_i)\|_2^2 = \text{tr} \left(\Delta^\top \left(\frac{1}{n} \sum_{i=1}^n \phi_w(\tilde{x}_i) \phi_w(\tilde{x}_i)^\top \right) \Delta \right) \geq \lambda_{\min}(\hat{\Sigma}_n) \|\Delta\|_F^2 = \lambda_{\min}(\hat{\Sigma}_n) \|\delta\|_2^2,$$

where the inequality follows from the fact that for any real symmetric matrix M , $\text{tr}(X^\top M X) \geq \lambda_{\min}(M) \|X\|_F^2$ for any compatible matrix X .

The proof is now complete. $\square$

Appendix D: Additional Details for Section 5.2

Pretraining The task-pretraining stage uses 2000 labeled comments sampled from the pretraining split of the augmented comment moderation data. The weak model is trained by cross-entropy for 500 optimizer steps with batch size 32, learning rate $5 \cdot 10^{-5}$, and weight decay 0.01. The strong model is trained by cross-entropy for 100 optimizer steps with batch size 32, learning rate $2 \cdot 10^{-5}$, and weight decay 0.01. In both cases, all encoder parameters and classifier-head parameters are trainable. The resulting weak model initializes the downstream weak-teacher runs, and the resulting strong model initializes the downstream strong-only and W2S runs.

Cost matrix The route-severity cost matrix used in the comment moderation experiment is

	$\xi = 0$	$\xi = 1$	$\xi = 2$
allow	0.000	1.000	3.000
remove	2.050	1.050	3.050
standard review	0.300	2.700	11.573
hold review	2.400	0.800	3.218
priority review	2.850	0.967	1.706

where $\xi = 0, 1, 2$ denote non-toxic, toxic, and severely toxic comments, respectively. The entries combine exposure cost, delay cost, human-review cost, user-friction cost from suppressing benign comments, and penalties for automatic toxic removals. For example, when $\xi = 2$, the matrix assigns cost 3.000 to automatic approval, 3.050 to automatic removal, and 1.706 to priority review. This reflects the following modeling choice: automatic approval leaves harmful content visible, while automatic removal avoids exposure but is penalized for making an enforcement decision without human review. Priority review is cheapest in this column because it handles the severe comment quickly while still using human review. The W2S framework does not rely on this particular numerical specification; other applications or moderation policies can be represented by replacing this matrix with the appropriate route-severity cost matrix.

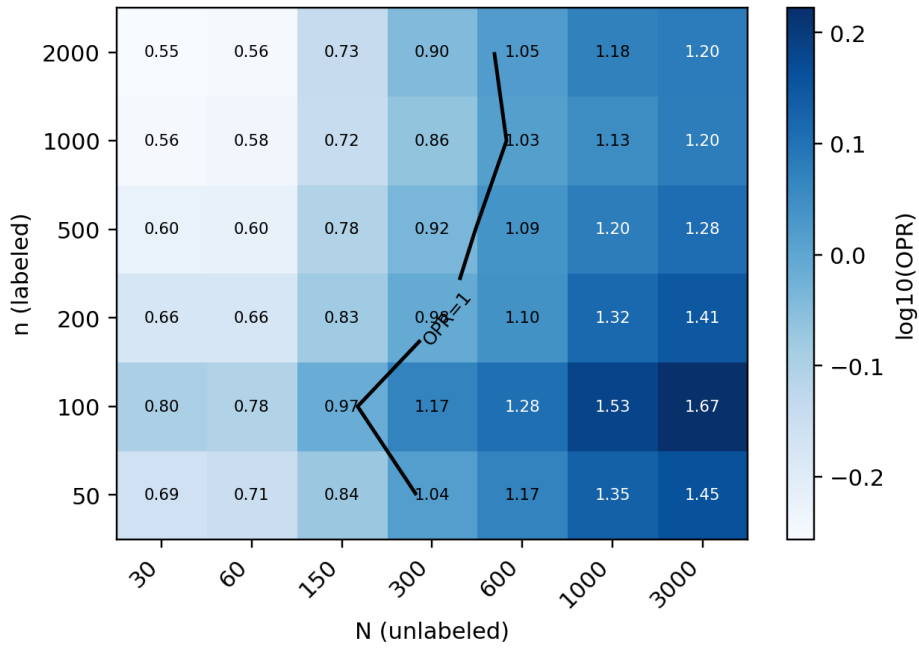


Figure 5 The heatmap reports the log-scaled empirical hard-route OPR over the labeled and unlabeled sample-size grid; the contour marks empirical hard-route OPR equal to one.

Appendix E: Standard Facts

LEMMA 15 (Corollary 4 in (Maurer 2016)). *Let $\mathcal{X}$ be any set, and fix samples $x_1, \dots, x_n \in \mathcal{X}$. Denote ℓ_2 the real Hilbert space of square-summable sequences equipped with the norm $\|u\|_2 := \left(\sum_{k \geq 1} u_k^2\right)^{1/2}$. For any function class $\mathcal{F}$ mapping $\mathcal{X}$ to ℓ_2 , we write $f(x) = (f_k(x))_{k \geq 1}$. Let $\mathcal{F}$ be any class of functions $f : \mathcal{X} \rightarrow \ell_2$. Let $h_i : \ell_2 \rightarrow \mathbb{R}$ for $i = 1, \dots, n$ be L -Lipschitz with respect to $\|\cdot\|_2$, i.e. $|h_i(u) - h_i(v)| \leq L\|u - v\|_2$ for all $u, v \in \ell_2$, and all i . Let $\varepsilon = (\varepsilon_i)_{i=1}^n$ be independent Rademacher random variables (each takes values in $\{-1, +1\}$ with probability $1/2$), and let $\varepsilon' = (\varepsilon_{ik})_{i=1, \dots, n; k \geq 1}$ be an independent doubly indexed family of Rademacher variables. Then*

$$\mathbb{E}_\varepsilon \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^n \varepsilon_i h_i(f(x_i)) \right] \leq \sqrt{2} L \mathbb{E}_{\varepsilon'} \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^n \sum_{k \geq 1} \varepsilon_{ik} f_k(x_i) \right]. \quad (200)$$

LEMMA 16 (**Theorem 5.39 in Vershynin (2010)**). Assume that A is an $N \times n$ real matrix whose rows A_i are independent sub-gaussian random vectors in $\mathbb{R}^n$ with second moment matrix $\Sigma = \mathbb{E}[A_i A_i^\top]$. Then for every $t \geq 0$, the following inequality holds with probability at least $1 - 2\exp(-ct^2)$:

$$\left\| \frac{1}{N} A^\top A - \Sigma \right\|_2 \leq \max(\delta, \delta^2), \quad \text{where } \delta = C \sqrt{\frac{n}{N}} + \frac{t}{\sqrt{N}}. \quad (201)$$

Here C and c are constants that depend only on the sub-gaussian norm of the rows A_i .

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